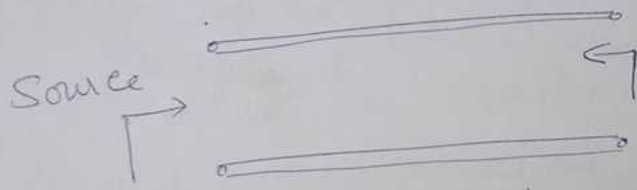
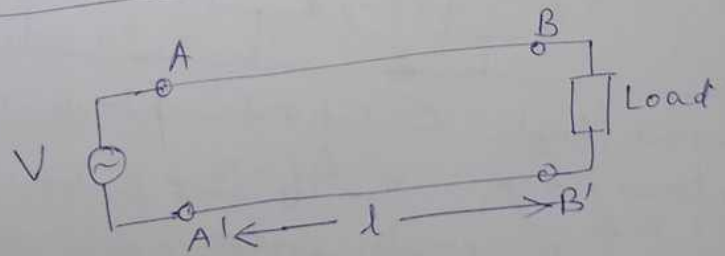
Unit I Transmission line

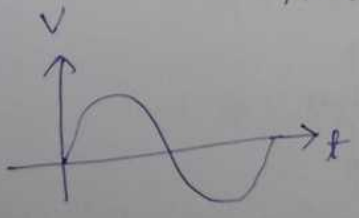
Transmission line is basically a means to transfer information from the source to destination. A source may be transmitter or destination may be an antenna or vice-versa. The transmission line is a bounded media like a waveguide. Transmission line is a two-conductor system. The common example of transmission line is coaxial cable, parallel wire transmission line, microstrip line



A simple two wire T-L

Why KVL and Ohm's law fails at high frequencies





frequency = f

Time Period

The voltage applied at AA' doesn't appears at BB' at same instant because no signal travel with the speed of light.

if T_r is the transit time $\quad T_r = \frac{l}{v}$ ← velocity of transmission

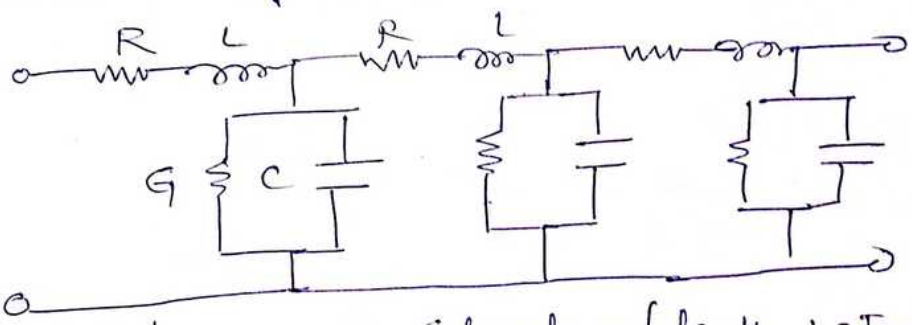
if Time period $T \gg T_r$ [we can ignore transit time effect]

$\Rightarrow T \gg \frac{l}{v}$

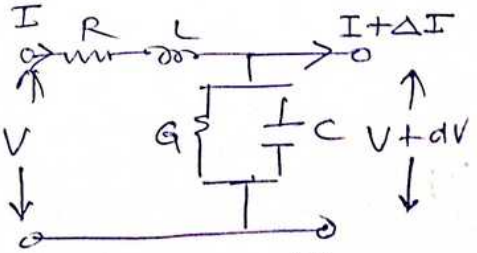
$\Rightarrow \frac{1}{f} \gg l$

$\Rightarrow$ $\boxed{\text{wavelength } \lambda = \frac{v}{f} \gg l}$ if this condition is satisfied we can neglect the transit time effect

Equivalent Circuit of a T.L.



R → Resistance/length . C = Capacitance/length. L = Inductance/length
G = Conductance/length. For a unit section



$\Rightarrow \Delta V = -(R+j\omega L)\Delta x I \Rightarrow \frac{\Delta V}{\Delta x} = -(R+j\omega L)I \Rightarrow \frac{dV}{dx} = -(R+j\omega L)$ ✓ — ①

and $\Delta I = -(G+j\omega C)\Delta x V \Rightarrow \frac{\Delta I}{\Delta x} = -(G+j\omega C)I \Rightarrow \frac{dI}{dx} = -(G+j\omega C)I$ — ②

differentiating eq^n ① and ② ⇒ we get.

$$\frac{d^2V}{dx^2} = \gamma^2 V \quad \text{——— (3)}$$

and $\frac{d^2I}{dx^2} = \gamma^2 I$, here, γ: Propagation Constant — (4)

$$\Rightarrow \gamma^2 = \left\{\sqrt{(R+j\omega L)(G+j\omega C)}\right\}^2$$

$$\Rightarrow \gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$$

The solution of eq^n (1) is

$$V(x) = V^+ e^{-\gamma x} + V^- e^{+\gamma x}$$

Here V^+ and V^- are complex constants in the forward and backward directions.

If we find out the instantaneous value then we have to multiply each term by $e^{j\omega t}$

so, $V(x,t) = V^+ e^{-\gamma x} e^{j\omega t} + V^- e^{\gamma x} e^{j\omega t}$

Now $\gamma = \alpha + j\beta$

α = Attenuation Constant (Neper/m) (dB/m)

and β = Phase Constant (rad/m)

$$\Rightarrow V(x,t) = V^+ e^{-\alpha x} e^{j(\omega t - \beta x)}$$ (Neglecting Back word wave)

Now for $\alpha = 0$

$$\Rightarrow V(x,t) = V^+ e^{j(\omega t - \beta x)}$$

The $V(x,t) = Re\{V^+ e^{j(\omega t - \beta x)}\}$

$V(x,t) = V^+ \cos(\omega t - \beta x)$

if Travelling wave phenomenon in the $+x$ direction

if Similarly for current

$I(x,t) = I^+ \cos(\omega t - \beta x)$

Again shows a travelling wave phenomen in current.

So voltages and currents exists in the form travelling waves on a transmission line

The characteristics of these waves are governe by the propagation constant γ.

Now, $\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$

$= \alpha + j\beta$

As we know the travelling wave in forward direct is $V^+ e^{-\gamma x}$

$= V^+ e^{-(\alpha+j\beta)x}$

$= |V^+| e^{-\alpha x} e^{-j\beta x}$

$e^{-\alpha x}$ ← Control the amp

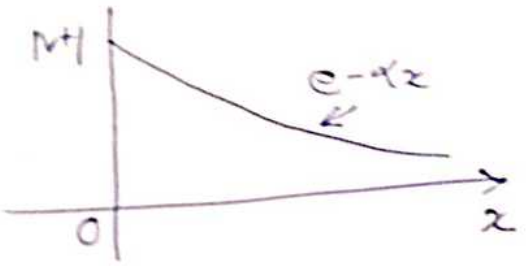
$e^{-j\beta x}$ → Control the Phase

differentiating eq (1) and (2) ↓

space phase $= -\beta x$

β = Phase Change / length $= \frac{2\pi}{\lambda}$

For amplitude



1 Nepel = 8.68 dB/m

Now $\frac{dV}{dx} = -(R+j\omega L)I$

$$\Rightarrow \frac{d}{dx}\{V^+e^{-\gamma x} + V^-e^{-\gamma x}\} = -(R+j\omega L)\{I^+e^{-\gamma x} + I^-e^{\gamma x}\}$$

$$\Rightarrow -\gamma V^+e^{-\gamma x} + \gamma V^-e^{\gamma x} = -(R+j\omega L)\{I^+e^{-\gamma x} + I^-e^{\gamma x}\}$$

So, $-\gamma V^+e^{-\gamma x} = -(R+j\omega L)I^+e^{-\gamma x}$ [For forward wave]

and $\gamma V^-e^{\gamma x} = -(R+j\omega L)I^-e^{\gamma x}$ [For backward wave]

$$\frac{V^+}{I^+} = \frac{(R+j\omega L)}{\gamma} = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

$$\frac{V^-}{I^-} = -\frac{(R+j\omega L)}{\gamma} = -\sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

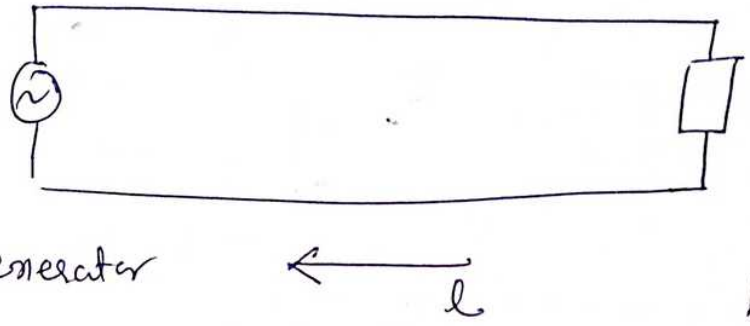
$\Rightarrow$ Characteristics Parameters of the line

So, Characteristics Impedance $Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$

Z_0 governs the energy flow on the T.L.

$$V = V^+ e^{-\gamma x} + V^- e^{\gamma x}$$

$$I = \frac{V^+}{Z_0} e^{-\gamma x} - \frac{V^-}{Z_0} e^{\gamma x}$$



As the l is increasing in the -ive direction of x.

so, $V = V^+ e^{\gamma l} + V^- e^{-\gamma l}$

and $I = \frac{V^+}{Z_0} e^{\gamma l} - \frac{V^-}{Z_0} e^{-\gamma l}$

$l = 0$ corresponds to the load

Boundary condition

At $l = 0$, $Z = Z_L$

$$Z_L = \left.\frac{V}{I}\right|_{l=0} = Z_0 \frac{V^+ + V^-}{V^+ - V^-}$$

Reflection Coefficient = Ratio of back travelling wave to the incident wave.

$$\Gamma(l) = \frac{V^- e^{-\gamma l}}{V^+ e^{\gamma l}}$$

At $l = 0$, $\Gamma(0) = \frac{V^-}{V^+}$

$$\Rightarrow Z_L = Z_0 \left\{ \frac{1 + \Gamma(0)}{1 - \Gamma(0)} \right\}$$

$Z_L [1 - \Gamma(0)] = Z_0 [1 + \Gamma(0)]$

$\Rightarrow Z_L - Z_L \Gamma(0) = Z_0 + Z_0 \Gamma(0)$

$\Rightarrow Z_L - Z_0 = Z_L \Gamma(0) + Z_0 \Gamma(0)$

$\Rightarrow \boxed{\Gamma(0) = \frac{Z_L - Z_0}{Z_L + Z_0}}$

differentiating eq (1) w.r.t. x,

at load $\Gamma_L \equiv \Gamma(0) = \dfrac{Z_L - Z_0}{Z_L + Z_0} = \dfrac{V^-}{V^+}$

$\Gamma_L = \dfrac{V^-}{V^+} = 0$, when $Z_L = Z_0$ (Matched Condition)

$$V(l) = V^+ e^{\gamma l}\{1 + \Gamma(l)\}$$

$$I(l) = \frac{V^+}{Z_0} e^{-\gamma l}\{1 - \Gamma(l)\}$$

$$\Rightarrow Z(l) = \frac{V(l)}{I(l)} = Z_0\left\{\frac{1 + \Gamma(l)}{1 - \Gamma(l)}\right\}$$

$$\Gamma(l) = \frac{Z(l) - Z_0}{Z(l) + Z_0}$$

and $$Z(l) = Z_0\left\{\frac{1 + \dfrac{V^- e^{-\gamma l}}{V^+ e^{\gamma l}}}{1 - \dfrac{V^- e^{-\gamma l}}{V^+ e^{\gamma l}}}\right\}$$

$$\Rightarrow Z(l) = Z_0\left\{\frac{1 + \Gamma_l e^{-2\gamma l}}{1 - \Gamma_l e^{-2\gamma l}}\right\} = Z_0\left\{\frac{1 + \left[\dfrac{Z_L - Z_0}{Z_L + Z_0}\right] e^{-2\gamma l}}{1 - \left[\dfrac{Z_L - Z_0}{Z_L + Z_0}\right] e^{-2\gamma l}}\right\}$$

$$\Rightarrow Z(l) = Z_0\left[\frac{Z_L \cosh\gamma l + Z_0 \sinh\gamma l}{Z_0 \cosh\gamma l + Z_L \sinh\gamma l}\right]$$

where $\cosh\gamma l = \dfrac{e^{\gamma l} + e^{-\gamma l}}{2}$ and $\sinh\gamma l = \dfrac{e^{\gamma l} - e^{-\gamma l}}{2}$

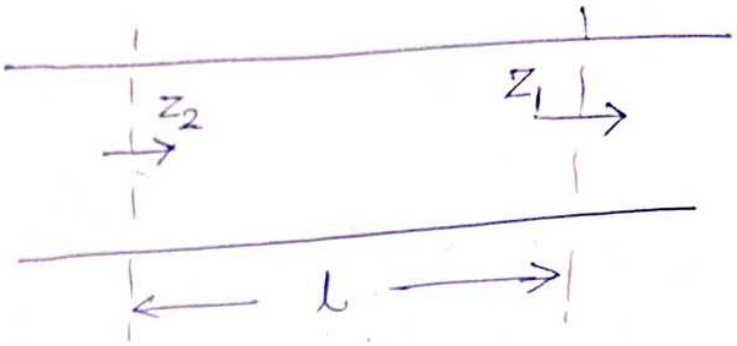
Impedance Transformation:-

$$\bar{Z} = \frac{Z}{Z_0}, \quad \bar{Z}_L = \frac{Z_L}{Z_0}, \quad \bar{Z}(l) = \frac{Z(l)}{Z_0}$$

$$\bar{Z}(l) = \left\{ \frac{\bar{Z}_L \cosh\gamma l + \sinh\gamma l}{\cosh\gamma l + \bar{Z}_L \sinh\gamma l} \right\}$$

If $\bar{Z}_L = 1$ then $Z_L = Z_0$

$\bar{Z}(l) = 1 \Rightarrow Z(l) = Z_0$



$$Z_2 = Z_0 \left[\frac{Z_1 \cosh\gamma l + Z_0 \sinh\gamma l}{Z_0 \cosh\gamma l + Z_1 \sinh\gamma l} \right]$$

$$Z_1 = Z_0 \left[\frac{Z_2 \cosh\gamma l - Z_0 \sinh\gamma l}{Z_0 \cosh\gamma l - Z_2 \sinh\gamma l} \right]$$

Input impedance of Short T.L., open circuited T.L and terminated with characteristics impedance:—

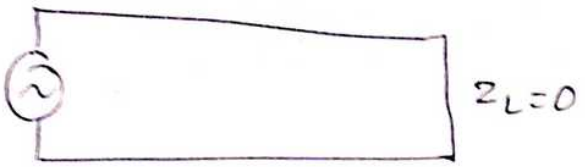
$$Z(l) = Z_{in} = Z_0 \left[\frac{Z_L \cosh\gamma l + Z_0 \sinh\gamma l}{Z_0 \cosh\gamma l + Z_L \sinh\gamma l} \right]$$

$$\Rightarrow \left[Z(l) = Z_0 \left[\frac{Z_L + Z_0 \tanh\gamma l}{Z_0 + Z_L \tanh\gamma l} \right] \right] \text{[For lossy T.L.]}$$

For a lossless line $\gamma = j\beta$ and $\tanh\gamma l = j\tan\beta l$

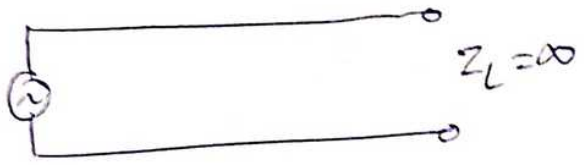
$$\Rightarrow \left[Z(l) = Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l} \right] \right] \text{[For lossless]}$$

For shorted line $Z_L = 0$



$$Z_{sc} = Z_{in} = jZ_0 \tan\beta l$$

For o/c line, $Z_L = \infty$



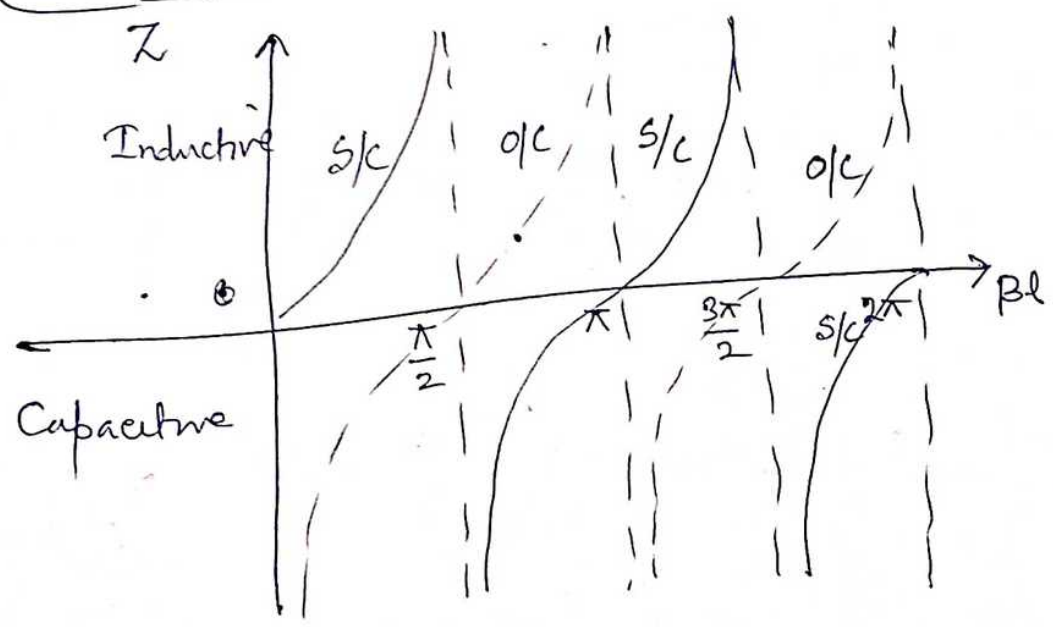
$$Z_{oc} = -jZ_0 \cot\beta l$$

The I/P impedance in both the cases can be inductive or capacitive or resistive, that's depends on the electrical length of the transmission line.

[$\beta l \rightarrow$ Electrical length]

For Matched T.L :- ($Z_L = Z_0$)

$$Z_{in} = Z_0 [x1] = Z_0$$



Smith Chart :-

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Rightarrow \Gamma_L = \frac{\frac{Z_L}{Z_0} - 1}{\frac{Z_L}{Z_0} + 1} = \frac{\bar{Z}_L - 1}{\bar{Z}_L + 1}, \text{ where } \bar{Z}_L = \frac{Z_L}{Z_0} = \text{Normal}$$

load

Lossless T.L. Z_0 is real.

Let $Z_L = R \pm jX$ ← +ive or −ve

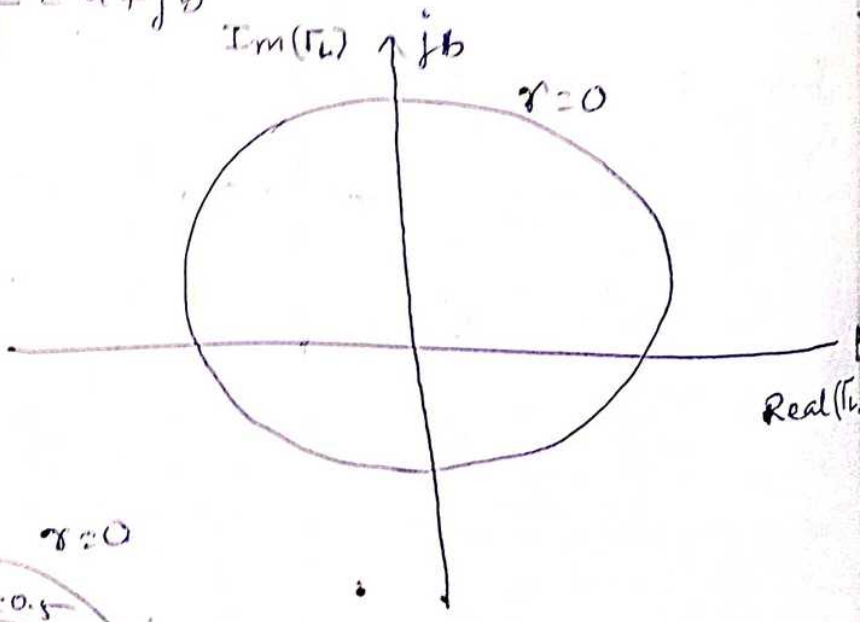
{+ive Passive}

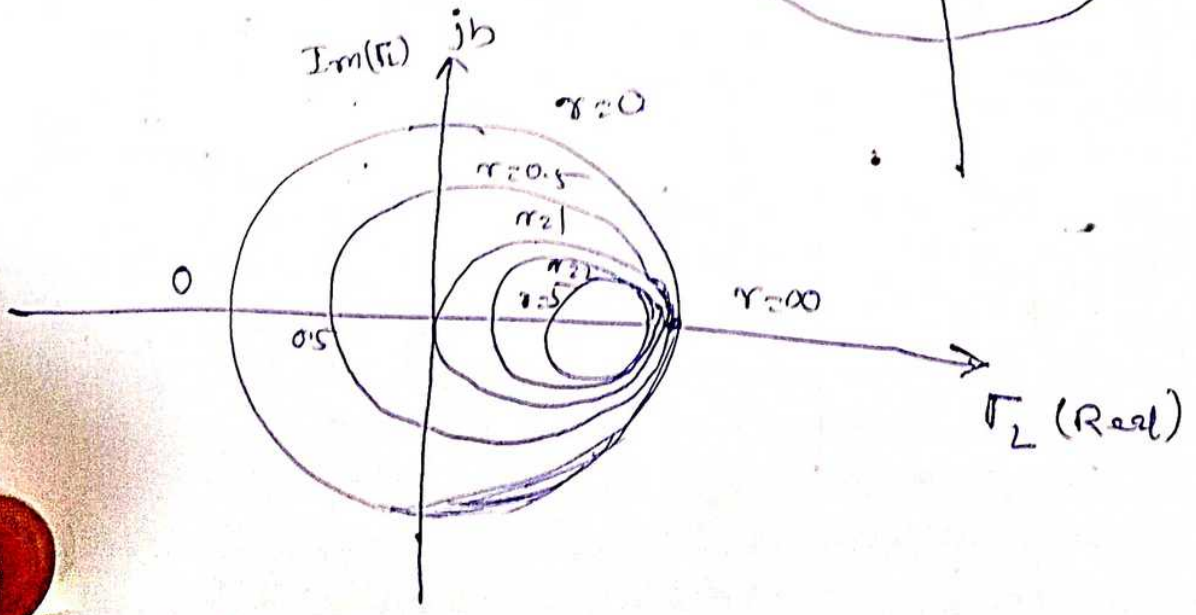
$$\Rightarrow \frac{Z_L}{Z_0} = \frac{R}{Z_0} \pm j\frac{X}{Z_0}$$

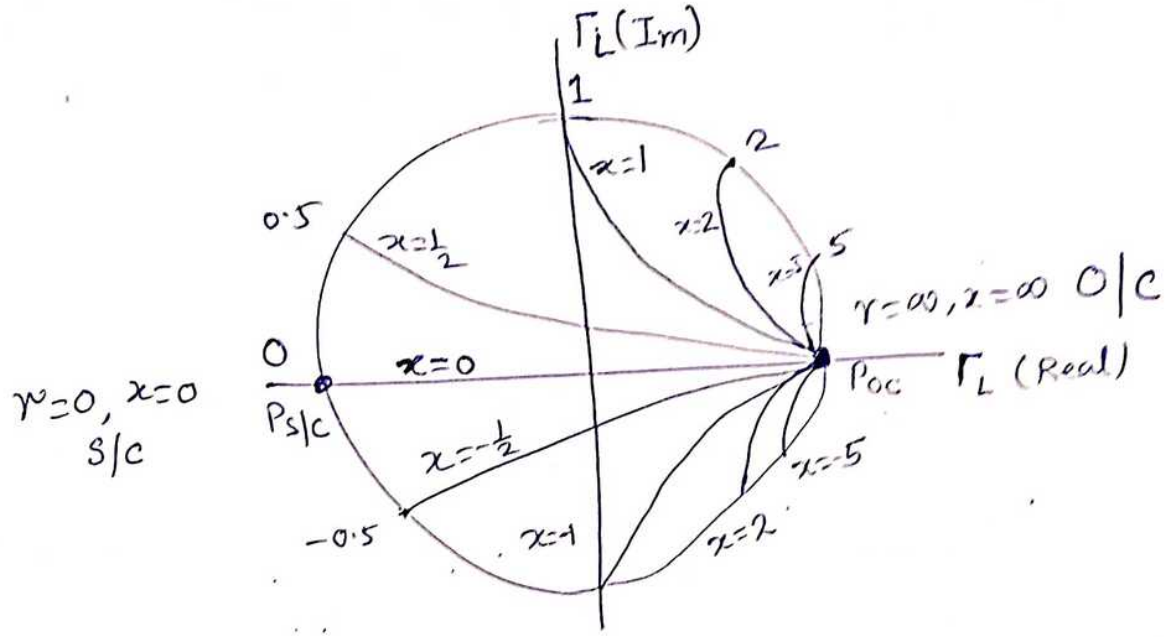
$$\Rightarrow \boxed{\frac{Z_L}{Z_0} = r \pm jx}$$

$$\Gamma_L = \frac{\bar{Z}_L - 1}{\bar{Z}_L + 1} = \frac{(r + jx) - 1}{(r + jx) + 1} = a + jb$$

$$= \frac{(r-1) + jx}{(r+1) + jx}$$







The points should be noted about smith chart :—

① At $r=\infty, x=\infty$, the Point is Poc and at $r=0, x=0$ the points is $P_{S/C}$

② A Complete revolution (360°) around the smith chart represents a length $\frac{\lambda}{2}$ on a transmission line. clockwise movement on the chart is regarded as moving toward the generator or (away from the load). Similarly counter-clockwise movement on the chart corresponds to moving toward the load (or away from the generator)

③ One wavelength (λ) distance on a transmission line corresponds to a 720° movement on the chart.

④ The point Poc ($r=\infty, x=\infty$) corresponds to V_{Max}, Z_{Max} and the point Psc ($r=0, x=0$) corresponds to V_{Min}, I_{Min}.

⑤ The Smith chart is used both as impedance chart ($Y=\frac{1}{Z}$) and admittance chart. As admittance chart $y=\frac{Y}{Y_0}=g+jb$, the g and b corresponds to r and x-circles respectively.

Q. A 30-m long transmission line with $Z_0 = 50\Omega$ operating at 2MHz is terminated with a load $Z_L = 60 + j40\ \Omega$. If $u = 0.6C$. find

(a) The reflection coefficient Γ

(b) The standing wave ratio S.

(c) The input impedance

Ans:- Method 1

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{60 + j40 - 50}{60 + j40 + 50} = \frac{10 + j}{110 + j}$$

$$\Rightarrow \Gamma = 0.3523 \angle 56^\circ$$

$$\Rightarrow u = \frac{\omega}{\beta} \text{ or } \beta = \frac{\omega}{u}$$

$$\Rightarrow \beta l = \frac{\omega l}{u} = \frac{2\pi (2\times 10^6) \times 30}{0.6 \times 3 \times 10^8} = \frac{2\pi}{3}$$

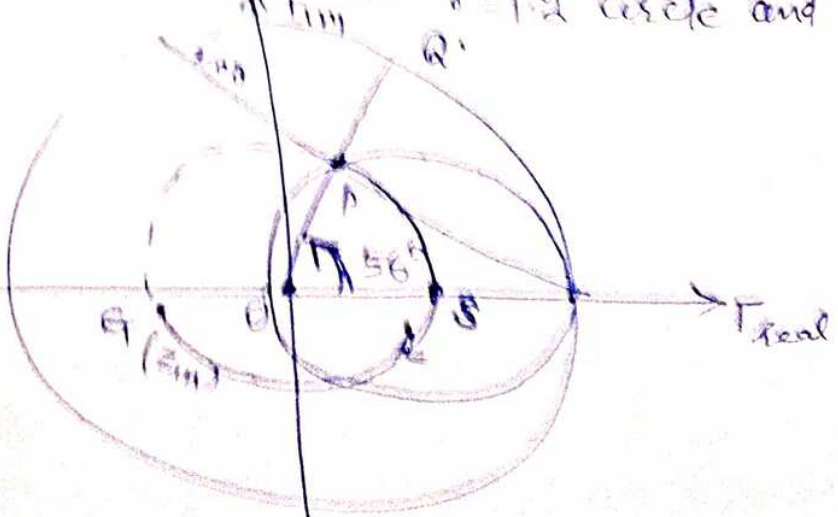
$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}\right] = 50\left[\frac{60 + j40 + j50 \tan\frac{2\pi}{3}}{50 + j(60 + j40)\tan\frac{2\pi}{3}}\right]$$

$$\Rightarrow Z_{in} = Z_0 \frac{(6 + j4 - j5\sqrt{3})}{[5 + 4\sqrt{3} - j6\sqrt{3}]} = 24.01 \angle 3.22^\circ$$

Method 2:- (Using Smith Chart)

$$\bar{Z}_L = Z_L / Z_0 = \frac{60 + j40}{50} = 1.2 + j0.8$$

Locate $\bar{Z}_L$ on smith chart where $r = 1.2$ circle and $x = 0.8$ meet



$$|\Gamma| = \frac{OP}{OQ} = \frac{3.2\text{cm}}{9\text{cm}}$$

$$\Rightarrow |\Gamma| = 0.3516$$

$$\boxed{\Gamma = 0.3516 \angle 56^\circ}$$

$S = r$
$S = 2.1$. } Draw a circle with radius OP and Center at O. This is a constant s or |Γ| circle.

Locate point S where the s-circles meets the Γ_r axis

For Z_{in} ?

$$\lambda = \frac{u}{f} = \frac{0.6 \times 3 \times 10^8}{2 \times 10^6} = 90m$$

$$l = 30m = \frac{30}{90}\lambda = \frac{\lambda}{3} \longrightarrow \frac{720^\circ}{3} = 240^\circ$$

For this move toward the generator (or away from the load in the clockwise direction 240° on the s-circles from the point P to G

$$\bar{Z}_{in} = 0.47 + j0.035$$

$$\Rightarrow Z_{in} = Z_0 Z_{in} = 50(0.47 + j0.035)$$

$$\Rightarrow \boxed{Z_{in} = 23.5 + j1.75\ \Omega}$$

Impedance Matching :-

① **Stub Matching :-** A stub is a part of transmission line, that is used for the impedance matching in transmission lines. Basically there are two types of transmission line stubs.

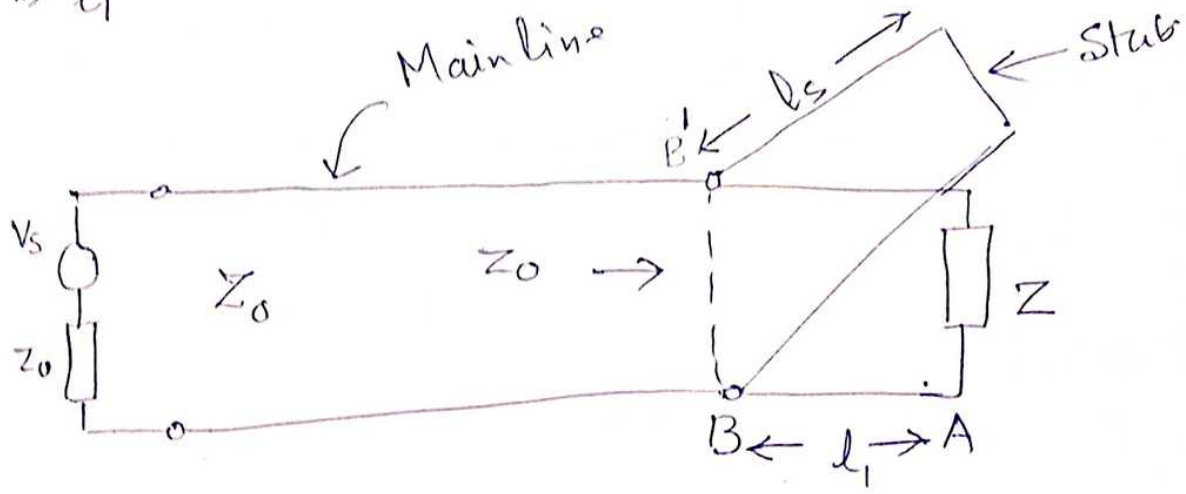
① Open circuit Stubs
② Short Circuit Stubs

O/c stubs are not used for the matching purpose

at high frequencies due to radiation losses
so short circuit stubs are preferred for the
matching purpose as compared to the o
circuit stubs. For this purpose a single
stub or double stub are connected par
to the mismatched transmission line

① Single Stub Matching :— The main dis
of QWT is th

it is a narrow band or frequency-sensative device. This draw back is eliminated by the single-stub-tuner. This single stub leng is considered as l_s and its distance from the end is l_1.



Choose the length l_1 in such as way that impedance transformed at B plane the resis part equal to the characteristics impedance the main line.

stub have input impedance only reactive.
the parallel combination of the stub and
impedance at B cancels the reactive impedance
then the impedance at the BB' is Z_0. This shows
that impedance is matched and there will be
no reflection at left of point B in the main line.

location of the Stub l_1

We know that $Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}\right]$

Now, $Y_0 = \frac{1}{Z_0}$ = Characteristics admittance

$\bar{Y}_L = \frac{1}{Z_L}$ = Load admittance

$Y_{in} = \frac{1}{Z_{in}}$ = Input impedance

$$\Rightarrow \frac{1}{Y_{in}} = \frac{1}{Y_0}\left[\frac{\frac{1}{\bar{Y}_L} + \frac{j}{Y_0}\tan\beta l}{\frac{1}{Y_0} + j\frac{1}{\bar{Y}_L}\tan\beta l}\right] = \frac{1}{Y_0}\left[\frac{Y_0 + jY_L\tan\beta l}{Y_L + jY_0\tan\beta l}\right]$$

$$\Rightarrow Y_{in} = Y_0\left[\frac{Y_L + jY_0\tan\beta l}{Y_0 + jY_L\tan\beta l}\right] = Y_0\left[\frac{\bar{Y}_L + j\tan\beta l}{1 + j\bar{Y}_L\tan\beta l}\right]$$

$$\Rightarrow \frac{Y_{in}}{Y_0} = \left[\frac{\bar{Y}_L + j\tan\beta l}{1 + j\bar{Y}_L\tan\beta l}\right] \Rightarrow \bar{Y}_{in} = \left[\frac{\bar{Y}_L + j\tan\beta l}{1 + j\bar{Y}_L\tan\beta l}\right]$$

$$\Rightarrow \bar{Y}_{in} = \left[\frac{\bar{Y}_L + j\tan\beta l}{1 + j\bar{Y}_L\tan\beta l}\right] \times \left[\frac{1 - j\bar{Y}_L\tan\beta l}{1 - j\bar{Y}_L\tan\beta l}\right]$$

$$\Rightarrow \bar{Y}_{in} = \frac{\bar{Y}_L(1 + \tan^2\beta l) + j\tan\beta l(1 - \bar{Y}_L^2)}{1 + (\bar{Y}_L\tan\beta l)^2} = g + jb$$

$$\Rightarrow g = \frac{\bar{Y}_L(1 + \tan^2\beta l)}{1 + (\bar{Y}_L\tan\beta l)^2}$$

$$b = \frac{\tan\beta l\,(1-\bar{Y}_L^2)}{1+(\bar{Y}_L\tan\beta l)^2}$$

But for no reflection $\bar{Y}_{in} = g + jb = 1 + j\cdot 0$

$$\Rightarrow g = 1 = \frac{\bar{Y}_L(1+\tan^2\beta l)}{1+(\bar{Y}_L\tan\beta l)^2}$$

$$\Rightarrow 1+\bar{Y}_L^2\tan^2\beta l = \bar{Y}_L(1+\tan^2\beta l)$$

Here, $l \to l_1$

$$\Rightarrow 1+\bar{Y}_L^2\tan^2\beta l_1 = \bar{Y}_L + \bar{Y}_L\tan^2\beta l_1$$

$$\Rightarrow \bar{Y}_L\tan^2\beta l_1(\bar{Y}_L-1) = (\bar{Y}_L-1)$$

$$\Rightarrow \tan^2\beta l_1 = \frac{1}{\bar{Y}_L} = \frac{1}{\frac{Y_L}{Y_0}}$$

$$\Rightarrow \tan\beta l_1 = \frac{1}{\sqrt{\frac{Y_L}{Y_0}}} \Rightarrow \beta l_1 = \tan^{-1}\sqrt{\frac{Y_0}{Y_L}}$$

$$\Rightarrow \boxed{l_1 = \frac{1}{\beta}\tan^{-1}\sqrt{\frac{Z_L}{Z_0}}}$$

$$b = \frac{\tan\beta l_1\,(1-\bar{Y}_L^2)}{1+(\bar{Y}_L\tan\beta l)^2} \Rightarrow b = \sqrt{\frac{Y_0}{Y_L}}\,(Y_0-Y_L)$$

Now at the attachment point the sum of Sum susceptance is zero

$$\sqrt{\frac{Y_0}{Y_L}}\,(Y_0-Y_L) + \frac{1}{jZ_0\tan\beta l} = 0$$

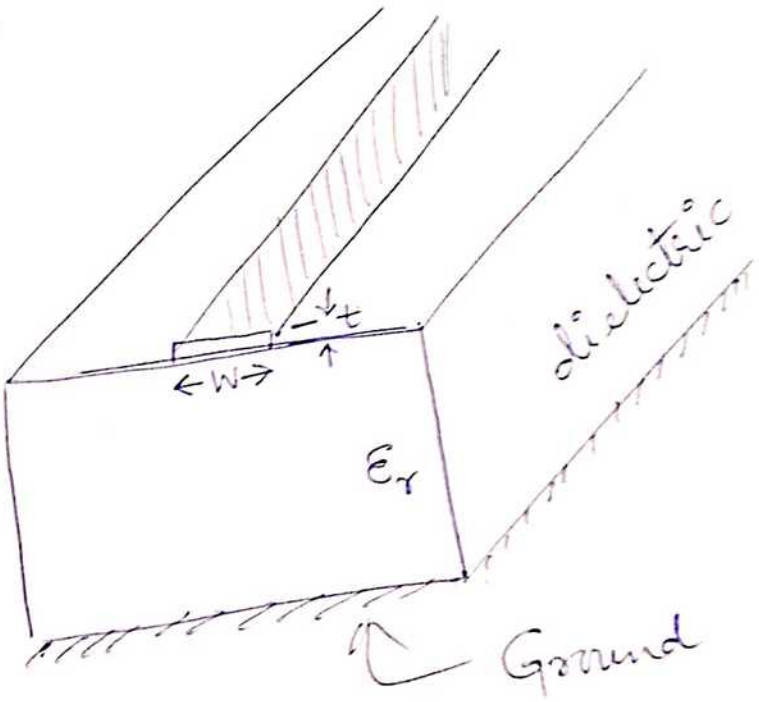
$$\Rightarrow \sqrt{\frac{Y_0}{Y_L}}\,(Y_0-Y_L) - Y_0\cot\beta l_s = 0$$

$$\cot\beta l_s = \left(\frac{Y_0 - Y_L}{Y_0}\right)\sqrt{\frac{Y_0}{Y_L}} = (Y_0 - Y_L)\cdot\frac{1}{\sqrt{Y_0 Y_L}}$$

2) $$\tan\beta l_s = \frac{\sqrt{Y_0 Y_L}}{Y_0 - Y_L} \Rightarrow l_s = \frac{1}{\beta}\tan^{-1}\frac{\sqrt{Y_0 Y_L}}{Y_0 - Y_L}$$

3) $$\boxed{l_s = \frac{\lambda}{2\pi}\tan^{-1}\frac{\sqrt{Z_L Z_0}}{Z_L - Z_0}}$$

Microstrip line :—



Microstrip line is a type of transmission line that is designed using PCB (printed circuit board) technology.

WAVE GUIDES

→ NO TEM wave exists in waveguide but TE and TM waves can exist.

Wave equations

Assume that all field vectors vary with respect to time 't' in a sinusoidal manner.

$$E = E_0 e^{j\omega t}$$

E_0 = max^m value of Electric field intensity.

$\omega = 2\pi f$, f = frequency of the sinusoidal variations.

$$\frac{\partial E}{\partial t} = E_0 j\omega e^{j\omega t} = E j\omega = j\omega E \quad \Rightarrow \frac{\partial}{\partial t} = j\omega$$

Differentiating again w.r.t 't',

$$\frac{\partial^2 E}{\partial t^2} = j\omega E_0 j\omega e^{j\omega t} = -\omega^2 E \quad \rightarrow \frac{\partial^2}{\partial t^2} = -\omega^2$$

Now consider a medium which does not contain any free charges and is also non-conducting, for example air or free space.

we have $\rho = 0$, $\sigma = 0$.

from Maxwell's 1st equation, $\nabla \times H = J + \frac{\partial D}{\partial t}$

$$\nabla \times H = \sigma E + \frac{\partial}{\partial t}(\varepsilon E)$$

$$= \frac{\partial}{\partial t}(\varepsilon E) = j\omega(\varepsilon E) = j\omega\varepsilon E.$$

$$\boxed{\nabla \times H = j\omega \varepsilon E}$$

From Maxwell's 2nd equation,

$$\nabla \times E = -\frac{\partial B}{\partial t} \Rightarrow \frac{-\partial}{\partial t}(\mu H) = -j\omega\mu H$$

$$\boxed{\nabla \times E = -j\omega\mu H}$$

Take curl of the above equation,

$$\nabla \times \nabla \times E = \nabla \times (-j\omega\mu H) = -j\omega\mu(\nabla \times H) = -j\omega\mu(j\omega\varepsilon E)$$

$$\nabla \times \nabla \times E = \omega^2 \mu\varepsilon E$$

From vector analysis,

$$\nabla \times \nabla \times E = \nabla(\nabla \cdot E) - \nabla^2 E \qquad (i)$$

$$\nabla(\nabla \cdot E) - \nabla^2 E = \omega^2 \mu \varepsilon E$$

From Maxwell's 3rd eqn.

$$\nabla \cdot D = \rho = 0$$

$$\varepsilon (\nabla \cdot E) = 0 \qquad \varepsilon \neq 0 \qquad \nabla \cdot E = 0$$

$\Rightarrow$ eqn (i) becomes, $\quad -\nabla^2 E = \omega^2 \mu \varepsilon E$

$$\nabla^2 E = -\omega^2 \mu \varepsilon E$$

Resolving E into 3 mutually ⊥lar directions we get,

$$\nabla^2 (\hat{i} E_x + \hat{j} E_y + \hat{k} E_z) = -\omega^2 \mu \varepsilon [\hat{i} E_x + \hat{j} E_y + \hat{k} E_z)$$

where $\hat{i}$, $\hat{j}$, $\hat{k}$ are unit vectors along x, y and z directions respectively.

equating coefficients of $\hat{i}$, $\hat{j}$ & $\hat{k}$ on both sides, we get,

$$\nabla^2 E_x = -\omega^2 \mu \varepsilon E_x$$

$$\nabla^2 E_y = -\omega^2 \mu \varepsilon E_y$$

$$\nabla^2 E_z = -\omega^2 \mu \varepsilon E_z$$

Similarly, $\quad \nabla^2 H = -\omega^2 \mu \varepsilon H$

$$\nabla^2 H_x = -\omega^2 \mu \varepsilon H_x$$

$$\nabla^2 H_y = -\omega^2 \mu \varepsilon H_y$$

$$\nabla^2 H_z = -\omega^2 \mu \varepsilon H_z$$

In general wave is written as, $\quad \nabla^2 E = -\omega^2 \mu \varepsilon E$

$$\nabla^2 H = -\omega^2 \mu \varepsilon H$$

replace '$-\omega^2$' by $\frac{\partial^2}{\partial t^2}$

$$\boxed{\nabla^2 E = \mu \varepsilon \frac{\partial^2 E}{\partial t^2}} \quad \text{and} \quad \boxed{\nabla^2 H = \mu \varepsilon \frac{\partial^2 H}{\partial t^2}}$$

$\varepsilon\mu = \mu_r \mu_0 \varepsilon_r \varepsilon_0 = \varepsilon_r \mu_r$ ($\mu_r = 1$, $\varepsilon_r = 1$ for air)

$\mu_0 = 4\pi \times 10^{-7}$ H/m

$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

$$\mu\varepsilon = \frac{1}{c^2}$$ c = velocity of light

$$\nabla^2 E = \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2}$$

$$\nabla^2 H = \frac{1}{c^2} \frac{\partial^2 H}{\partial t^2}$$

These 2 equations represent waves propagating through free space with a velocity equal to that of light and are also known as Helmholtz's wave equations.

If we consider a wave travelling in positive z direction then wave eq. take the form,

$\nabla^2 E_z = -\omega^2 \mu\varepsilon E_z \rightarrow$ for a TM wave, $H_z = 0$, $E_z \neq 0$

and $\nabla^2 H_z = -\omega^2 \mu\varepsilon H_z \rightarrow$ for a TE wave $E_z = 0$, $H_z \neq 0$

Then $E_z = E_{0z} e^{-\gamma z}$

E_{0z} = max^m. value of Electric field along z direction

$\gamma = \alpha + j\beta$

γ = propagation constant

α = attenuation constant

β = phase constant $= \left(\frac{2\pi}{\lambda}\right)$

The condition for wave propagation is that γ must be imaginary.

$$\frac{\partial E_z}{\partial z} = E_{0z} e^{-\gamma z} (-\gamma)$$

$$E_{0z} e^{-\gamma z} = E_z$$

$$\frac{\partial E_z}{\partial z} = -E_z \cdot \gamma = -\gamma E_z \qquad \Rightarrow \frac{\partial}{\partial z} = -\gamma$$

Diff. w.r.t. 'z' $\Rightarrow \frac{\partial^2 E_z}{\partial z^2} = E_{0z} e^{-\gamma z} (-\gamma)(-\gamma)$

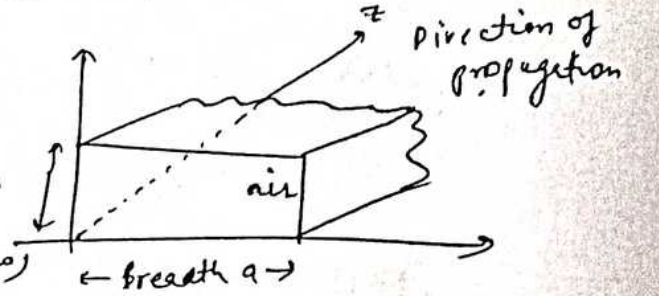
$$= \gamma^2 E_{0z} e^{-\gamma z} = \gamma^2 E_z$$

$$\Rightarrow \boxed{\frac{\partial^2}{\partial z^2} = \gamma^2}$$

Propagation of waves in Rectangular Waveguides

Consider a Rectangular waveguide situated in the Rectangular coordinate system with its breadth along x-axis, width along y axis and the wave is assumed to propagate along the z-direction

ii) waveguide is filled with air as dielectric.



Wave equation for TE and TM waves are given by

$\nabla^2 H_z = -\omega^2 \mu \varepsilon H_z$ for TE wave ($E_z = 0$)

$\nabla^2 E_z = -\omega^2 \mu \varepsilon E_z$ for TM wave ($H_z = 0$)

Expand $\nabla^2 E_z$ in Rectangular Coordinate system,

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2} = -\omega^2 \mu \varepsilon E_z$$

since the wave is propagating in the 'z' direction we have the operator $\frac{\partial^2}{\partial z^2} = \gamma^2$

$$\Rightarrow \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \gamma^2 E_z = -\omega^2 \mu \varepsilon E_z$$

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + (\gamma^2 + \omega^2 \mu \varepsilon) E_z = 0$$

Let $\gamma^2 + \omega^2 \mu \varepsilon = h^2$, h a constant

$$\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + h^2 E_z = 0 \quad \text{for TM wave}$$

Similarly, $\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + h^2 H_z = 0$ for TE wave

By solving the above partial diff. eqn., we get sol. for E_z and H_z. Using Maxwell's eq., its possible to find the various components along x and y directions (E_x, E_y, H_y, H_x).

From Maxwell's 1st eq^n. $\nabla \times H = j\omega \varepsilon E$

expand $\nabla \times H$ $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = j\omega\varepsilon [\hat{i}E_x + \hat{j}E_y + \hat{k}E_z]$

replace $\frac{\partial}{\partial z} = -\gamma$ $\rightarrow$ $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ H_x & H_y & H_z \end{vmatrix} = j\omega\varepsilon [\hat{i}E_x + \hat{j}E_y + \hat{k}E_z]$

equat. the coefficients, $\hat{i}$, $\hat{j}$ and $\hat{k}$ we get,

$$\frac{\partial H_z}{\partial y} + \gamma H_y = j\omega\varepsilon E_x \quad (1)$$

$$\frac{\partial H_z}{\partial x} + \gamma H_x = -j\omega\varepsilon E_y \quad (2)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega\varepsilon E_z \quad (3)$$

Similarly from Maxwell's 2nd eq^n. $\nabla \times E = -j\omega\mu H$

$$\nabla \times E = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ E_x & E_y & E_z \end{vmatrix}$$

$$= -j\omega\mu [\hat{i}H_x + \hat{j}H_y + \hat{k}H_z]$$

expand. and equating coefficients of $\hat{i}$, $\hat{j}$ & $\hat{k}$, we get

$$\frac{\partial E_z}{\partial y} + \gamma E_y = -j\omega\mu H_x \quad (4)$$

$$\frac{\partial E_z}{\partial x} + \gamma E_x = +j\omega\mu H_y \quad (5)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z \quad (6)$$

Combining eq^n: ① and ⑤ to eliminate H_y, we get an eq^n for E_x.

from eq^n ⑤ $\longrightarrow$ $H_y = \frac{1}{j\omega\mu}\frac{\partial E_z}{\partial x} + \frac{1}{j\omega\mu}\gamma E_x$ — ⑥

put ⑥ in ①,

$$\frac{\partial E_z}{\partial x} + \gamma E_x = +j\omega\mu\left(\frac{1}{j\omega\mu}\frac{\partial E_z}{\partial x} + \frac{\gamma}{j\omega\mu}E_x\right)$$

$$= \frac{\partial E_z}{\partial x} + \gamma E_x$$

$$\frac{\partial H_z}{\partial y} + \gamma\left(\frac{1}{j\omega\mu}\frac{\partial E_z}{\partial x} + \frac{\gamma}{j\omega\mu}E_x\right) = j\omega\varepsilon E_x$$

$$\Rightarrow \frac{\partial H_z}{\partial y} + \frac{\gamma}{j\omega\mu}\frac{\partial E_z}{\partial x} = j\omega\varepsilon E_x - \frac{\gamma^2}{j\omega\mu}E_x$$

$$= E_x\left(j\omega\varepsilon - \frac{\gamma^2}{j\omega\mu}\right)$$

$$j\omega\mu\frac{\partial H_z}{\partial y} + \gamma\frac{\partial E_z}{\partial x} = E_x\left[-\omega^2\mu\varepsilon - \gamma^2\right)$$

$$E_x\left[-(\gamma^2+\omega^2\mu\varepsilon)\right] = j\omega\mu\frac{\partial H_z}{\partial y} + \gamma\frac{\partial E_z}{\partial x}$$

put $\gamma^2+\omega^2\mu\varepsilon = h^2$ (any constant)

divide through out by h^2,

$$E_x = -\frac{\gamma}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y} \quad ⑦$$

similarly $E_y = -\frac{\gamma}{h^2}\frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$ ⑧

$$H_x = -\frac{\gamma}{h^2}\frac{\partial H_z}{\partial x} + \frac{j\omega\varepsilon}{h^2}\frac{\partial E_z}{\partial y} \quad ⑨$$

$$H_y = -\frac{\gamma}{h^2}\frac{\partial H_z}{\partial y} - \frac{j\omega\varepsilon}{h^2}\frac{\partial E_z}{\partial x} \quad ⑩$$

Propagation of TEM waves

For a TEM wave, $E_z = 0$ and $H_z = 0$

put these values in eq^n (7), (8), (9), (10)

E_x, E_y, H_x, H_y vanish hence a TEM wave cannot exist inside a wave guide.

TE and TM modes

Propagation of TM waves in rectangular waveguide

For TM wave, $H_z = 0$; $E_z \neq 0$

The wave eq^n of a TM wave is, $\frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + h^2 E_z = 0$ (A)

This is a partial differential eq^n. which can be solved to get the different field components E_x, E_y, H_x and H_y by 'separation of variables method'.

let $E_z = X \cdot Y$

X = pure function of 'x' only
Y = pure function of 'y' only

Since X and Y are independent variables,

$$\left.\begin{aligned} \frac{\partial^2 E_z}{\partial x^2} &= \frac{\partial^2 (XY)}{\partial x^2} = Y\frac{\partial^2 X}{\partial x^2} \\ \frac{\partial^2 E_z}{\partial y^2} &= \frac{\partial^2 (XY)}{\partial y^2} = X\frac{\partial^2 Y}{\partial y^2} \end{aligned}\right\}$$ use these two equations in (A)

$$\Rightarrow Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial y^2} + h^2 XY = 0$$

Divide by XY,

$$\underbrace{\frac{1}{X}\frac{\partial^2 X}{\partial x^2}}_{\text{pure fun}^n \text{ of } x \text{ only}} + \underbrace{\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2}}_{\text{pure fun}^n \text{ of } y \text{ only}} + h^2 = 0 \qquad (B)$$

The sum of these is a constant. Hence each term must be equal to a constant separately since X and Y are independent variables.

We use separation of variables method to solve the eq^n (B)

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -B^2 \quad \text{(i)} \qquad \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} = -A^2 \quad \text{(ii)}$$

where $-A^2$ & $-B^2$ are constants.

$-B^2 - A^2 + h^2 = 0 \Rightarrow \boxed{h^2 = A^2 + B^2}$

eq^n (i) & (ii) are ordinary 2nd order differential equations the solutions of which are given by,

$$X = C_1 \cos Bx + C_2 \sin Bx$$
$$Y = C_3 \cos Ay + C_4 \sin Ay$$

C_1, C_2, C_3, C_4 are constants and are solved by boundary conditions.

Complete solution is $E_z = X \cdot Y$

$$E_z = (C_1 \cos Bx + C_2 \sin Bx)(C_3 \cos Ay + C_4 \sin Ay) \quad (I)$$

Boundary Conditions

→ The entire surface of the rectangular waveguide acts as a short circuit or ground for Electric field, $E_z = 0$, all along the boundary walls. There are 4 boundary walls.

I boundary condition (Bottom plane or Bottom wall)

$E_z = 0$, all along the bottom wall

$E_z = 0$ at $y = 0 \; \forall \; x \rightarrow 0$ to a

$\forall \rightarrow$ for all $x \rightarrow a$ means x varying b/w 0 to a,

2nd Boundary condition (Left side plane or left side wall)

$E_z = 0$ at $x = 0 \; \forall \; y \rightarrow 0$ to b

3rd boundary condition (Top plane or top wall)

$E_z = 0$ at $y = b \; \forall \; x \rightarrow 0$ to a

4th boundary condition

$E_z = 0$ at $x = a \; \forall \; y \rightarrow 0$ to b

→ Substitute boundary conditions in equation (I)

$$E_z = [C_1 \cos Bx + C_2 \sin Bx][C_3 \cos Ay + C_4 \sin Ay]$$

we have, $E_z = 0$ at $y = 0 \; \forall \; x \rightarrow 0$ to a

$$0 = (C_1 \cos Bx + C_2 \sin Bx) C_3$$

This is true for all $(\forall) x \rightarrow 0$ to a

$$\boxed{C_1 \cos Bx + C_2 \sin Bx \neq 0}; \quad \boxed{C_3 = 0}$$

$$\therefore E_z = (C_1 \cos Bx + C_2 \sin Bx)(C_4 \sin Ay) \quad \text{II}$$

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S.C. Gupta

ii) Put 2nd boundary condition in eq^n II,

$E_z = 0 = C_1 C_4 \sin Ay$ $\forall\ y \to 0$ to b

$\sin Ay \neq 0$, $C_4 \neq 0$ $\Rightarrow C_1 = 0$

use this in eq^n II, $E_z = C_2 C_4 \sin Bx \sin Ay$ (III)

iii) Put 3rd boundary condition in eq^n (III) we get

$E_z = 0 = C_2 C_4 \sin Bx \sin Ay$ [at $y = b$ for all $x \to 0$ to a]

since $\sin Bx \neq 0$, $C_4 \neq 0$, $C_2 \neq 0$ $\sin Ab = 0$

$\Rightarrow Ab$ is a multiple of π

where n is a constant

$n = 0, 1, 2 \ldots$

$Ab = n\pi$

$$A = \frac{n\pi}{b}$$

iv) Substitute 4th boundary cond^n

$E_z = 0 = C_2 C_4 \sin Ba \sin Ay$ [at $x = a$, $y \to 0$ to b]

since $\sin Ay \neq 0$, $C_4 \neq 0$, $C_2 \neq 0$

$\sin Ba = 0 \Rightarrow Ba = m\pi$

where m is another constant, $m = 0, 1, 2 \ldots$

$$B = \frac{m\pi}{a}$$

so, $E_z = C_2 C_4 \sin\left(\frac{m\pi}{a}\right) x \sin\left(\frac{n\pi}{b}\right) y \cdot e^{-\gamma z} \cdot e^{j\omega t}$

where $e^{-\gamma z}$ = propagation constant along 'z' direction

$e^{j\omega t}$ = sinusoidal variation w.r.t 't'.

$$E_z = C \sin\left(\frac{m\pi}{a}\right) x \sin\left(\frac{n\pi}{b}\right) y \, e^{j\omega t - \gamma z}$$

since E_z is known, E_x, E_y, H_x and H_y are given by following equations

$$\frac{\partial E_z}{\partial y} + \gamma E_y = -j\omega\mu H_x$$

$$\frac{\partial E_z}{\partial x} + \gamma E_x = +j\omega\mu H_y$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z$$

$$E_x = \frac{-\gamma}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}$$

$\Rightarrow$ $E_x = \frac{-\gamma}{h^2}\frac{\partial E_z}{\partial x}$ (as for TM wave $H_z = 0$)

$$E_x = \frac{-\gamma}{h^2} C\left(\frac{m\pi}{a}\right) \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \; e^{j\omega t - \gamma z}$$

$$E_y = -\frac{\gamma}{h^2}\frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$

$$E_y = -\frac{\gamma}{h^2} C\left(\frac{n\pi}{b}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \; e^{j\omega t - \gamma z}$$

$$H_x = -\frac{\gamma}{h^2}\frac{\partial H_z}{\partial x} + \frac{j\omega\varepsilon}{h^2}\frac{\partial E_z}{\partial y}$$

$$H_x = \frac{j\omega\varepsilon}{h^2} C\left(\frac{n\pi}{b}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \; e^{j\omega t - \gamma z}$$

$$H_y = -\frac{\gamma}{h^2}\frac{\partial H_z}{\partial y} + \frac{j\omega\varepsilon}{h^2}\frac{\partial E_z}{\partial x}$$

$$H_y = \frac{j\omega\varepsilon}{h^2} C\left(\frac{m\pi}{a}\right) \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \; e^{j\omega t - \gamma z}$$

TM Modes in Rectangular Waveguides

Various TM_{mn} Modes

TM_{00} mode : $m=0$ and $n=0$

If $m=0$ and $n=0$ E_x, E_y, H_x, H_y vanish hence TM_{00} mode cannot exist.

TM_{01} mode : $m=0$ and $n=1$,,

TM_{10} mode : $m=1$ and $n=0$,,

TM_{11} mode : $m=1$ and $n=1$ TM_{11} mode exists and all other higher modes exist.

Cut off frequency of a waveguide (waveguide as a High pass filter)

$$h^2 = \gamma^2 + \omega^2\mu\varepsilon = A^2 + B^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\gamma^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\varepsilon$$

$$\gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\varepsilon} = \alpha + j\beta$$

at lower frequencies, $\omega^2\mu\varepsilon < \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

γ becomes real and positive and equal to attenuation constant 'α' i.e the wave is completely attenuated and there is no phase change. Hence the wave cannot propagate.

At higher frequencies, $\omega^2\mu\varepsilon > \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

γ becomes imaginary, there will be phase change β and hence the wave propagates. At the transition, γ becomes zero and propagation just starts. The frequency at which γ just becomes zero is defined as the cutoff frequency (or threshold frequency) f_c.

At $f = f_c$, $\gamma = 0$, $\omega = 2\pi f = 2\pi f_c$

$$0 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega_c^2\mu\varepsilon$$

$$f_c = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

The cutoff wavelength (λ_c) is $\lambda_c = \frac{c}{f_c} = \frac{c}{\frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}}$

$$\lambda_{c_{m,n}} = \frac{2ab}{\sqrt{m^2b^2+n^2a^2}}$$

All wavelengths greater than λ_c are attenuated and those less than λ_c are allowed to propagate inside the waveguide.

Guide wavelength (λ_g)

→ It is defined as the distance travelled by the wave in order to undergo a phase shift of 2π radians.

$$\lambda_g = \frac{2\pi}{\beta}$$

→ The wavelength in the waveguide is different from the wavelength in free space. It (λ_g) is related to free space wavelength λ_0 and cutoff wavelength λ_c

$$\frac{1}{\lambda_g^2} = \frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{1-(\lambda_0/\lambda_c)^2}}$$

This equation is true for any mode in a waveguide of any cross section, provided λ_c corresponds to the mode and the cross section of the waveguide.

If $\lambda_0 > \lambda_c \rightarrow \lambda_g \rightarrow$ imaginary $\Rightarrow$ no propagation in waveguide

Phase velocity (v_p)

→ wave propagates in a waveguide when $\lambda_g > \lambda_0$.

→ velocity of propagation is the product of λ and f, so in a waveguide $v_p = \lambda_g f$, where v_p is the phase velocity.

But speed of light $c = \lambda_0 f$. Thus $v_p > c$ since $\lambda_g > \lambda_0$

$$v_p = \lambda_g \cdot f = \lambda_g \frac{(2\pi f)}{2\pi} = \frac{2\pi f}{(2\pi/\lambda_g)} = \frac{2\pi f}{\beta} = \frac{\omega}{\beta}$$

$$\beta = \frac{2\pi}{\lambda_g}$$

V_p → defined as the rate at which wave changes its phase in terms of guide wavelength.

Group Velocity (V_g)

Rate at which wave propagates through the waveguide. $\boxed{V_g = \frac{d\omega}{d\beta}}$

An EM wave can have 2 velocities, one with which it propagates and the other with which it changes its phase.
But in free space they are same and equal to velocity of light.

$V_p = \frac{\omega}{\beta}$ $\qquad h^2 = \gamma^2 + \omega^2\mu\varepsilon = \alpha^2 + \beta^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$

$\gamma = \alpha + j\beta$

for wave propagation, $\gamma = j\beta$

$$\gamma^2 = (j\beta)^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \omega^2\mu\varepsilon \quad (1)$$

at $f = f_c$, $\omega = \omega_c$, $\gamma = 0$. $\qquad \omega_c^2\mu\varepsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$ (2)

put eq^n (2) in (1),

$$\gamma^2 = (j\beta)^2 = \omega_c^2\mu\varepsilon - \omega^2\mu\varepsilon \Rightarrow \beta = \sqrt{\omega^2\mu\varepsilon - \omega_c^2\mu\varepsilon}$$

$$\beta = \sqrt{\mu\varepsilon(\omega^2 - \omega_c^2)} \Rightarrow \sqrt{\mu\varepsilon}\sqrt{\omega^2 - \omega_c^2}$$

$$V_p = \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\mu\varepsilon}\sqrt{\omega^2 - \omega_c^2}} = \frac{1}{\sqrt{\mu\varepsilon}} \cdot \frac{1}{\sqrt{1 - (\omega_c/\omega)^2}}$$

$$\boxed{V_p = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}} \qquad f = \frac{c}{\lambda_0}$$

λ_0 = free space wavelength
f_c = cut off frequency $= \frac{c}{\lambda_c}$

$$\frac{f_c}{f} = \frac{c}{\lambda_c} \cdot \frac{\lambda_0}{c} = \frac{\lambda_0}{\lambda_c}$$

$$\boxed{V_p = \frac{c}{\sqrt{1 - (\lambda_0/\lambda_c)^2}}}$$

II. Expression for v_g $\quad \boxed{v_g = \frac{d\omega}{d\beta}}$

$$\beta = \sqrt{\mu\varepsilon(\omega^2 - \omega_c^2)}$$

$$\frac{d\beta}{d\omega} = \frac{1}{2\sqrt{(\omega^2-\omega_c^2)\mu\varepsilon}} \cdot 2\omega\mu\varepsilon$$

$$\frac{d\beta}{d\omega} = \frac{\sqrt{\mu\varepsilon}}{\sqrt{1-(\omega_c/\omega)^2}} = \frac{\sqrt{\mu\varepsilon}}{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}$$

$$v_g = \frac{d\omega}{d\beta}$$

$$= \frac{\sqrt{1-\left(\frac{f_c}{f}\right)^2}}{\sqrt{\mu\varepsilon}}$$

$$\boxed{v_g = c\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}}$$

$$v_p \cdot v_g = \frac{c}{\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}} \cdot c\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2} \Rightarrow c^2$$

$$\boxed{v_p \cdot v_g = c^2}$$

Relation b/w λ_g, λ_0 and λ_c

$$v_p = \lambda_g \cdot f = \lambda_g \cdot \frac{c}{\lambda_0}$$

$$v_p = \frac{c}{\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}} \Rightarrow \frac{c}{\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}} = \frac{\lambda_g}{\lambda_0} \cdot c$$

$$\boxed{\lambda_g = \frac{\lambda_0}{\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}}}$$

TM Mode in Rectangular waveguides

TM_{11} mode Minimum possible mode $m=1,\ n=1$

$$\lambda_{cmn} = \frac{2ab}{\sqrt{m^2b^2+n^2a^2}} = \frac{2ab}{\sqrt{a^2+b^2}} = \lambda_{c11}$$

2) TM_{12} mode $\quad \lambda_{c12} = \frac{2ab}{\sqrt{b^2+4a^2}}$

3) TM_{21} mode $\quad \lambda_{c21} = \frac{2ab}{\sqrt{4b^2+a^2}}$

$\Rightarrow \lambda_{c11} > \lambda_{c12}$

$\lambda_{c11} > \lambda_{c21}$

and so on

Propagation of TE waves in a Rectangular waveguide (M)

The TE_{mn} modes in a Rectangular waveguide are characterised by $E_z = 0$.

Helmholtz equation for TE wave,

$$\nabla^2 H_z = -\omega^2 \mu\varepsilon H_z$$

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + \frac{\partial^2 H_z}{\partial z^2} = -\omega^2 \mu\varepsilon H_z$$

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + \gamma^2 H_z = -\omega^2 \mu\varepsilon H_z$$

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + (\gamma^2 + \omega^2 \mu\varepsilon) H_z = 0$$

$$\frac{\partial^2 H_z}{\partial x^2} + \frac{\partial^2 H_z}{\partial y^2} + h^2 H_z = 0$$

→ This is a partial differential eqn. whose sol? can be assumed.

Let solution $H_z = XY$.

X = pure function of x only

Y = pure function of y only

$$Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial y^2} + h^2 XY = 0$$

→ Divide by XY.

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + h^2 = 0$$

here, $\frac{1}{X}\frac{\partial^2 X}{\partial x^2}$ is purely a function of x

$\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2}$ is purely a function of y.

→ equating each of these items to a constant, we get

$$\frac{1}{X}\frac{\partial^2 X}{\partial x^2} = -B^2$$

$$\frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} = -A^2$$

$$\Rightarrow -B^2 - A^2 + h^2 = 0 \Rightarrow \boxed{h^2 = A^2 + B^2}$$

Solving for X & Y by separation of variable method

$$X = C_1 \cos Bx + C_2 \sin Bx$$

$$Y = C_3 \cos Ay + C_4 \sin Ay$$

Hence, the complete solution is $H_z = XY$

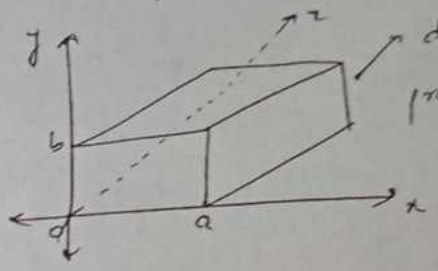
$$H_z = (C_1 \cos Bx + C_2 \sin Bx).(C_3 \cos Ay + C_4 \sin Ay) \quad \text{I}$$

where C_1, C_2, C_3 & C_4 are constants which can be evaluated by applying boundary conditions.

Boundary Conditions:

$E_z = 0$ all along the bottom and top walls of the waveguide

$E_y = 0$ all along left and right walls of the waveguide



I) $E_x = 0$ at $y = 0$ $\forall x \rightarrow 0$ to a (bottom wall)

II boundary condition
$E_x = 0$ at $y = b$ $\forall x \rightarrow 0$ to a (Top wall)

III boundary condition
$E_y = 0$ at $x = 0$ $\forall y \rightarrow 0$ to b (left side wall)

IV boundary condition
$E_y = 0$ at $x = a$ $\forall y \rightarrow 0$ to b (right side wall)

i) Substituting 1st Boundary condition

$E_x = 0$ at $y = 0$ $\forall x \rightarrow 0$ to a

previously, $E_x = \frac{-\gamma}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}$

since $E_z = 0$, for TE wave $\quad E_x = -\frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}$

$$E_x = -\frac{j\omega\mu}{h^2}\frac{\partial}{\partial y}\left((C_1\cos Bx + C_2\sin Bx)(C_3\cos Ay + C_4\sin Ay)\right)$$

$$\Rightarrow -\frac{j\omega\mu}{h^2}\left(C_1\cos Bx + C_2\sin Bx\right)\left(-AC_3\sin Ay + AC_4\cos Ay\right)$$

put 1st boundary condition,

$$0 = -\frac{j\omega\mu}{h^2}\left(C_1\cos Bx + C_2\sin Bx\right)(AC_4)$$

here $C_4 = 0$ -- (i)

$\Rightarrow$ put (i) in (I) $\Rightarrow H_z = (C_1\cos Bx + C_2\sin Bx)(C_3\cos Ay)$

ii) 3rd boundary condition

$E_y = 0$ at $x = 0$ $\forall y \to 0$ to b

$$E_y = \frac{-\gamma}{h^2}\frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$ for a TE wave $E_z = 0$

$$E_y = \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x} \quad \text{or} \quad E_y = +\frac{j\omega\mu}{h^2}\frac{\partial}{\partial x}(C_1\cos Bx + C_2\sin Bx)(C_3\cos Ay)$$

$$E_y = \frac{j\omega\mu}{h^2}[(-BC_1\sin Bx + BC_2\cos Bx)C_3\cos Ay)$$

put 3rd boundary cond^n., $x = 0$ $\forall y \to 0$ to b

$$0 = \frac{j\omega\mu}{h^2}(BC_2)(C_3\cos Ay) \quad \Rightarrow C_2 = 0 \quad \text{(ii)}$$

put this value eq^n (ii) in eq^n II.

$$H_z = C_1C_3\cos Bx\cos Ay$$

iii) 2nd Boundary Condition

$E_x = 0$ at $y = b$ $\forall x \to 0$ to a

Again $E_x = \frac{-\gamma}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}$

$$= -\frac{j\omega\mu}{h^2}\frac{\partial}{\partial y}[C_1C_3\cos Bx\cos Ay)$$

$$= +\frac{j\omega\mu}{h^2}C_1C_3A\cos Bx\sin Ay$$

put 2nd boundary condition. we get,

$$0 = \frac{j\omega\mu}{h^2}C_1C_3A\cos Bx\sin Ab$$

$\cos Bx \neq 0$, $C_1 \neq C_3 \neq 0$

$\sin Ab = 0$ or $Ab = n\pi$ where $n = 1, 2, \cdots \infty$

$$\boxed{A = \frac{n\pi}{b}}$$

iv) 4th Boundary Condition

$E_y = 0$ at $x = a$ $\forall y \rightarrow 0$ to b

$$E_y = -\frac{\gamma}{h^2}\frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$

Since for a TE wave $E_z = 0$

$$\Rightarrow E_y = \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$

$$= -\frac{j\omega\mu}{h^2}\left(C_1 C_3 B \sin Bx \cos Ay\right) \quad \left(\begin{matrix} E_z = 0 \text{ and} \\ H_z = C_1 C_3 \cos Bx \cos Ay \end{matrix}\right.$$

put boundary cond^n.

$$0 = -\frac{j\omega\mu}{h^2} C_1 C_3 B \sin Bx \cos Ay$$

$\cos Ay \neq 0$, $C_1, C_3 \neq 0$ $\Rightarrow \sin Ba = 0$

$Ba = m\pi$

$$\boxed{B = \frac{m\pi}{a}}$$

$$H_z = C_1 C_3 \cos Bx \cos Ay \rightarrow \text{complete solution}$$

$$H_z = C_1 C_3 \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y$$

$$\boxed{H_z = C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \; e^{j\omega t - \gamma z}}$$

Thus it can be seen that for TM wave E_z has sine-sine components and for a TE wave H_z has cosine cosine components.

Field components

$$E_x = -\frac{\gamma}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}$$

For a ~~TM~~ TE wave, $E_z = 0$ $\rightarrow E_x = -\frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}$

$$E_x = -\frac{j\omega\mu}{h^2} C\left(\frac{n\pi}{b}\right) \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \; e^{j\omega t - \gamma z}$$

$$E_y = \frac{1}{h^2}\frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$

For a TE wave $E_z = 0$

$$E_y = \frac{-j\omega\mu}{h^2} C\left(\frac{m\pi}{a}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \; e^{(j\omega t - rz)}$$

similarly, $$\boxed{H_x = \frac{-r}{h^2}\frac{\partial H_z}{\partial x} + \frac{j\omega\varepsilon}{h^2}\frac{\partial E_z}{\partial y}}$$

$$H_x = \frac{+r}{h^2} C\left(\frac{m\pi}{a}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \; e^{(j\omega t - rz)}$$

$$\boxed{H_y = \frac{-r}{h^2}\frac{\partial H_z}{\partial y} - \frac{j\omega\varepsilon}{h^2}\frac{\partial E_z}{\partial x}}$$

put $E_z = 0$ for a TE wave

$$H_y = \frac{-r}{h^2} C\left(\frac{m\pi}{a}\right)\left(\frac{n\pi}{b}\right) \cos\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \; e^{(j\omega t - rz)}$$

TE modes in Rectangular Waveguide

a) TE_{00} mode $m=0$, $n=0$
All field components ~~exist~~ vanish hence it cannot exist.

b) TE_{01} mode $m=0$, $n=1$
$E_y = 0$, $H_x = 0$, E_x and H_y exist

c) TE_{10} mode $m=1$ $n=0$
$E_x = 0$, $H_y = 0$, E_y and H_x exist hence TE_{10} mode exists.

d) TE_{11} mode $m=1$ $n=1$
TE_{11} exists & even higher modes exist.

Dominant Mode

i) That mode for which cutoff wavelength (λ_c) assumes a maximum value

ii) $\lambda < \lambda_c$ → propagate through waveguide

iii) $\lambda > \lambda_c$ → do not propagate through waveguide

$$\boxed{\lambda_{cm,n} = \frac{2ab}{\sqrt{m^2b^2 + n^2a^2}}}$$

TE_{01} mode $\lambda_{c01} = \frac{2ab}{a} = 2b$

TE_{10} mode $\lambda_{c10} = \frac{2ab}{b} = 2a$

TE_{11} mode $\lambda_{c11} = \frac{2ab}{\sqrt{a^2+b^2}}$

of these λ_{c10} has the max^m value since a is the larger dimension Hence TE_{10} mode is the dominant mode in rectangular waveguides.

$$\beta = \frac{2\pi}{\lambda_g} = \sqrt{\omega^2\mu\varepsilon - \omega_c^2\mu\varepsilon} \quad \leftarrow \text{eqn.}$$

$$v_p = \frac{c}{\sqrt{1-(\lambda_0/\lambda_c)^2}} \qquad v_g = c\sqrt{1-(\lambda_0/\lambda_c)^2}$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{1-(\lambda_0/\lambda_c)^2}}$$

Wave Impedance (Z_z) in TM and TE waves

Wave impedance - ratio of the strength of electric field in one transverse direction to the strength of the magnetic field along other transverse direction.

$$Z_z = \frac{E_x}{H_y} = \frac{-E_y}{H_x} \qquad Z_z = \frac{\sqrt{E_x^2 + E_y^2}}{\sqrt{H_x^2 + H_y^2}}$$

1. $$Z_z = Z_{TM} = \frac{E_x}{H_y} = \frac{\frac{-\gamma}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}}{\frac{-\gamma}{h^2}\frac{\partial H_z}{\partial y} - \frac{j\omega\mu}{h^2}\frac{\partial E_z}{\partial x}}$$

For a TM wave $H_z = 0$ and $\gamma = j\beta$

$$Z_{TM} = \frac{\gamma}{j\omega\mu} = \frac{j\beta}{j\omega\varepsilon} = \frac{\beta}{\omega\varepsilon}$$

$$\boxed{Z_{TM} = \frac{\beta}{\omega\varepsilon}}$$

$$\beta = \sqrt{\omega^2\mu\varepsilon - \omega_c^2\mu\varepsilon}$$

$$Z_{TM} = \frac{\sqrt{\omega^2\mu\varepsilon - \omega_c^2\mu\varepsilon}}{\omega\varepsilon} = \frac{\sqrt{\mu\varepsilon}\sqrt{\omega^2-\omega_c^2}}{\omega\varepsilon} = \sqrt{\frac{\mu}{\varepsilon}}\sqrt{1-\left(\frac{\omega_c}{\omega}\right)^2}$$

$$\Rightarrow \sqrt{\frac{\mu}{\varepsilon}}\sqrt{1-\left(\frac{f_c}{f}\right)^2} = \sqrt{\frac{\mu}{\varepsilon}}\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2} \qquad \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0\mu_r}{\varepsilon_0\varepsilon_r}}$$

$$\Rightarrow \sqrt{\frac{\mu_0\mu_r}{\varepsilon_0\varepsilon_r}}\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2} \qquad \Rightarrow \sqrt{\frac{4\pi\times10^{-7}}{8.854\times10^{-12}}}$$

$$\Rightarrow 377\,\Omega$$

$$\boxed{Z_{TM} = \eta\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}} \qquad = \eta$$

Since λ_0 is always less than λ_c for wave propagation $Z_{TM} < \eta$

So wave impedance for a TM wave is always less than free space impedance.

$$Z_z = Z_{TE} = \frac{E_x}{H_y} = \frac{\frac{-\gamma}{h^2}\frac{\partial E_z}{\partial x} - \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial y}}{\frac{-\gamma}{h^2}\frac{\partial H_z}{\partial y} - \frac{j\omega\mu}{h^2}\frac{\partial E_z}{\partial x}}$$

For a TE wave, $E_z = 0$ and $\gamma = j\beta$

$$Z_{TE} = \frac{j\omega\mu}{\gamma} = \frac{j\omega\mu}{j\beta} = \frac{\omega\mu}{\beta} = \frac{\omega\mu}{\sqrt{\mu\varepsilon}\sqrt{\omega^2 - \omega_c^2}} = \sqrt{\frac{\mu}{\varepsilon}}\frac{1}{\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}}$$

$$\Rightarrow \sqrt{\frac{\mu}{\varepsilon}}\frac{1}{\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}} = \frac{\eta}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}}$$

Therefore $Z_{TE} > \eta$ as $\lambda_0 < \lambda_c$ for wave propagation.

This shows that wave impedance for a TE wave is always greater than free space impedance.

Power Transmission in Rectangular waveguides

→ can be calculated by Complex Poynting Theorem.

$$P_{transmitted} = \int P \cdot ds = \int \frac{1}{2} (E \times H^*) ds$$

For a lossless dielectric, the time average power flow through a Rectangular waveguide is

$$P_{tr} = \frac{1}{2Z_z}\int_a |E|^2 da = \frac{Z_z}{2}\int_a |H|^2 da$$

$$Z_z = \frac{E_x}{H_y} = \frac{-E_y}{H_x}$$

$$|E|^2 = |E_x|^2 + |E_y|^2$$

$$|H|^2 = |H_x|^2 + |H_y|^2$$

For TM_{mn} mode, the average power transmitted through a Rectangular waveguide of dimensions a and b is

$$P_{tr} = \frac{1}{2\eta\sqrt{1-(\lambda_0/\lambda_c)^2}} \int_0^a \int_0^b |E_x|^2 + |E_y|^2 \, dx \, dy$$

for TE_{mn} modes, $Z_{TE} = \frac{\eta}{\sqrt{1-(\lambda_0/\lambda_c)^2}}$

$$P_{tr} = \frac{\sqrt{1-(\lambda_0/\lambda_c)^2}}{2\eta} \int_0^b \int_0^a |E_x|^2 + |E_y|^2 \, dx \, dy$$

Dominant and degenerate modes in Rectangular waveguides

Degenerate Modes - Some of the higher order modes having the same cut off frequency are called degenerate modes.

For a rectangular waveguide TE_{mn}/TM_{mn} modes for which both $m \neq 0$ $n \neq 0$ will always be degenerate modes.

Intrinsic Impedance (η)

When E is transferred to H and vice versa, the ratio of transformation depends on the medium and this is called η

$$\eta = \frac{E}{H} = \frac{\text{volts/m}}{\text{Amp/m}} = \text{ohms.}$$

using Maxwell's equation, $\nabla \times E = -j\omega\mu H$

$$\frac{\partial E}{\partial z} = -j\omega\mu H \Rightarrow -\gamma E = -j\omega\mu H \Rightarrow \frac{E}{H} = \frac{j\omega\mu}{\gamma} = \frac{j\omega\mu}{\sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}}$$

Case I

η's real part is the cause of attenuation
η's imaginary part is the cause of phase shift

i.e in the Transformation there is loss in amplitude and change in phase.

$$\gamma = \alpha + j\beta$$
$$\eta = R + jX$$

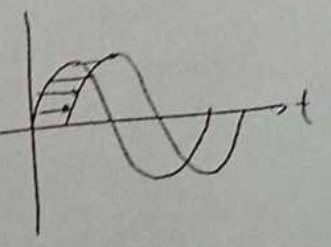
Case I: EM wave propagation in free space

($\sigma = 0$, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$)

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} \xrightarrow{\sigma = 0} \gamma = \sqrt{j\omega\mu\, j\omega\varepsilon} \Rightarrow j\beta = j\sqrt{\omega^2\mu\varepsilon}$$

$$\beta = \omega\sqrt{\mu\varepsilon} \Rightarrow \omega\sqrt{\mu_0\varepsilon_0}, \quad \alpha = 0$$

Result: So EM wave is not attenuated in free space.
Only propagation and phase shift occurs.



Velocity $V_f = \frac{1}{T}$

V_p = distance travelled by any in-phase point per unit time.

$$V_p = \lambda f$$
$$= \frac{2\pi}{2\pi}\lambda f \quad \left(\because f = \frac{1}{T}\right) \quad = \frac{\omega}{\beta}$$

$$V_p = \frac{\omega}{\omega\sqrt{\mu_0\varepsilon_0}} \quad \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3 \times 10^8 \text{ m/sec.}$$

Q. Determine the cut off wavelength for the Dominant mode in a rectangular waveguide of breadth 10 cm. for a 2.5 GHz signal propagated in this waveguide is the dominant mode calculate the Guide wavelength, v_p & v_g.

In a Rect waveguide Dominant mode is TE_{10} mode

λ_c for TE_{10} mode $= 2a = 2 \times 10 = 20$ cm.

$f = 2.5$ GHz.

$$\lambda_g = \frac{\lambda_0}{\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}} \qquad \lambda_0 = \frac{c}{f} = \frac{3\times10^{10}\ \text{cm/sec}}{2.5\times10^9} = 12\ \text{cm}$$

$$\lambda_g = \frac{12}{\sqrt{1-\left(\frac{12}{20}\right)^2}} = 15\ \text{cm}$$

$$v_p = \frac{c}{\sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2}} \qquad \sqrt{1-\left(\frac{\lambda_0}{\lambda_c}\right)^2} = \sqrt{1-\left(\frac{12}{20}\right)^2} = 0.8$$

$$v_p = \frac{3\times10^{10}}{0.8} = 3.75\times10^{10}\ \text{cm/sec}.$$

$$\boxed{v_p v_g = c^2}$$

$$v_g = \frac{c^2}{v_p} = 2.4\times10^{10}\ \text{cm/sec}.$$

Cavity Resonators

Expression for f_0 in a Rectangular Cavity Resonator

for a Rectangular waveguide

$$h^2 = \gamma^2 + \omega^2\mu\varepsilon = A^2 + B^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$\omega^2\mu\varepsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 - \gamma^2$$

for wave propagation $\gamma = j\beta$

$$\omega^2\mu\varepsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \beta^2$$

If a wave has to exist in a cavity Resonator there must be phase change correspond to a given guide wavelength ($\beta = \frac{2\pi}{\lambda_g}$

The cond: for the Resonator to Resonate is

$$\beta = \frac{p\pi}{d}$$

p = a constant = 1, 2, 3, --- so that indicates half wave variation of either electric or magnetic field along the z direction.

d = length of the Resonator

$$\omega_0^2\mu\varepsilon = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2$$

$$f_0 = \frac{1}{2\pi\sqrt{\mu\varepsilon}}\left(\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2\right)^{\frac{1}{2}}$$

$$\therefore f_0 = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

General mode of propagation in a cavity resonator is TE_{mnp} or TM_{mnp}. for both TE & TM the resonant frequency is the same in a Rectangular cavity Resonator.

Applications of Cavity Resonator

i) Tuned circuits

ii) Cavity measurement in ᵐᵉᵗ of frequency.

Expⁿ for fo in a Circular Cavity Resonator

Here circular end plates are used to short both ends

a = radius of the waveguide and

d = length or height of the cylindrical waveguide.

Condition for resonance $\beta = \frac{p\pi}{d}$

for a circular waveguide section, we know that,

$$h^2 = \gamma^2 + \omega^2\mu\varepsilon = h_{nm}^2 = \left(\frac{p_{nm}}{a}\right)^2$$

$$\omega^2\mu\varepsilon = \left(\frac{p_{nm}}{a}\right)^2 - \gamma^2$$

for wave propagation $\gamma = j\beta$ and for resonance $\beta = \frac{p\pi}{d}$ and $\omega = \omega_0$

$$\omega_0^2\mu\varepsilon = \left(\frac{p_{nm}}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2$$

$$f_0 = \frac{1}{2\pi\sqrt{\mu\varepsilon}}\left(\left(\frac{p_{nm}}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2\right)^{\frac{1}{2}}$$

$$f_0 = \frac{c}{2\pi}\sqrt{\left(\frac{p_{nm}}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2}$$

This is for TM_{nmp} mode similarly for TE_{nmp} mode

$$f_0 = \frac{c}{2\pi}\sqrt{\left(\frac{p'_{nm}}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2}$$

Field Expressions for TM_{mnp} and TE_{mnp} modes in a Rectangular Cavity Resonator.

$$E_z = C_3 C_4 \sin\left(\frac{m\pi}{a}\right)x \cdot \sin\left(\frac{n\pi}{b}\right)y \, e^{j\omega t} e^{-\gamma z}$$

for wave propagating along +z direction.

$$E_z = C_3 C_4 \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \, e^{j\omega t} e^{\gamma z}$$

for wave propagating in negative z direction.

When wave propagates $\gamma = j\beta$. Adding the fields of two travelling waves i.e. one in +z direction & other in negative z direction.

$$E_z = C_3 C_4 \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \, e^{j(\omega t \pm \beta z)}$$

Let A^+ be the amplitude constant for the wave propagating in the +ve z direction and A^- be the amplitude constant for the wave propagating in the negative direction.

$$E_z = (A^+ e^{-j\beta z} + A^- e^{j\beta z})$$

$$E_z = C_2 C_4 A (e^{-j\beta z} + e^{+j\beta z}) \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \, e^{j\omega t}$$

$$e^{-i\theta} + e^{+i\theta} = 2\cos\theta$$

$$e^{-j\beta d} + e^{+j\beta d} = 2\cos\beta z$$

$$E_z = C_2 C_4 2A \cos\beta z \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \, e^{j\omega t} \quad (I)$$

$E_z = 0$ all along the surface of the resonator

$$0 = C_2 C_4 2A \cos\beta z \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \, e^{j\omega t}$$

Since $C_2 C_4 \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \neq 0$ $A \neq 0$ $\cos\beta z = 0$

$$\beta = \frac{p\pi}{d}$$

β in (I) becomes,

$$E_z = C \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cos\left(\frac{p\pi}{d}\right)z \, e^{j\omega t - \gamma z}$$

$$C = 2 C_2 C_4 A$$

$$E_z = C \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cos\left(\frac{p\pi}{d}\right)z \, e^{j\omega t - \gamma z}$$

$$E_z(TM_{mnp}) = C \sin\left(\frac{m\pi}{a}\right)x \sin\left(\frac{n\pi}{b}\right)y \cos\left(\frac{p\pi}{d}\right)z \, e^{j\omega t - \gamma z}$$

$m = 0, 1, 2, 3 - -$ → no. of half wave variations in x direction

$n = 0, 1, 2, 3 - -$ no of " " y direction

$p = 0, 1, 2, 3, - -$ no of half wave variations in z direction

II) For TE waves: ($E_z = 0$, $H_z \neq 0$)

$$H_z = C_1 C_3 \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \, e^{j(\omega t - \gamma z)}$$

since $\gamma = j\beta$ for wave propagation along the positive z direction

$$H_z = C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \, e^{j(\omega t - j\beta z)}$$

and for wave propagating along the -ve z direction, $C = C_1 C_3$.

$$H_z = C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \, e^{j(\omega t - j\beta z)}$$

The Amplitude constant along the positive z direction is represented by A^+ & that along the $-ve_z$ direction by A^-. Add^n. the two travelling waves to obtain the fields of standing wave we have, $H_z = \left(A^+ e^{-j\beta z} + A^- e^{+j\beta z}\right) \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \, e^{j\omega t}$

To make E_y vanish at $z = 0$ and $z = d$ we must make $A^+ = -A^-$ or $A^- = -A^+$ we know,

$$E_y = \frac{-\gamma}{h^2}\frac{\partial E_z}{\partial y} + \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$

for TE waves, $E_z = 0$

$$E_y = \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$

$$E_y = \frac{j\omega\mu}{h^2}\frac{\partial H_z}{\partial x}$$

$$= \frac{j\omega\mu}{h^2}\left(\frac{\partial}{\partial x}\left(A^+ e^{-j\beta z} + A^- e^{+j\beta z}\right) \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y\right) e^{j\omega t}$$

since E_y vanishes at $z = 0$ and $z = d$, we have

$$0 = \left(\left(A^+ e^{-j\beta z} + A^- e^{+j\beta z}\right)(-)\left(\frac{m\pi}{a}\right) \sin\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y\right) e^{j\omega t} \quad (A)$$

$\sin\left(\frac{m\pi}{a}\right)x$ and $\cos\left(\frac{n\pi}{b}\right)y \neq 0$

$$A^+ e^{-j\beta z} + A^- e^{+j\beta z} = 0$$

To make $E_y = 0$ choose $A^- = -A^+$. put these in eq^n (A)

$$\left(A^+ e^{-j\beta z} + A^- e^{+j\beta z}\right) = 0$$

$$A^+ e^{-j\beta z} - A^+ e^{+j\beta z} = A^+\left(e^{-j\beta z} - e^{+j\beta z}\right)$$

$$e^{-i\theta} - e^{+i\theta} = -2i \sin\theta$$

$$0 = -2j A^+ \sin\beta z$$

since $A^+ \neq 0$ $\quad \sin\beta d = 0$ $\quad$ with $z = d$

$$\beta = \frac{\pi}{d}$$

$$\beta = p\frac{\pi}{d} \quad \text{where } p = 1, 2, 3 - - -$$

$$H_z = -2j A^+ \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \sin\left(\frac{p\pi}{d}\right)z \; e^{j\omega t - \gamma z}$$

$-2jA^+ = C$, constant

$$H_z = C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \sin\left(\frac{p\pi}{d}\right)z \; e^{j\omega t - \gamma z}$$

For TEmnp mode $\quad H_z = C \cos\left(\frac{m\pi}{a}\right)x \cos\left(\frac{n\pi}{b}\right)y \sin\left(\frac{p\pi}{d}\right)z \; e^{j\omega t - \gamma z}$

Q. Calculate the lowest Resonant frequency of a rectangular cavity Resonator of dimension $a = 2$ cm, $b = 1$ cm, $d = 3$ cm. → length of waveguide.

Lowest Resonant frequency is obtained for dominant mode

TE_{10}

for dominant mode TEmnp = TE_{101}

$$f_0 = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$$

$$\boxed{f_0 = 9 \text{ GHz}}$$

Q factor of a cavity Resonator

Q → measure of frequency selectivity of a Resonant or antiresonant circuit

$$Q = 2\pi \left(\frac{\text{maximum energy stored}}{\text{energy dissipated per cycle}}\right) = \frac{\omega W}{P} \qquad (I)$$

W = max^m energy stored or max^m stored energy

P = average Power loss

→ At Resonant frequency, the Electric and Magnetic Energies are equal and in time quadrature.

→ When the electric energy is maximum, the magnetic energy is zero and vice versa.

→ The total energy stored in the Resonator is obtained by integrating the energy density over the volume of the Resonator.

$$W_e = \int_v \frac{\varepsilon}{2} |E|^2 \, dv = W_m = \int \frac{\mu}{2} |H^2| \, dv = W \quad \cdots (1)$$

Where |E| and |H| are the peak values of the field intensities.

The average power loss in the Resonator can be evaluated by integrating the power density

$$P = \frac{R_s}{2} \int |H_t|^2 \, da \quad - (2)$$

$R_s = \sqrt{\frac{\omega\mu}{2\sigma}}$ → skin effect and magnitude of conductor surface resistance.

sub eq^n (1) & (2) in (I)

$$Q = \frac{\omega\mu}{R_s} \frac{\int_v |H|^2 dv}{\int_s |H_t|^2 da}$$

$$|H|^2 = |H_t|^2 + |H_n|^2$$

H_n = peak value of the normal magnetic intensity

$|H_t|^2 =$

$$Q = \frac{\omega\mu \text{ (volume)}}{2R_s \text{ (surface area)}}$$

- quality factor of a cavity resonator $Q = \frac{\omega_0 W}{P}$

W = max" energy stored

P = average power loss (dissipated power)

ω_0 = resonant frequency.

$Q = 2\pi \left(\frac{\text{max" energy stored per cycle}}{\text{energy dissipated per cycle}}\right)$

i) Q → of a perfect or ideal cavity resonator is infinite since in a perfect conductor forming the cavity, R would be zero. and also once energized it would resonate for ever

→ $Q_0 = \frac{\omega_0 L}{R}$ → a cavity resonator that is resonant at one frequency only

→ If there is more than one resonant frequency, there will be different values of Q for the various values of frequencies.

→ Coupling loops are used to couple the energy in & out of a cavity resonator.

→ The coupling has the effect of an imperfect by reflecting wall and so is the finite termination or load of the cavity.

→ This also changes the value of Q

→ This Q takes into account the coupling between the cavity and coupling paths is known as loaded Q_L

$$\frac{1}{Q_L} = \frac{1}{Q_0} + \frac{1}{Q_{ext}}$$

Q_0 = Q of an unloaded cavity

Q_{ext} = Q due to external ohmic losses

for An unloaded resonator represented by a series or a parallel resonant ckt. The unloaded Q is given by

$$Q_0 = \frac{\omega_0 L}{R}$$

$$Q_{external} = \frac{Q_0}{k} = \frac{\omega_0 L}{R k}$$

k = coupling coefficient of cavity

for critically coupled cavity (Resonator matched to Generator)
$k = 1$ hence loaded Q_L $\qquad Q_L = \frac{Q_0}{2}$

for under coupled cavity, $k < 1$ → cavity terminals are at voltage minimum and I/P impedance is reciprocal of SWR.

$k = \frac{1}{\beta}$ → SWR $\qquad \boxed{Q_L = \frac{\beta}{1+\beta} \cdot Q_0}$

for over coupled cavity, $k > 1$, cavity terminals are at voltage maximum and the impedance is standing wave ratio. $k = \beta$

$$Q_L = \left(\frac{Q_0}{1+\beta}\right)$$

Unloaded quality factor

$$Q_0 = \frac{\text{volume of cavity}}{(\text{skin depth})(\text{surface area of cavity})}$$

Q ↑ves with the ↑se in frequency
as skin depth ↓ with the ↑se in frequency.

Coupling coefficients

1) Critical coupling

If the resonator is matched to Generator then

$K = 1$

$$\frac{1}{Q_l} = \frac{1}{Q_0} + \frac{1}{Q_{external}}$$

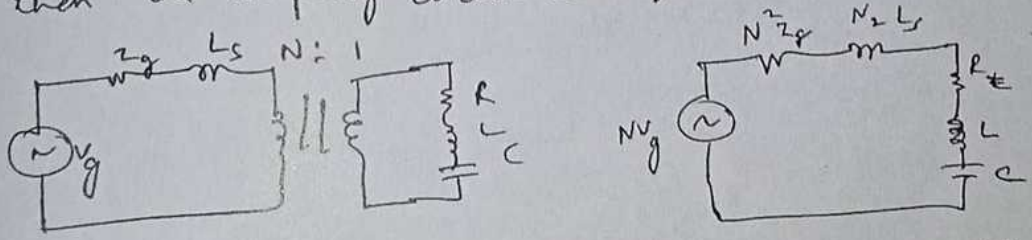
Q_l — loaded quality factor; Q_0 — unloaded Q

$$Q_{ext} = \frac{Q_0}{K}$$

unloaded Resonator → either a series or parallel Resonant circuit

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \qquad Q_0 = \frac{\omega_0 L}{R}$$

Let the cavity is coupled by means of an ideal $N:1$ transformer & a series inductance L_s to a generator having internal impedance Z_g then the coupling circuit & its equivalent.



a) coupling circuit

b) Equivalent ckt.

Cavity coupled to Generator

loaded Q

$$Q_L = \frac{\omega_0 L}{R + N^2 Z_g}$$

$$Q_{ext} = \frac{Q_0}{K}$$

In case of cavity Resonator R_s

loaded $Q_L = \frac{1}{2} Q_{ext}$

2) Overcoupling ($K > 1$)

Cavity terminals are at voltage maximum in the input line at Resonance.

The normalised impedance at the voltage max^m is SWR

$K = J$

loaded

$$Q_l = \frac{Q_0}{1+\beta}$$

3) Undercoupling (K<1)

Coupling terminals are at a voltage minimum and the input terminal impedance is equal to the Reciprocal of SWR.

$$K = \frac{1}{S}$$

loaded Q_l $\quad \boxed{Q_l = \frac{S}{S+1} \cdot Q_0}$

Multiple Questions Bank (Gate and and various public sectors) – Transmission lines and Waveguide and Antenna Design

1. Maxwell's equation for electromagnetic waves in a waveguide is
a) $\nabla \times E = -j\omega\mu$(vector H)
b) $\nabla \times E = -j\omega\mu$(vector E)
c) $\nabla \times H = -j\omega\mu$(vector H)
d) $\nabla \times H = j\omega\mu$(vector H)

Answer: a
Explanation: Maxwell's equation governs the propagation of waves in a waveguide. This equation describes the relation between electric field and magnetic field inside the waveguide.

2. If the wavelength of a signal is 10 mm, then the wavenumber of the material when a waveguide is filled with that material is:
a) 628
b) 345
c) 123
d) None of the mentioned

Answer: a
Explanation: The wave number is given by the expression $2\pi/\lambda$. Substituting the given wavelength, wave number is 628.

3. If a waveguide is filled with a lossy material then the expression for ϵ for that material is
a) $\epsilon = \epsilon\epsilon_r(1 - j\tan\delta)$
b) $\epsilon = \epsilon_o\ \epsilon_r\ (1/j\tan\delta)$
c) $\epsilon = \epsilon_o\ \epsilon_r/(1 + j\tan\delta)$
d) $\epsilon = \epsilon_o\ \epsilon_r/(1 - j\tan\delta)$

Answer: a
Explanation: When a waveguide is filled with a dielectric of or any lossy material, the relative permittivity is given by $\epsilon = \epsilon\epsilon_r(1 - j\tan\delta)$.

4. If a waveguide is filled with a lossless material of relative permeability 2, then the wave impedance in the TEM mode is:
a) 188.5 Ω
b) 170 Ω
c) 123 Ω
d) 345 Ω

Answer: a
Explanation: The wave impedance is given by the expression $\sqrt{\mu}/\sqrt{\epsilon}$ is the intrinsic impedance of the medium substituting the given values in the above expression, wave impedance is 188.5 Ω.

5. If the wave impedance of a medium is 200 Ω, then what is the relative permittivity of that medium?
a) 1.885
b) 2
c) 2.2
d) 2.5
Answer: a

Explanation: The wave impedance is given by the expression /√∈. is the intrinsic impedance of the medium substituting the given values in the above expression, the relative permittivity is 1.885.

6. If p=0.3 and the wave number of air in TM mode is 16, then the intrinsic impedance of air in TM mode given wave number is 125 is:
a) 1 Ω
b) 0.9 Ω
c) 0.8 Ω
d) 2 Ω
Answer: b
Explanation: Intrinsic impedance for TM mode of propagation is given by β/k. substituting the given values in the above equation; the intrinsic impedance is 0.9 Ω.

7. If the intrinsic impedance of a medium is 0.8 Ω, with wave number 125 and β being 0.2, then the relative permeability of the medium is:
a) 1.326
b) 2.34
c) 4.5
d) 6.7

Answer: a
Explanation: Intrinsic impedance for TM mode of propagation is given by β/k. substituting the given values in the above equation; the permeability of the medium is 1.326.

8. The losses that occur in a transmission line is:
a) Conduction losses
b) Di-electric loss
c) Both of the mentioned
d) None of the mentioned

Answer: c
Explanation: Both conductor loss and dielectric loss occur in a transmission medium. Conductor loss is due to the property of the transmission line while dielectric loss is due to the medium inside the transmission line.

9. Which of the following is true regarding attenuation?
a) Conductor loss
b) Di-electric loss
c) Sum of both conductor loss and dielectric loss
d) Attenuation is different from the losses

Answer: c
Explanation: Attenuation of a propagating wave is due to both the irregularities in the waveguide and as the property of the dielectric material. Hence attenuation of a propagating wave is due to both conductor loss and dielectric loss.

10. If the wave number of a medium is 20 and loss tangent is 0.4 , then the dielectric loss caused by the medium is:
a) 4
b) 2
c) 3

d) 6

Answer: a
Explanation: Dielectric loss in a medium is given by the expression k tan δ/2. Substituting the given value of loss tangent and wave number dielectric loss is 4 Np/m.

11. When a lossless line is terminated with an arbitrary load impedance Z_L, then it :
a) causes wave reflection on transmission lines
b) transmits the entire supplied power
c) causes loss in transmission line
d) none of the mentioned

Answer: a
Explanation: When a line is terminated with a impedance other the characteristic impedance of the transmission line, It results in reflection of waves from the load end of the transmission line hence resulting in wave reflection in the transmission line.

12. We say a transmission line is matched when:
a) $Z_L=Z_0$
b) $Z_L=\sqrt{Z_0}$
c) $Z_L=Z_0/2$
d) $Z_L=2Z_0$
Answer: a
Explanation: We say a line is matched only when the characteristic impedance of the transmission line is equal to the terminating load impedance. Hence condition for a line to be matched is $Z_L=Z_0$.

13. Voltage reflection coefficient can be defined as:
a) ratio of amplitude of reflected voltage wave to the transmitted voltage wave
b) ratio of amplitude of reflected voltage to the incident voltage wave
c) ratio of load impedance to the characteristic impedance of the transmission line
d) none of the mention

Answer: b
Explanation: From transmission line theory, reflection co-efficient of a transmission line is defined as the ratio of amplitude of reflected voltage to the incident voltage wave.

14. Expression for a voltage reflection co-efficient in terms of load impedance and characteristics impedance is:
a) $(Z_L- Z_0)/(Z_L+ Z_0)$
b) $(Z_L+ Z_0)/(Z_L- Z_0)$
c) $Z_L. Z_0/(Z_L+ Z_0)(Z_L-Z_0)$
d) $(Z_L+ Z_0)(Z_L-Z_0)/ Z_L. Z_0$
Answer: a
Explanation: The amplitude of the reflected voltage wave at the load end is equal to the difference between the load and the characteristic impedance, incident voltage is proportional to the sum of the load and characteristic impedance.

15. If a transmission line of a characteristics impedance 50 Ω is terminated with a load impedance of 100 Ω, then the reflection co efficient is

a) 0.3334

b) 0.6667

c) 1.6

d) 1.333

Answer: a

Explanation: Expression for reflection co-efficient of a transmission line is $(Z_L - Z_0)/(Z_L + Z_0)$ .substituting the given values of load and characteristic impedance, we get reflection co-efficient equal to 0.3334.

16. Return loss for a transmission line in terms of its reflection co efficient is given by

a) -20logl ┌l in dB where ┌ is the reflection coefficient.

b) -10logl ┌l in dB where ┌ is the reflection coefficient

c) -10log (1/l ┌l) in dB where ┌ is the reflection coefficient

d) -20log (1/l ┌l) in dB where ┌ is the reflection coefficient

Answer: a

Explanation: Return loss signifies the amount of energy reflected back from the load which is proportional to the reflection co-efficient of the line. Return loss in dB is given by the logarithm of the reflection co-efficient.

17. If the reflection coefficient for transmission line is 0.24, then the return lossin dB is

a) 12.39dB

b) 15dB

c) -12.39dB

d) -15.2dB

Answer: a

Explanation: The return loss of a transmission line, given the reflection co-efficient is -20logl ┌l in dB where ┌ is the reflection co-efficient. Substituting for reflection co-efficient in the above equation, return loss is 12.39dB.

18. Expression for VSWR in terms of reflection co-efficient is:

a) (1+| ┌|)/(1-| ┌|)

b) (1-| ┌|)/(1+| ┌|)

c) 1/| ┌|

d) 1/1+| ┌|

Answer: a

Explanation: VSWR is the ratio of maximum amplitude of the standing wave formed to the minimum amplitude of the standing wave, when these voltages are expressed in terms of reflection co-efficient, we get the expression (1+| ┌|)/(1-| ┌|).

19. If the reflection co-efficient for a transmission line is 0.3, then the VSWR is

a) 0.5384

b) 1.8571

c) 0.4567

d) 3.6732
Answer: b
Explanation: VSWR (voltage standing wave ratio) in terms of reflection co-efficient is given by $(1+|\Gamma|)/(1-|\Gamma|)$.substituting $\Gamma=0.3$ in this equation we get, VSWR=1.8571.

20. If a transmission line of characteristic impedance 50 Ω is terminated with a load impedance of 150 Ω then VSWR is:
a) 0.75
b) 0.5
c) 2
d) none of the mentioned
Answer: a
Explanation: VSWR (voltage standing wave ratio) in terms of load and characteristic impedance is given by $Z_L - Z_0 / Z_L + Z_0$. Substituting for Z_L and Z_0 in the above equation, VSWR is 0.5.

21. Expression for input impedance of a transmission line in terms of load impedance and characteristic impedance is:
a) $Z_0 (Z_L+j Z_0 \tan \beta l)/ (Z_0+j Z_L \tan \beta l)$
b) $(Z_0+j Z_L \tan \beta l)/ (Z_L+j Z_0 \tan \beta l)$
c) $Z_0 (Z_L-j Z_0 \tan \beta l)/ (Z_0-j Z_L \tan \beta l)$
d) $(Z_0-j Z_L \tan \beta l)/ (Z_L-j Z_0 \tan \beta l)$
Answer: a
Explanation: Representing the input voltage as the ratio of voltage at current, representing voltage and currents in hyperbolic function form and simplifying, we get $Z_0 (Z_L+j Z_0 \tan \beta l)/ (Z_0+j Z_L \tan \beta l)$.

22. Input impedance of a short circuited transmission line is
a) $-jZ_0 \tan \beta l$
b) $jZ_0 \tan \beta l$
c) $jZ_0 \cot \beta l$
d) $-jZ_0 \cot \beta l$
Answer: b
Explanation: Since the load impedance of a short circuited transmission line is zero, substituting ZL=0 in the expression for input impedance of a transmission line $Z_0 (Z_L+j Z_0 \tan \beta l)/ (Z_0+j Z_L \tan \beta l)$, input impedance of the transmission line comes out to be $jZ_0 \tan \beta l$.

23. Input impedance of a transmission line can be represented in terms of this simple trigonometry function.
a) sine function
b) cosine function
c) cotangent function
d) tangent function
Answer: d
Explanation: The input impedance of a transmission line is expressed in the standard form as $Z_0 (Z_L+j Z_0 \tan \beta l)/ (Z_0+j Z_L \tan \beta l)$ which is represented in terms of a tangent function.

24. If a λ /3 transmission line is short circuited that has a characteristic impedance of 50 Ω, then its input impedance is:
a) -j100Ω
b) 50Ω
c) 86.60Ω
d) –j86.60Ω
Answer: d
Explanation: For a short circuited transmission line, the input impedance is given by $jZ_0\tan\beta l$.substituting for characteristic impedance and 'l' in the above equation, input impedance is –j86.60Ω.

25. Expression for input impedance of an Open circuited transmission line is:
a) $-jZ_0\tan\beta l$
b) $jZ_0\tan\beta l$
c) $jZ_0\cot\beta l$
d) $-jZ_0\cot\beta l$
Answer: d
Explanation: Since the load impedance of a open circuited transmission line is infinity, substituting ZL=infinity ($1/ Z_L = 0$) in the expression for input impedance of a transmission line $Z_0 (Z_L+j Z_0\tan \beta l)/ (Z_0+j Z_L\tan \beta l)$, input impedance of the open circuited transmission line comes out to be- $jZ_0\cot\beta l$.

26. Input impedance of a open circuited transmission line is represented using this trigonometric function:
a) sine function
b) cosine function
c) cotangent function
d) tangent function
Answer: c
Explanation: The input impedance of a transmission line is expressed in the standard form as $Z_0 (Z_L+j Z_0\tan \beta l)/ (Z_0+j Z_L\tan \beta l)$. With Z_L equal to infinity for open circuit termination, $1/ Z_L$ equal to 0, substituting this, we get input impedance in terms of a cotangent function.

27. For a λ/2 transmission line, if the characteristic impedance of the line is 50 Ω and the terminated with a load of 100 Ω, then its input impedance is:
a) 100Ω
b) 50Ω
c) 88.86Ω
d) none of the mentioned
Answer: a
Explanation: Input impedance of a transmission line is given by $Z_0 (Z_L+j Z_0\tan \beta l)/ (Z_0+j Z_L\tan \beta l)$. Substituting $\beta=2\pi/\lambda$, and $l=\lambda/2$, we get input impedance of the transmission line equal to the load impedance or the terminated load.

28. If a λ/3 transmission line is open circuited and has characteristic impedance of 50 Ω then the input impedance is:
a) j28.86Ω
b) 50Ω

c) j50Ω
d) 28.86Ω
Answer: a
Explanation: Input impedance of an open circuited transmission line is given by $-jZ_0\cot\beta l$. Substituting $l=\lambda/3$ and $\beta=2\pi/\lambda$ in the above equation, input impedance is j28.86Ω.

29. Expression for a transmission co-efficient of a transmission line is :
a) $2Z_L/(Z_L+Z_0)$
b) $(Z_L-Z_0)/(Z_L+Z_0)$
c) $2Z_0/(Z_L+Z_0)$
d) $(Z_L+Z_0)/(Z_L-Z_0)$
Answer: a
Explanation: $T=\Gamma+1$, where T is the transmission co-efficient and Γ is the reflection co-efficient substituting $\Gamma=(Z_L-Z_0)/(Z_L+Z_0)$ in the equation for transmission co-efficient, we get $2Z_L/(Z_L+Z_0)$.

30. For a transmission line, if the reflection coefficient is 0.4, then the transmission coefficient is:
a) 0.4
b) 1.4
c) 0.8
d) 2.8
Answer: b
Explanation: $T=\Gamma+1$, where T is the transmission co-efficient and Γ is the reflection co-efficient substituting $\Gamma=0.4$ in the above equation, transmission co-efficient is equal to 1.4.

31. If the transmission coefficient of a transmission line is 1.6, then the reflection co efficient is:
a) 0.8
b) 0.6
c) 0.4
d) 0.3

Answer: b
Explanation: $T=\Gamma+1$, where T is the transmission co-efficient and Γ is the reflection co-efficient substituting T=1.6, we get $\Gamma=0.6$.

32. For a transmission line, if the transmission coefficient is 1.4, then the insertion loss in dB is:
a) -2.922dB
b) 29dB
c) 1.46dB
d) -29dB
Answer: a
Explanation: Insertion loss for a transmission line is given by the expression $-20\log|T|$ in dB. Substituting T=1.4 and taking logarithm to base 10, insertion loss is -2.922dB.

33. The relation between nepers and decibels is
a) 1 Np= 8.686 dB

b) 1 dB=8.868 dB
c) Np≥dB
d) dB≥Np
Answer: a
Explanation: 1 Np=10log e^2 dB. Substituting e=2.718 in the above equation , 1Np=8.686 dB.

34. In a two wire transmission line, if the distance between the lines is 20 mm and the radii is 5 mm the the inductance of the line is:
a) 0.1 μH
b) 0.526 μH
c) 0.9 μH
d) 1 μH
Answer: b
Explanation: The inductance of a two wire transmission line is given by the equation μ*ln(b/a)/2π. Substituting the given values in the above equation, inductance is 0.526 μH.

35. In a two wire transmission line , if the distance between the lines is 60mm and the radii is 10 mm , then the capacitive reactance of the line when operated at 12.5 GHz is
a) 20 pF
b) 21.13 pF
c) 23 pF
d) 12 pF
Answer: b
Explanation: The capacitive reactance of a two wire transmission line is π∈/$\cosh^{-1}$(D/2a). substituting the given values in the above expression, the capacitive reactance is 21.13 pF.

36. For a parallel plate type of a transmission line, then expression for conductance of the line is:
a) ∈(ω)w/d
b) $2R_s/x$
c) μ/2π ln (w/d)
d) μ/π $\cosh^{-1}$(w/2d)
Answer: a
Explanation: The conductance of a parallel plate waveguide is dependent on the complex value of the permittivity, width of the waveguide and the distance between the waveguide plates.

37. One of the Maxwell's curl equation that is satisfied inside a coaxial line is:
a) ∇×E =-jωμ (vector H)
b) ∇×E =-jωμ(vector E)
c) ∇×H=-jωμ(vector H)
d) ∇×H=jωμ(vector H)
Answer: a
Explanation: ∇×E = -jωμ (vector H).This is the Maxwell's equation satisfied by the electric and magnetic fields inside a waveguide.

38. The wave impedance of air for a wave propagating in it is:
a) 377 Ω

b) 345 Ω
c) Insufficient data
d) None of the mentioned
Answer: a
Explanation: Intrinsic impedance is the impedance offered by air for a wave propagating in it. This is a standard value and is 377 Ω.

39. Wave impedance of a wave travelling in a medium of a relative permittivity 2 and permeability 4 is
a) 188.5 Ω
b) 200 Ω
c) 300 Ω
d) None of the mentioned
Answer: a
Explanation: Intrinsic impedance of a medium is given by the expression $\sqrt{\mu/\epsilon}$. Substituting the given values in the above expression, the wave impedance is 188.5 Ω.

40. For a parallel plate transmission line, if w= 12 mm and the distance between the plates is 2 mm, then the inductance of the transmission line is:
a) 0.2 μH
b) 0.1 μH
c) 0.3 μH
d) 0.4 μH

Answer: a
Explanation: The inductance of a parallel plate transmission line is given by μd/W. substituting the given values in the above expression, the inductance is 0.2 μH.

41. Expression for capacitance of a two wire transmission line is
a) $\epsilon*\pi/\cosh^{-1}(D/2a)$
b) $\mu/\pi*\cosh^{-1}(D/2a)$
c) $2\pi\epsilon/\ln (D/2a)$
d) $\epsilon*\pi\omega/\cosh^{-1}(D/2a)$
Answer: a
Explanation: The expression for capacitance of a two wire transmission line is $\epsilon\pi/\cosh^{-1}(D/2a)$.
Capacitance of a two wire transmission line is dependent on the distance between the two lines and the radius of the line.

42. If the distance between the 2 wires in a 2 wire transmission line is 10 mm and the radii 2 mm, then the inductance of the transmission line is:
a) 0.62 μH
b) 1 μH
c) 2 μH
d) None of the mentioned
Answer: a

Explanation: The inductance of a two wire transmission line is given by the expression $\mu\cosh^{-1}(D/2a)/\pi$. Substituting the given values in the above expression, the inductance is 0.62 μH.

43. For a parallel plate transmission line, if the complex part of permittivity is 2.5, if the width is 100 mm and the distance between the plates is 10 mm, then the conductance of the transmission line is:
a) 25 Ʊ
b) 30 Ʊ
c) 45 Ʊ
d) None of the mentioned
Answer: a
Explanation: Conductance of a parallel plate transmission line is ∈W/d. substituting the given values in the above expression, the conductance of the transmission line is 25 Ʊ.

44. If the dielectric loss of a medium is 0.2 Np/m with a wave number of 12, then the value of loss tangent is
a) 0.0334
b) 0.05
c) 0.08
d) 0.09
Answer: a
Explanation: Dielectric loss in a medium is given by the expression k tan δ/2. Substituting the given values, the loss tangent of the medium is 0.0334

45. The key difference between circuit theory and transmission line theory is
a) circuit elements
b) voltage
c) current
d) electrical size
Answer: d
Explanation: Circuit theory assumes physical dimensions of the network smaller than electrical wavelength, while transmission lines may be considerable fraction of wavelength.

46. Transmission line is a_____parameter network.
a) Lumped
b) distributed
c) active
d) none of the mentioned
Answer: b
Explanation: Since no lumped elements like resistors, capacitors are used at microwave frequencies, only transmission lines are used. Hence they are called distributed parameter network.

47. For transverse electromagnetic wave propagation, we need a minimum of
a) 1 conductor
b) 2 conductors
c) 3 conductors
d) bunch of conductors

Answer: b
Explanation: With a single conductor, transverse electromagnetic wave propagation is not possible. Hence we need a minimum of 2 conductors.

48. To model a transmission line of infinitesimal length Δz, the lumped element that is not used is
a) resistor
b) inductor
c) capacitor
d) transistor
Answer: d
Explanation: In the lumped element circuit model of a transmission line, we use only resistor, capacitor and inductor. Hence no transistor is used.

49. _____and_____contribute to the impedance of a transmission line in the lumped element representation.
a) resistor, inductor
b) resistor, capacitor
c) capacitor, inductor
d) transistor, capacitor
Answer: a
Explanation: $Z=R+j\omega L$. Hence, both resistor and inductor contribute to the impedance of the transmission line.

50. _____and_____contribute to the admittance of a transmission line in the lumped element representation.
a) conductance G, capacitor
b) conductance, inductor
c) resistor, capacitor
d) resistor, inductor
Answer: a
Explanation: $Y=G+j\omega C$. Hence, both conductance and capacitance contribute to the admittance of the transmission line.

51. Characteristic impedance of a transmission line is:
a) impedance Z of a transmission line
b) impedance which is a constant at any point on the transmission line
c) reciprocal of admittance of a transmission line
d) none of the mentioned
Answer: b
Explanation: Characteristic impedance is defined as that impedance of a line which is a constant when measured at any point on the line, Hence B.

52. Propagation constant γ is a :
a) real value
b) none of the mentioned
c) imaginary value

Unit - IV

Antenna Basics

1) Maxwell's Equations

By Gauss's law $\nabla \cdot \vec{D} = \rho_v$ — (1)

$\nabla \cdot \vec{B} = 0$ — (2)

By Faraday's law $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\dot{\vec{B}}$ — (3)

The source of electric field is the changing magnetic field.

By Ampere's law

$$\nabla \times \vec{H} = \vec{J_c} + \frac{\partial \vec{D}}{\partial t}$$

$$\Rightarrow \nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t}$$

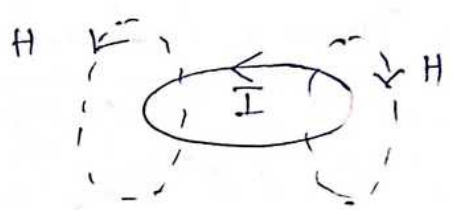
This means source of magnetic field is changing electric field.

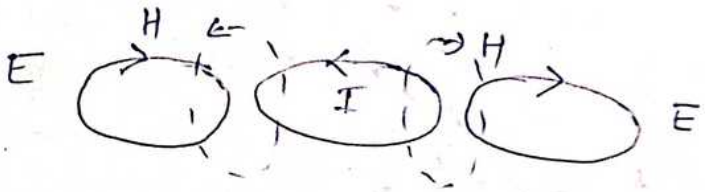
Another important eq^{n}s is continuity equations that is

$$\nabla \cdot \vec{J} = \frac{-\partial \rho_v}{\partial t}$$

It says that changing charge density produces $\vec{J}$.

But the source of electric field is charge (ρ_v) and magnetic field is $\vec{J}$. but $\vec{J}$ and ρ_v are related then, so time varying em field can be studied from the current density alone.





Defination of an Antenna :-

(i) By Webster's dictionary "A usually metallic (as a rod or wire) for radiating or receiving radio waves.

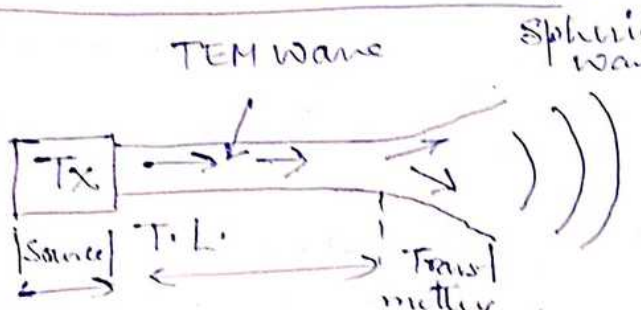
(ii) IEEE standard difinition — "A means for radiating or receiving radio waves.

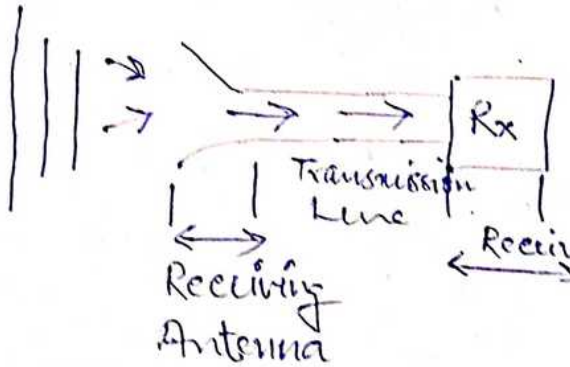
(iii) An antenna is a transitional structure b/w a free space and a guiding structure or vice-ve

Fuctions of an antenna :-

* In the transmitting mode, it is used to radiate EM
* In the receiving mode, it is used as a sensor of EM wave.
* Functions as a transducer which converts electrical energy into EM wave at the transmitter side and it convert EM energy into electrical energy at receiving side
* Acts as a impedance matching device

Radio Communication Link :-

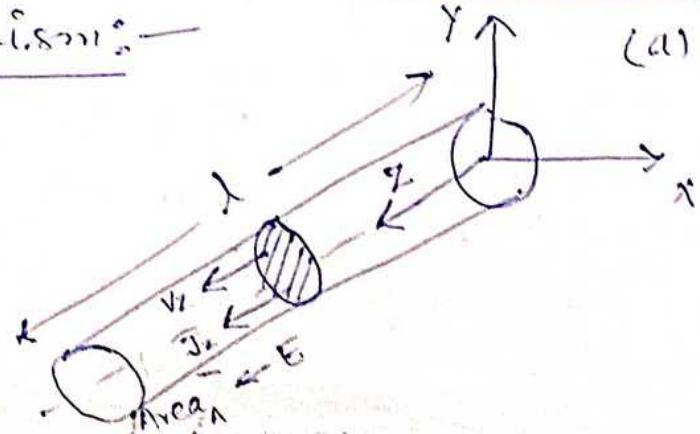




Radio Communication link.

Radiation Mechanism :-

(a) Simple Wire



$\rho_v v_z$ —(i)

J_z = Current Density $\left(\frac{A}{m^2}\right)$ $\Rightarrow \frac{A}{m^2} = \left(\frac{C}{m^3} \cdot \frac{m}{s}\right) = \frac{C}{m^2 s}$

ρ_v : volume charge density $\left(C/m^3\right)$

v_z = velocity of charge Q. $\left(\frac{m}{s}\right)$

When the wire is ideal conductor, current flow on the surface of conductor, resulting in surface charge density $\left(\frac{\rho_s}{m^2}\right)$

$$J_s = \rho_s v_z \quad \text{—(ii)}$$

J_s = Surface current density $\left(C/m^2\right)$

v_z = velocity of charge carrier $\left(m/s\right)$

Now if the wire is ideally thin, then instead of current density, we will express current in z-direction

$$I_z = \rho_l v_z \Rightarrow \left(\frac{C}{m} \times \frac{m}{s}\right) \quad \text{—(iii)}$$

if current is time varying, then differentiating (iii)

then $\frac{dI_z}{dt} \rho_l \frac{dv_z}{dt} = \rho_l a_z$ —(iv)

For the wire of length L, multiplying L by (iv)

$$L \frac{dI_z}{dt} = L \rho_L a_z$$

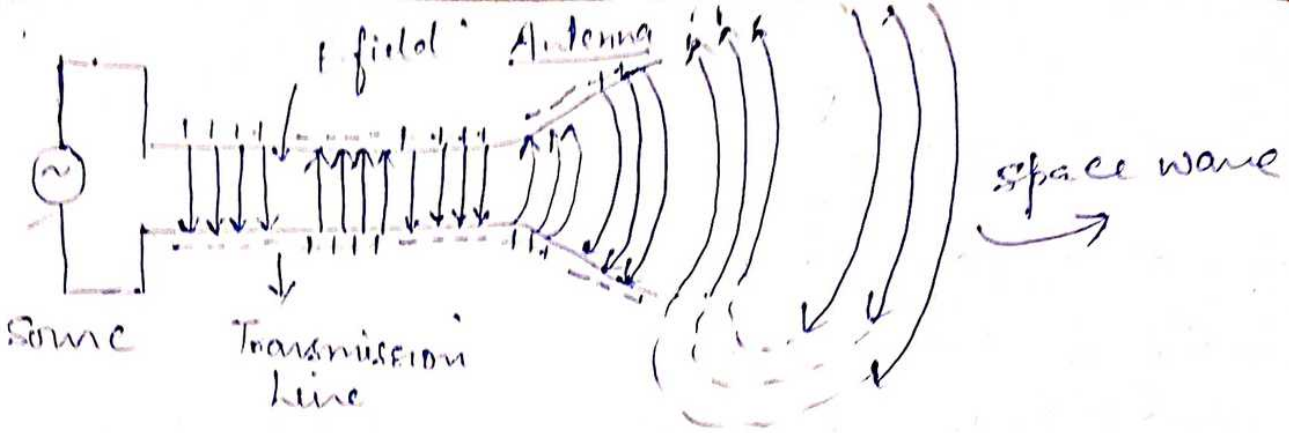
$$\Rightarrow \boxed{L\dot{I} = Q\dot{V}}$$ Basic Radiation Equation

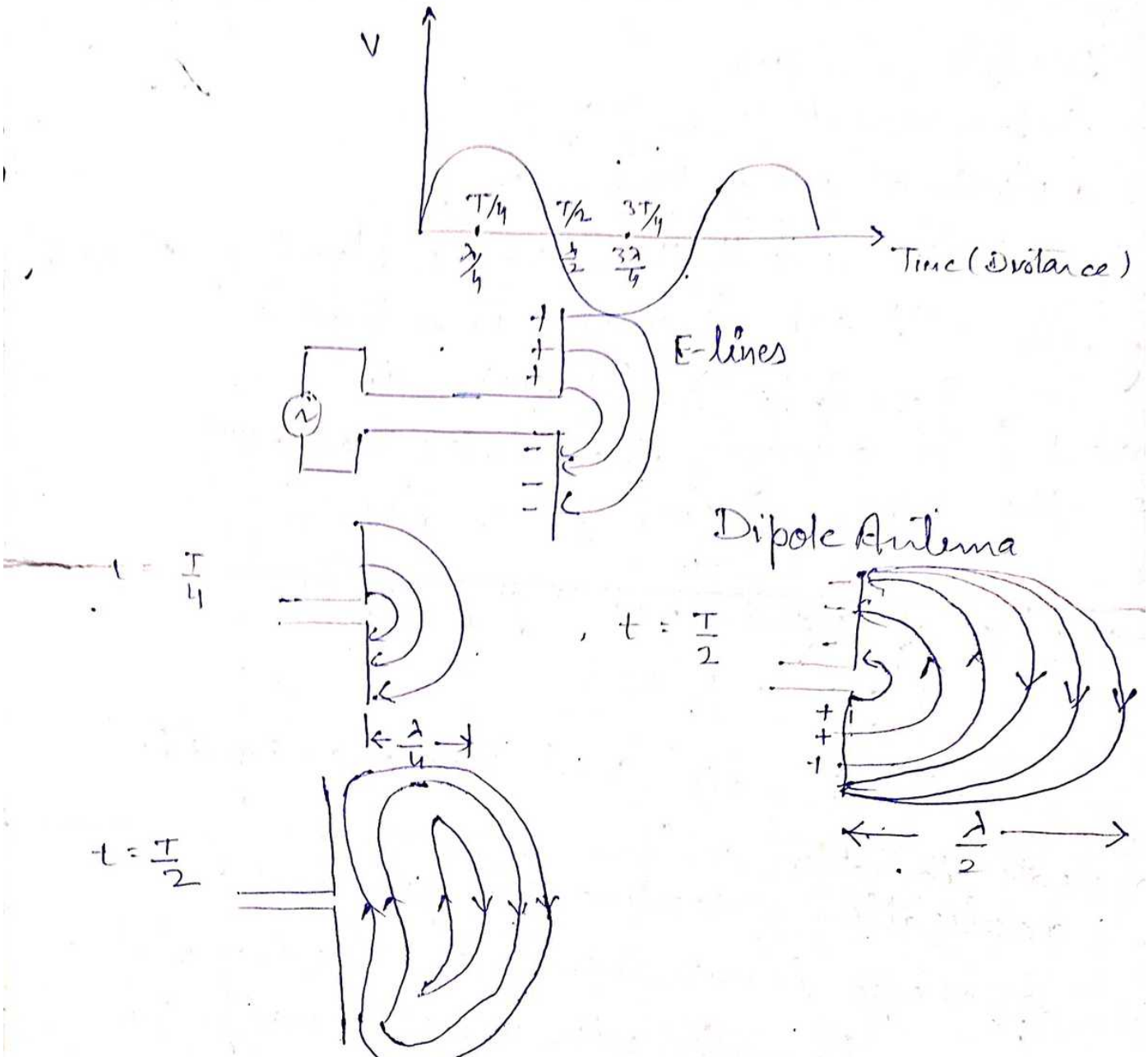
So to create radiation, there must be time varying current or an acceleration or deceleration of charge.

Case I: if charge is not moving $v = 0$, Radiation $= 0$

Case II: if charge is moving with uniform velocity then acceleration is zero $\Rightarrow$ Radiation $= 0$

Double Wires as a source of radiation:—

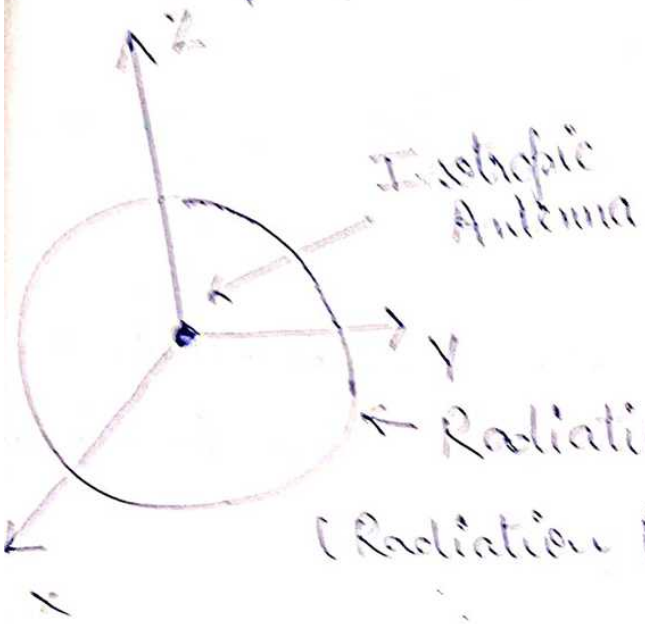




Antenna Parameters :- Irrespective of type of application antenna, all antennas posses cer basic properties which are as follows.

(I) Radiation Pattern
(II) Half-Power Beam Width (HPBW), First Null Beam Width (F
(III) Beam Area (IV) Beam Efficiency (V) Directivity D or ga
(VI) Effective aperture Ae etc.

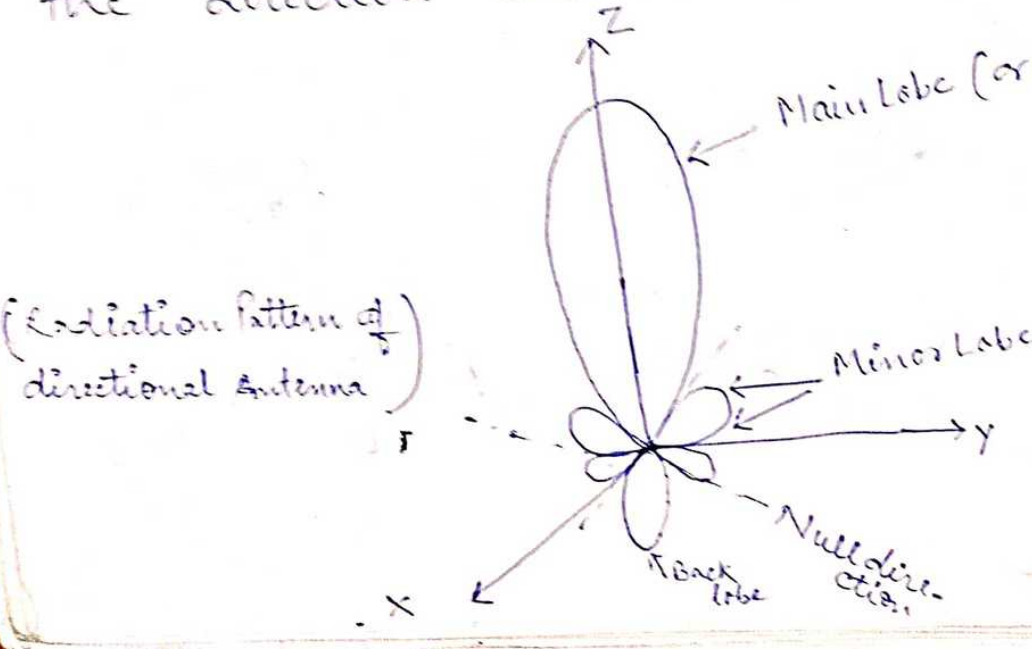
iation Pattern :- The radiation pattern is a graphical representation of the ation of the antenna as a function of direction



(Radiation pattern of isotropic antenna)

) Isotropic Antenna :- It is an antenna which radiates in all directions. It a point source of radiation. It is not possible n practice.

ii) Directional Antenna :- (Non-Isotropic Antenna) That antenna radiates more in a particular direction and less in other directions. is called the direction antenna.



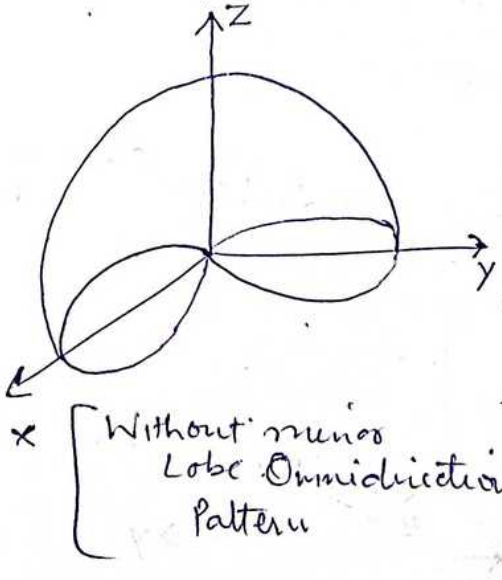
* Major Lobe :- The major Lobe also called the major lobe. and it is defined as the radiation lobe containing the direction of maximum radiation.

* Minor Lobe :- It is any lobe except the major lobe.

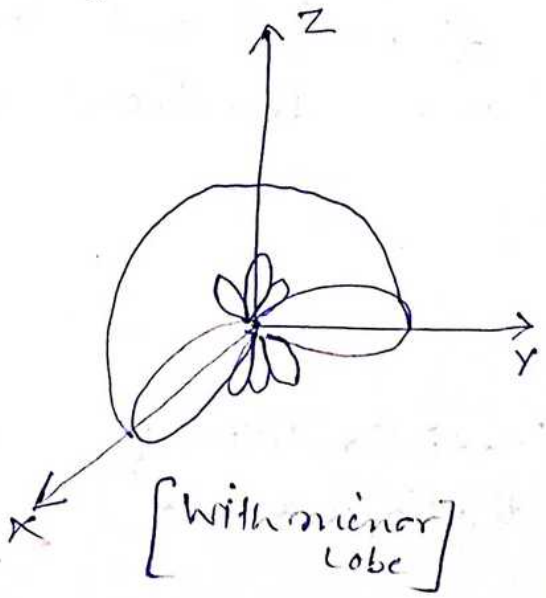
* Side Lobe :- A side lobe is adjacent to the main lobe and occupies the hemisphere in the direction of the main lobe.

* Back Lobe :- It is the radiation lobe whose axis is at 180° with respect to the major beam of an antenna.

Omni directional Antenna :- It is defined as an antenna having an essentially non-directional pattern in given plane and a directional pattern in any orthogonal plane.



[Without minor Lobe Omnidirectional Pattern]

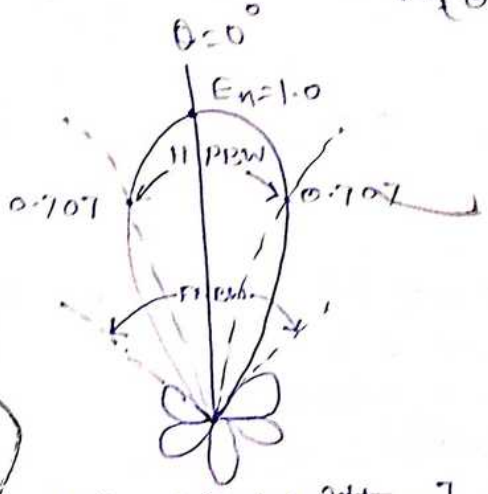


[With minor Lobe]

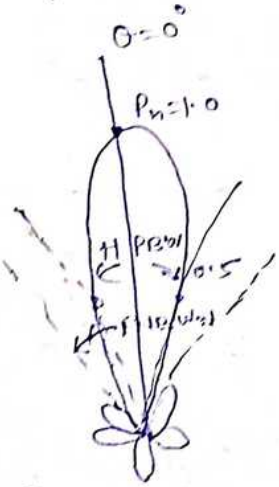
Normalized Pattern :-

$$E_0(\theta,\phi)_n = \frac{E_0(\theta,\phi)}{E_0(\theta,\phi)_{max}}$$

$$P_0(\theta,\phi)_n = \frac{W(\theta,\phi)}{W(\theta,\phi)_{max}}$$



[Normalized Field Pattern]

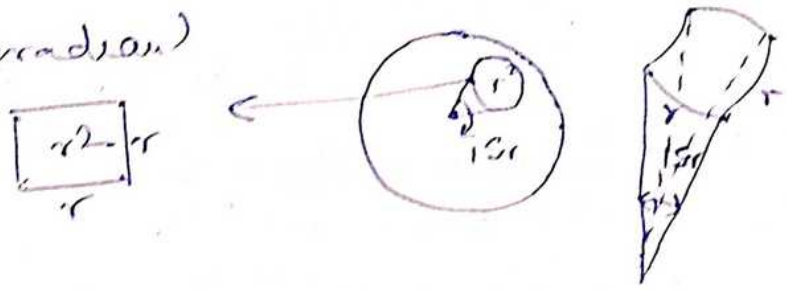


[Normalized Power Pattern]

Half-Power Beam Width (HPBW) :- HPBW is defined as the angular distance b/w two points where the power is half of the maximum value and the electric field is 0.707 ($\frac{1}{\sqrt{2}}$) of its maximum value.

First Null Beam Width (FNBW) :- FNBW is defined as the angular distance b/w points on each side of major lobe where the radiation drops to zero.

Beam Area (Ω_A) :- Beam Area is defined in terms of solid angle (steradian)



$$ds = r^2 \sin\theta \, d\theta \, d\phi$$

$$d\Omega = \frac{ds}{r^2} = \sin\theta \, d\theta \, d\phi$$

[r^2 subtends 1 Str, then ds subtends an angle $d\Omega$ Str]

$$\Omega_A = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} P_n(\theta, \phi) \, d\Omega \quad (Sr)$$

$$\Omega_A = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} P_n(\theta, \phi) \sin\theta \, d\theta \, d\phi$$

$$\boxed{\Omega_A \cong \theta_{HP} \phi_{HP} \, Sr}$$

Power

Radiation Density: $\vec{W} = \vec{E} \times \vec{H}$

So Instantaneous Power $\therefore P = \oint_S \vec{W} \cdot d\vec{s}$

$W_{av} = \frac{1}{2} Re[\vec{E} \times \vec{H}^*]$

So Power Radiated $P_{rad} = P_{av} = \oint_S \vec{W}_{rad} \cdot d\vec{s} = \frac{1}{2} \oint Re[\vec{E} \times \vec{H}^*] \cdot d\vec{s}$

Radiation Intensity (U) — The Power radiated from an antenna per unit solid angle is called the radiation intensity $U (W/Sr)$

It is obtained by multiplying the radiation density (W_{rad}) by the square of distance

$$U = r^2 W_{rad}$$

The Power Radiated $P_{rad} = \oint U \, d\Omega = \oint U \sin\theta \, d\theta \, d\phi$ watts

Ex-1 For isotropic radiator, find the radiation intensity U_0 when the power radiated is P_{rad}.

Ans

$$P_{rad} = \oint U_0 \, d\Omega = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} U_0 \sin\theta \, d\theta \, d\phi = U_0 (2\pi)(-\cos\theta)_0^{\pi} = 4\pi U_0$$

So $\boxed{U_0 = \frac{P_{rad}}{4\pi}}$

ctivity :- The directivity of an antenna is defined as the ratio of the radiation intensity in given direction from the antenna to the radiation intensity averaged over all directions.

$$D = \frac{U_{\text{given direction}}}{U_{av}} = \frac{U}{U_{av}}$$

$$D = \frac{4\pi U_{max}}{P_{rad}} = \frac{4\pi U_{max}}{\oint U \, d\Omega}$$

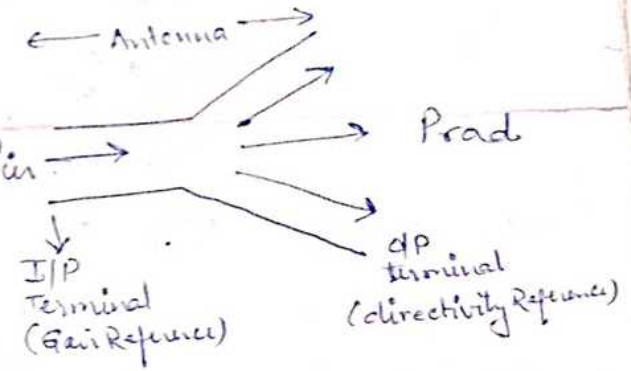
$$= \frac{4\pi}{\oint \frac{U}{U_{max}} \sin\theta \, d\theta \, d\phi} \cdot \frac{4\pi}{\Omega_A}$$

$$U_{av} = \frac{P_{rad}}{4\pi} \Rightarrow D = \frac{U}{P_{rad}/4\pi} = \frac{4\pi U}{P_{rad}} = \frac{U}{U_0}$$

Another defination

$$D = \frac{\text{Radion intensity in max. direction}}{\text{Radiation intensity of isotropic source}}$$

Gain of an Antenna :-



$$G = \frac{4\pi U}{P_{in}}$$

The gain is an actual or realized quantity which is less than the directivity 'D' due to ohmic loss in the antenna.

$$G = \eta D$$

Here η = Efficiency of the antenna

Beam Efficiency :- The beam area is the total beam angle which includes major lobe and minor lobes beam areas.

$$\Omega_A = \Omega_M + \Omega_m$$

Here, Ω_A = Total beam area

Ω_M = Major lobe beam area

Ω_m = minor lobe beam area.

So, the beam efficiency $\varepsilon_M = \frac{\Omega_M}{\Omega_A}$ (9)

... ...magnitudes and $\psi_1, \psi_2, \ldots$ are the phase (1)

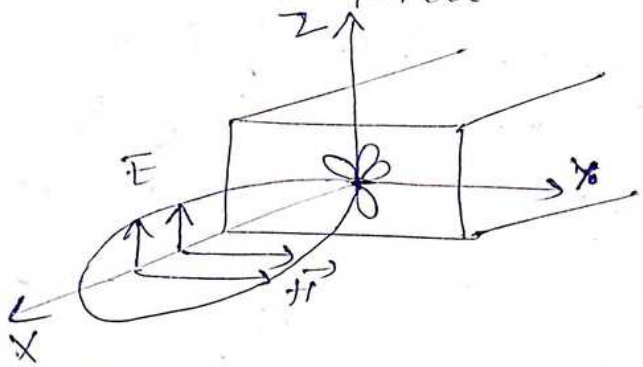
and stay factor $\varepsilon_{on} = \frac{\Omega_A}{\Omega_A}$

Principal Pattern :- E and H Plane patterns are known as called principle Pattern

$E_\theta(\theta, \phi)$. $E_\phi(\theta, \phi)$, $\delta_\theta(\theta, \phi)$; $\delta_\phi(\theta, \phi)$

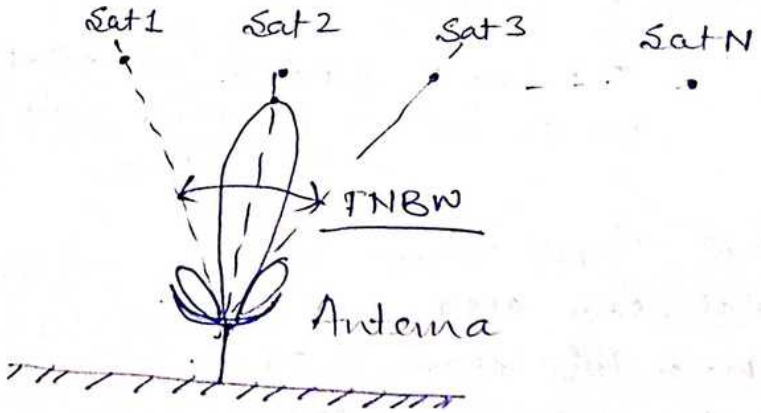
$$E_\theta(\theta, \phi)_n = \frac{E_\theta(\theta, \phi)}{E_\theta(\theta, \phi)_{max}}$$

$$P_n(\theta, \phi) = \frac{W(\theta, \phi)}{W(\theta, \phi)_{Max}}$$



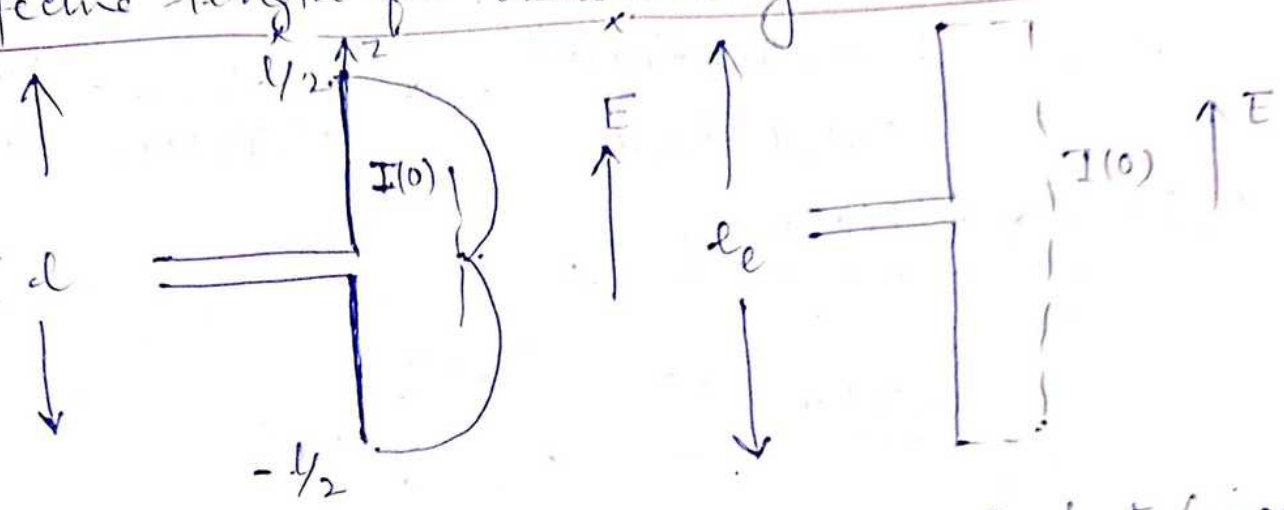
Antenna Resolution :- It is defined as the half the beam widths between first nulls.

$$\text{Resolution} = \frac{FNBW}{2}$$



ective Height & Effective Length (l_e) :- The term Effective length indicates how much antenna is effective a radiator or collector of electromagnetic energy.

* Effective length for transmitting antenna :-



The effective length is that length of an equivalent linear antenna that has a current $I(0)$ at all the points along its length and that radiates the same field strength as the actual antenna in the direction perpendicular to its length.

$$\int_{-l/2}^{l/2} I(z)\,dz = I(0)\,l_e \quad \text{or} \quad l_e = \frac{1}{I(0)} \int_{-l/2}^{l/2} I(z)\,dz$$

$$\Rightarrow \boxed{l_e = \frac{I_{av}\, l}{I(0)}}$$

For transmitting case

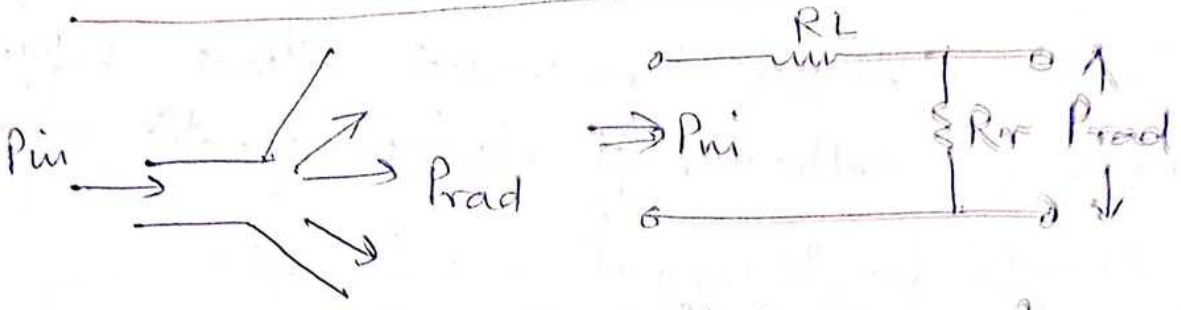
l_e = Effective length (m)

l = physical or actual length (m)

I_{av} = Avg current (A)

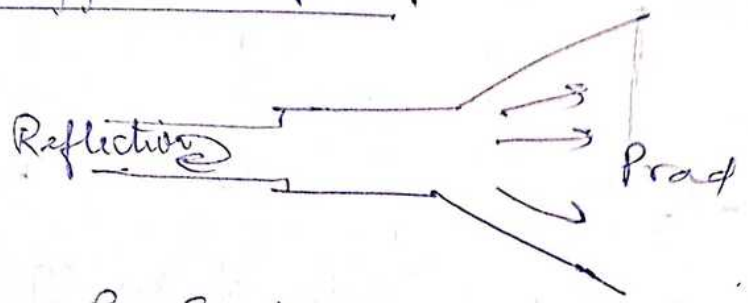
* Effective length for Receiving Antenna :- Consider a vertical dipole of length(l) is immersed in an incident field E then some voltage V_{oc} will get at the open terminal of the antenna so effective length $\boxed{l_e = \frac{V_{oc}}{E}}$

Antenna Radiation Efficiency (e_{cd})



$$e_{cd} = \frac{\text{Power Radiated}}{\text{Total Power}} = \frac{I^2 R_r}{I^2 (R_r + R_L)} = \frac{R_r}{R_r + R_L}$$

Antenna Efficiency



$$e_t = e_r e_{cd} = (1 - |\Gamma|^2) e_{cd}$$

$$\Gamma = \frac{V_r}{V_i} \Rightarrow \Gamma^2 = \frac{P_r}{P_i}$$

$$e_r = \frac{P_{rad}}{P_{in}} = \frac{P_{in} - P_{ref}}{P_{in}} = 1 - \frac{P_{ref}}{P_{in}} = 1 - |\Gamma|^2$$

$$e_t = e_r e_{cd} = [1 - |\Gamma|^2] e_{cd}.$$

Num 1:- An antenna has a directivity of 20 and radiation efficiency of 90%. Compute gain in dB.

$$G = e_{cd} D$$

$$\Rightarrow G = 0.9 \times 20 = 18$$

$$G = 10 \log_{10} 18$$

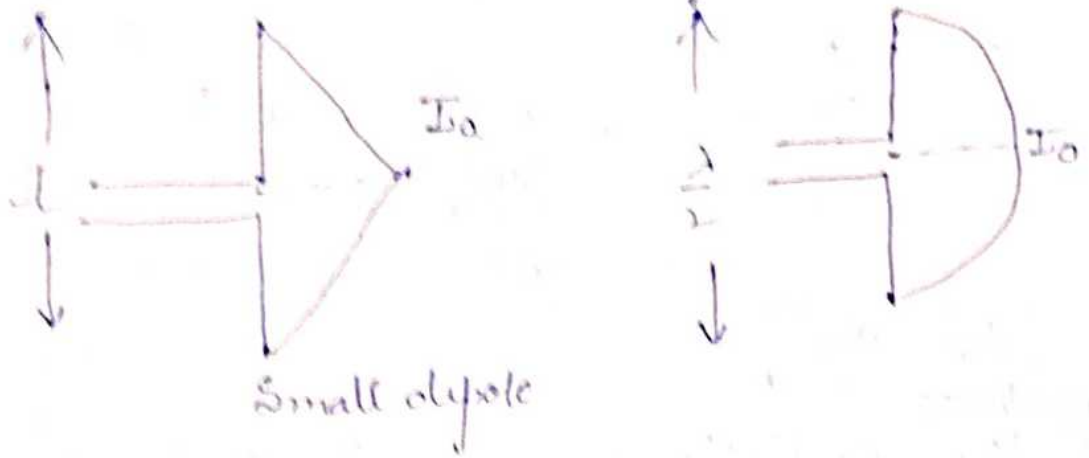
Num 2: The Radiation resistance of an antenna is 72 Ω and the loss resistance is 8 Ω. What is the directivity in dB if the Power gain is 16

$$e_{cd} = \frac{R_r}{R_r + R_L} = \frac{72}{80} = 0.9 \rightarrow 90\%$$

$$\Rightarrow D = \frac{16}{0.9} = 17.77$$

$$D(dB) = 10 \log (17.77) = 12.49$$

sign shows that potential increases in opposite direction to the electric field



Small dipole

For small dipole

$$\text{Area} = \frac{1}{2} I_0 l$$

I_{av} current over length $= \frac{1}{l} \times \frac{I_0 l}{2} = \frac{I_0}{2}$

$$I(0) = I_0$$

So $$l_e = \frac{I_{av}}{I(0)} l = \frac{I_0}{2 I_0} l = \frac{l}{2} = 0.5\, l \ (\text{m})$$

For $\frac{\lambda}{2}$ dipole $I_{av} = \frac{2 I_0}{\pi}$

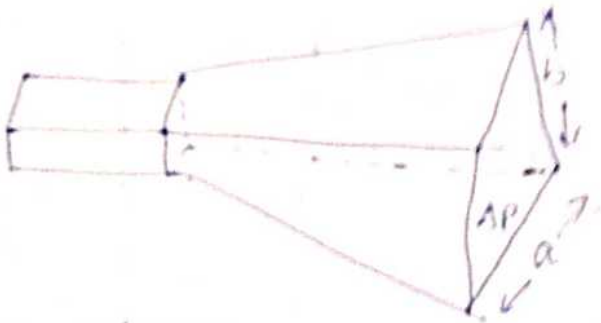
$$I(0) = I_0$$

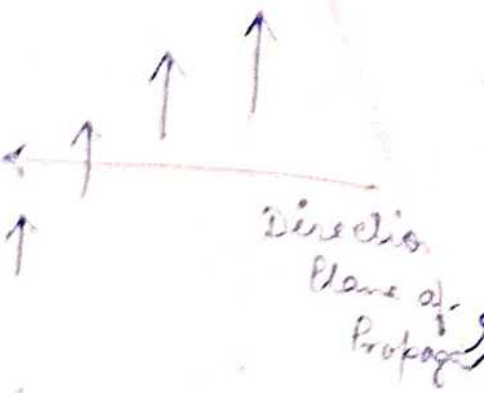
$$\Rightarrow l_e\left(\tfrac{\lambda}{2}\text{dipole}\right) = \frac{2 I_0}{\pi I_0} l = \frac{2}{\pi} \frac{\lambda}{2} = \frac{\lambda}{\pi} \text{ meter}$$

Antenna Aperture :- With each antenna there are no of equivalent areas (apertures). These are used to describe the power capturing characteristics of the antenna when a wave impinges on it. These areas or apertures are distinguished as

(i) Physical aperture (ii) Effective aperture (iii) Scattering aperture (iv) Loss aperture (v) Collecting aperture

1. Physical Aperture:—





$$A_p = ab$$

Effective Aperture:—

Let the Radiation Power density of the wave is $W_i \left(\frac{\text{Watts}}{m^2}\right)$.

$$P = W A_p$$

When we measure this power it is less than the calculated one the field response of the horn is not uniform over the opening. The actual power received is

$$P_R = W_i A_e$$

A_e = effective aperture

Thus the ratio of Power available at the antenna terminal to the power per unit area of the appropriately Polarized incident wave.

$$A_e = \frac{P_R}{W_i} : \frac{\text{Available Power at antenna terminal}}{\text{Incident wave Power density}}$$

Scattering Aperture:—

$$A_s : \frac{\text{Power Scattered}}{\text{Incident Power Density}}$$

Loss Aperture:—

$$A_L = \frac{\text{Power Loss}}{\text{Incident Power density}}$$

Relation b/w Max Directivity and Max. effective area



Case-I: Ant 1 is transmitting and 2 is receiving

Ant 1 : I_T, D_T , Ant 2 : I_R, D_R

Radiated :

$$W = \frac{P_T}{4\pi R^2} \frac{W}{m^2}$$

$$W_c = \frac{P_T D_T}{4\pi R^2}$$

The Power collected at Ant 2 is $P_r = W_c A_r$

$$\Rightarrow P_r = \frac{P_T D_T A_r}{4\pi R^2}$$

$$\Rightarrow \frac{P_r}{P_T}(4\pi R^2) = D_T A_r \quad -(i)$$

If now 2 is used as a transmitting & 1 is receiving

then $D_r A_T = \frac{P_r}{P_T}(4\pi R^2)$ — (2)

$$\Rightarrow D_T A_r = D_r A_T$$

$$\Rightarrow \boxed{\frac{D_T}{A_T} = \frac{D_r}{A_r}}$$

Increasing directivity increasing area

$$\Rightarrow \frac{D_{TO}}{A_{TM}} = \frac{D_{RO}}{A_{RM}} \quad \left[\text{Receiving antenna is short dipole}\right]$$

$$\Rightarrow A_{TM} = \frac{D_{TO} A_{RM}}{D_{RO}} = D_{TO}\left[\frac{3}{8\pi}\lambda^2\right] = 1.5\frac{\lambda^2}{4\pi} D_{TO} \quad (VI)$$

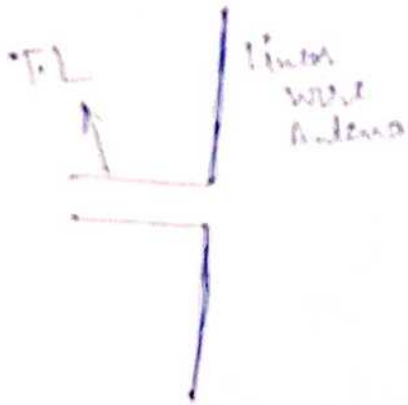
If $W_1 = W_2$ ... then the below ...

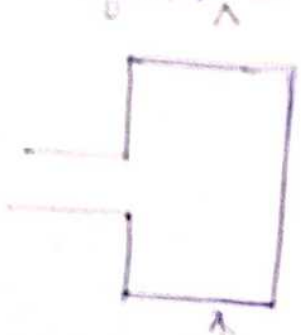
$$D = \frac{4\pi A_e}{\lambda^2}$$

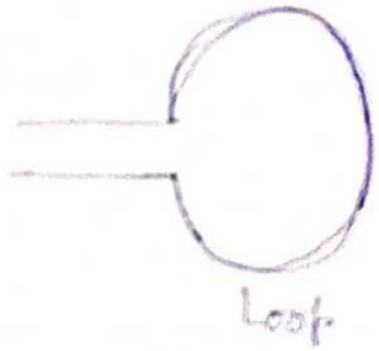
Types of Antennas:— (1) Wire Antennas

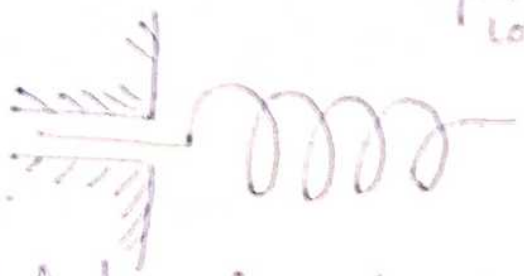
1) Linear Wire Antenna:— Antenna which are in the form of straight linear wire are called linear wire antenas.

2) Loop Antenas:— When the wire is used in the form of loop it is called ~~linear wire~~ loop ant



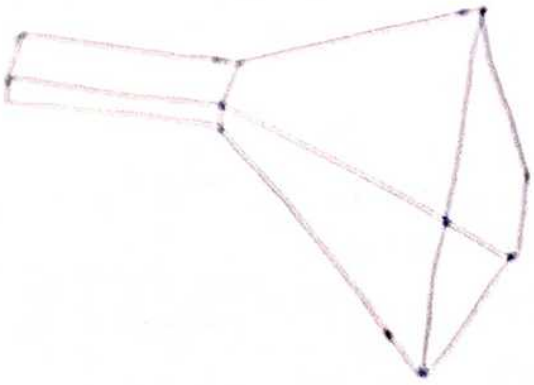


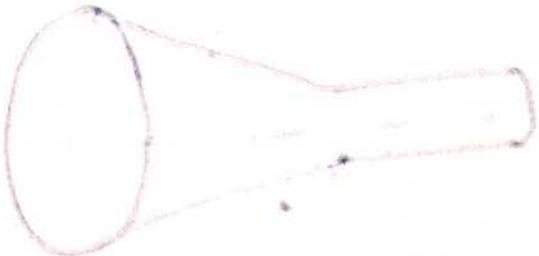




3 Helical Antenas:— The two wires is bend in the form of helix.

(2) Aperture Antenas:— Rectangular and circular horn are examples of Reflector antenas.





m b/w effective length and effective aperture:

$$A_{em} = \frac{|V_T|^2}{8W_i}\left[\frac{1}{R_r + R_L}\right]$$

V_T = Peak value of the terminal voltage received.

$$A_{em} = \frac{(\sqrt{2}V_{rms})^2}{8W_i}\left[\frac{1}{R_r+R_L}\right] = \frac{V_{rms}^2}{4W_i}\left[\frac{1}{R_r+R_L}\right]$$

$$\Rightarrow V^2 = V_{rms}^2 = 4W_i A_{em}[R_r + R_L]$$

Incident Power density $W_i = \frac{E^2}{Z}$ $[Z = 120\pi]$

$$\Rightarrow V^2 = 4\frac{E^2}{Z} A_{em}[R_r + R_L]$$

when $R_L = 0$

$$\Rightarrow \frac{V}{E} = \frac{2}{\sqrt{Z}}\sqrt{A_{em} R_r}$$

$$\Rightarrow I_c = \frac{V}{E} = \frac{2}{\sqrt{Z}}\sqrt{A_{em} R_r} \Rightarrow \boxed{A_{em} = \frac{I_e^2 Z}{4R_r}}$$

N Calculate the ~~effective~~ radiation resistance of an antenna which is drawing 15A current when radiating 5KW

$$\Rightarrow I_{rms} = 15A, P_{rad} = 5KW, \quad P_{rad} = I_{rms}^2 \times R_r$$

$$\Rightarrow R_r = \frac{P_{rad}}{I_{rms}^2} = \frac{5\times10^3}{15^2} = 22.22\Omega$$

2. Calculate the maximum effective aperture half wave dipole having radiation resistance (R_r)

$$A_{em} = \frac{l_e^2 Z}{4 R_r} = \frac{\left(\frac{\lambda}{\pi}\right)^2 \times 120\pi}{4 \times 73} = \underline{0.13\lambda^2 \ (m^2)}$$

Transmission Eq^n :—

$W = \frac{P_T}{4\pi r^2}$ $\qquad D = \frac{4\pi A_e}{\lambda^2}$

$W_{(directive\ antenna)} = \frac{P_T G_T}{4\pi r^2}$

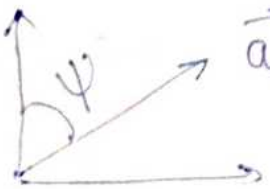
Power Received $P_r = \frac{P_T G_T A_e}{4\pi r^2}$

$= \frac{P_T G_T}{4\pi r^2} \frac{G_r \lambda^2}{4\pi}$

$\vec{E_i} = E_i \vec{a_w}$

$$\boxed{P_r = \frac{P_T G_T G_r \lambda^2}{(4\pi r)^2}}$$

E_i



$\vec{a_a}$ [Unit vector of ~~ant~~ electric field of antenna]

$PLF = |\vec{a_w} \cdot \vec{a_a}|^2 = |\cos\psi_p|^2$

$$P_r = \frac{P_T G_T G_r \lambda^2}{(4\pi r^2)} PLF$$

N The transmitting and receiving antenna are operating at 3GHz with gains of 25 and 20dB respectively are seperated by a distance of 500m. find the Power delined to load when I/P Power is 100W

Assume PLF = 1

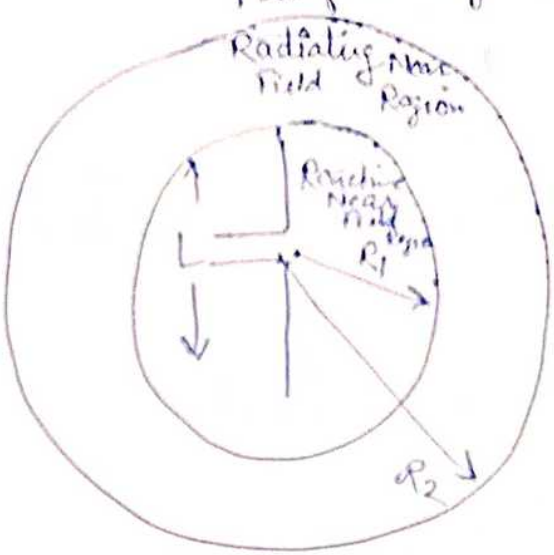
$G_T = 25dB = 10 \log G_T \Rightarrow G_T = 10^{2.5} = 316.24$

$G_r = 100$

PLF = 1 , $P_r = \frac{100 \times 316.26 \times 100}{} \left(\frac{0.1}{4\pi \times 500}\right)^2 = 0.01 \times 10^{-4} W$

(19) (20)

Antenna Radiation Regions :-



(I) Reactive Near field Region

The outer boundary of this region $R < 0.62\sqrt{\frac{L^3}{\lambda}}$

But for a very short dipole $R < \frac{\lambda}{2\pi}$

$$R < 0.62\sqrt{\frac{D^3}{\lambda}}$$

(II) Radiating Near field Region (Fresnel) field :-

$$0.62\sqrt{\frac{L^3}{\lambda}} \leq R < \frac{2L^2}{\lambda}$$ For wire antenna

$$0.62\sqrt{\frac{D^3}{\lambda}} \leq R < \frac{2D^2}{\lambda}$$ For aperture antenna

In this region antenna pattern is taking shape but not finely formed.

(III) Far-field (Fraunhofer) Region

$$R > \frac{2L^2}{\lambda}, \text{ or } R > \frac{2D^2}{\lambda}$$

The radiation pattern is completely formed and does not vary with distance.

t Impedance (Antena Impedance)

ery antena is connected to a trasmetter via a
ismission line. The applied radio frequency
stablishes a current distribution on the antena
structure. This causes power radiated into free space
and a small part of I/P Power is dissipated due
to conductor-dielectric loss (Ohmic loss) in the
antena. Also the applied voltage establishes a reactive
field in the near field of an antena.

$$Z_A = R_A + jX_A$$

Z_A = Input impedance of an antena = Z_{in}

R_A = Input resistance of an antena = R_{in}

X_A = Input reactance of an antena = X_{in}

$$P_{in} = \frac{1}{2} R_{in} |I_m|^2$$

This power is dissipated into two parts.

(1) Power radiated (P_r)

(11) Power dissipating (losses) P_L

$$P_{in} = P_r + P_L$$

$$\Rightarrow P_{in} = \frac{1}{2} R_r |I_m|^2 + \frac{1}{2} R_L |I_m|^2$$

$$= \frac{1}{2} [R_r + R_L] I_m^2$$

$$\Rightarrow \frac{1}{2} R_{in} I_n^2 = \frac{1}{2} [R_r + R_L] I_m^2$$

$$\Rightarrow \boxed{R_{in} = R_r + R_L}$$

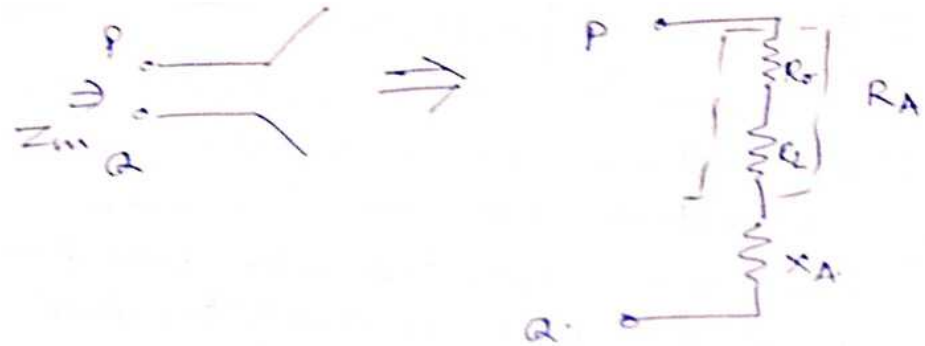
$$R_r = \frac{2P_r}{I_n^2}, \quad R_L = \frac{2P_L}{I_n^2} = \frac{2(P_{in} - P_r)}{I_n^2}$$

(22)

- is generalized

, $E_2 e^{j\psi_2}$ - - -

are the magnitudes and $\psi_1, \psi_2, - -$ are the phases (11)

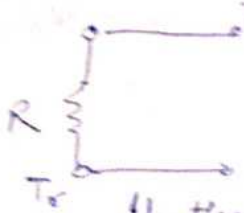


(Antenna and its equivalent)

Antenna Temperature :— The noise power per unit bandwidth available at the terminals of a resistor (R) at a temperature T_r is given by

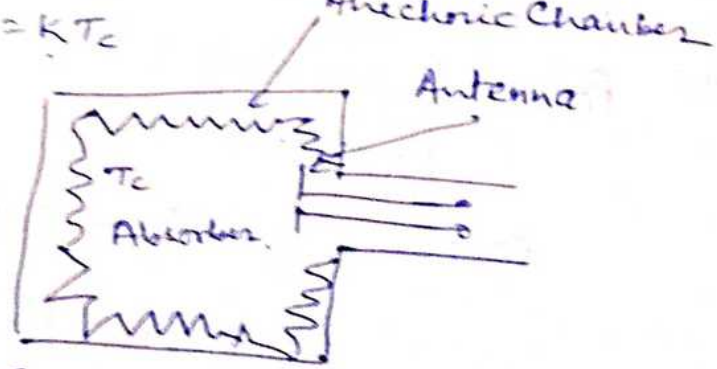
$$P = K T_r$$

P = Power per unit b/width W/Hz
K = Boltzman's Constant J/K
T_r = Absolute temp

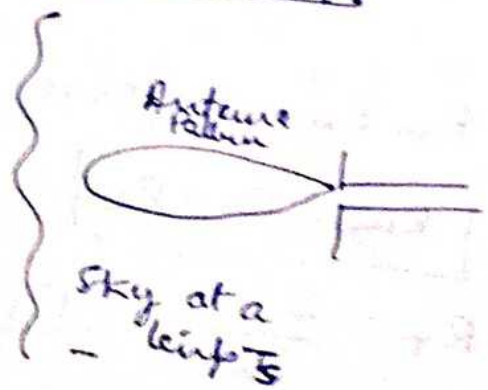


If the resistor is replaced by a lossless antenna of radiation resistance R in an anechoic chamber at a temperature T_c. Then Noise Power

$$P = K T_c$$

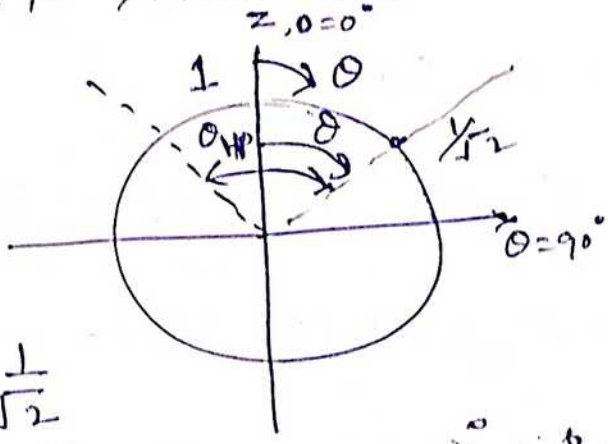


The noise temp T_A of the antenna is equal to the sky temp T_s and not to the physical temperature. So T_A is antenna temp or temperature of antenna radiation pattern receives, determined by sky temperature at which antenna beam is directed.



An antenna is having a field pattern given by

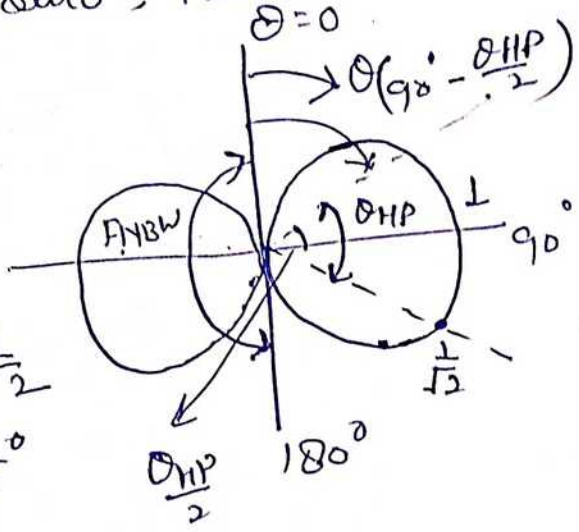
$E(\theta) = \cos\theta \Rightarrow 0^\circ \le \theta \le 90^\circ$, Find HPBW



$\Rightarrow E\left(\frac{\theta_{HP}}{2}\right) = \cos\left(\frac{\theta_{HP}}{2}\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow \frac{\theta_{HP}}{2} = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ \Rightarrow \theta_{HP} = 2 \times 45^\circ = 90^\circ = HPBW$

2 For an antenna $E(\theta) = \sin\theta$, Find HPBW, FNBW



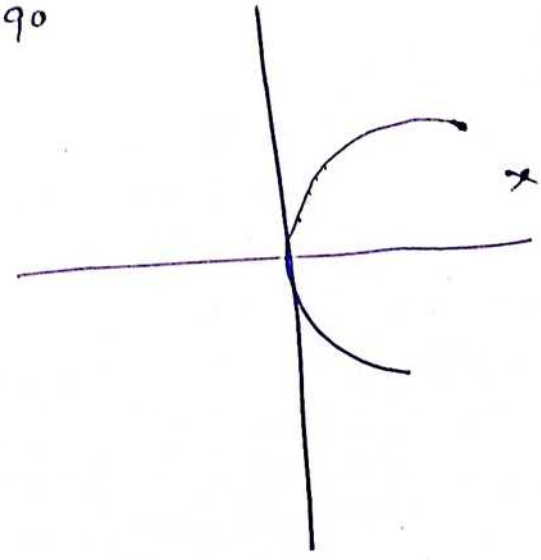
$E\left(90^\circ - \frac{\theta_{HP}}{2}\right) = \sin\left(90^\circ - \frac{\theta_{HP}}{2}\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow 90^\circ - \frac{\theta_{HP}}{2} = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$

$\Rightarrow \frac{\theta_{HP}}{2} = 45^\circ \Rightarrow \theta_{HPBW} = 90^\circ$

$\Rightarrow FNBW = 180^\circ$

3 $E(\theta) = \cos^2\theta$, for $0^\circ < \theta < 90^\circ$



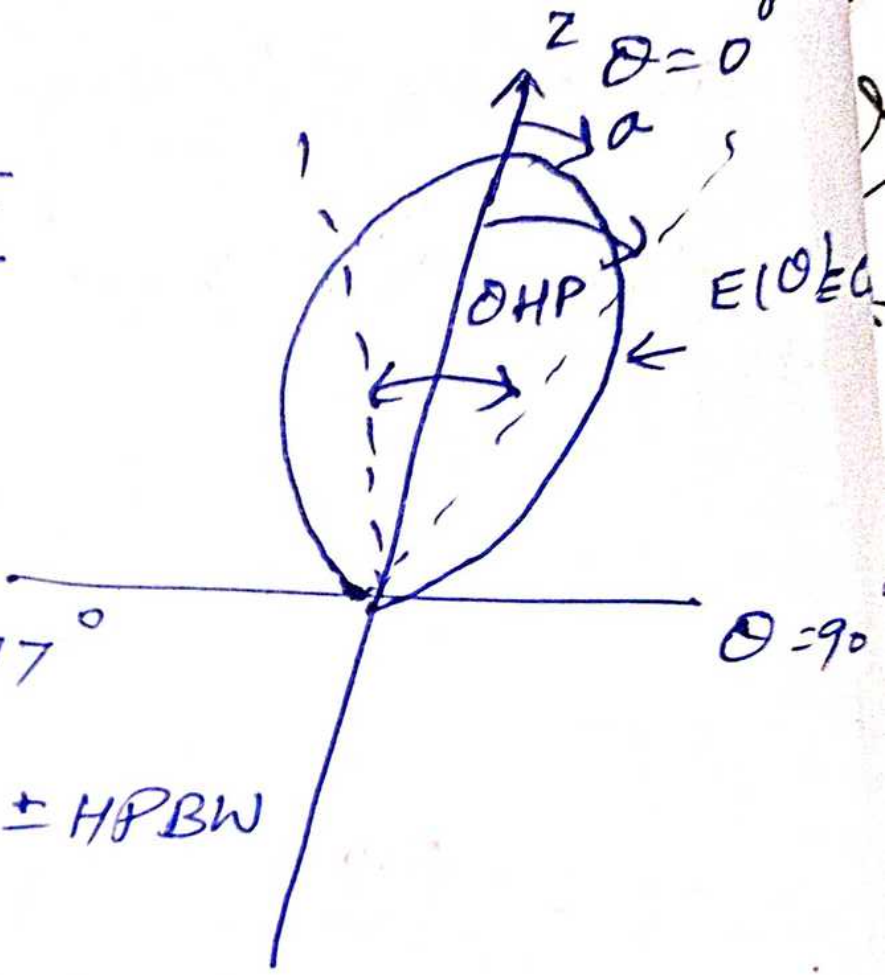
ma is ope

$$E\left(\frac{\theta_{HP}}{2}\right) = \cos^2\left(\frac{\theta_{HP}}{2}\right) = \frac{1}{\sqrt{2}}$$

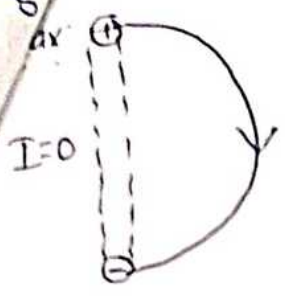
$$\Rightarrow \cos\frac{\theta_{HP}}{2} = \sqrt{0.707}$$

$$\Rightarrow \frac{\theta_{HP}}{2} = \cos^{-1}\left(\sqrt{0.707}\right) = 32.77^\circ$$

$$\Rightarrow \theta_{HP} = 2(32.77) = 65.54^\circ, \pm HPBW$$



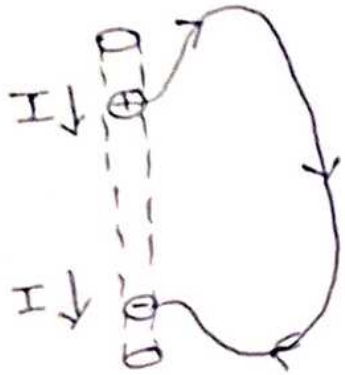
Fields from oscillating dipole :—



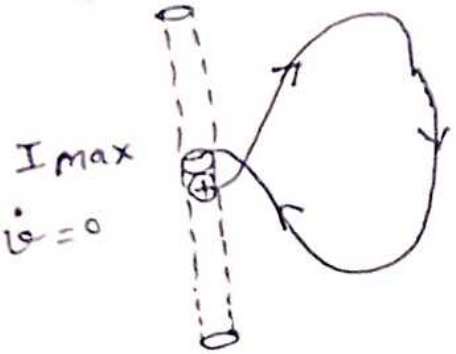
l_0

$t = 0$

Electric field lines or wave front with charges at ends of dipole

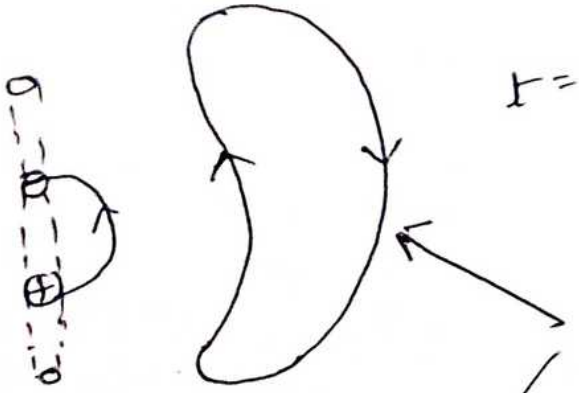


Field line $t = \frac{T}{8}$. Wave front moves out as charges go in



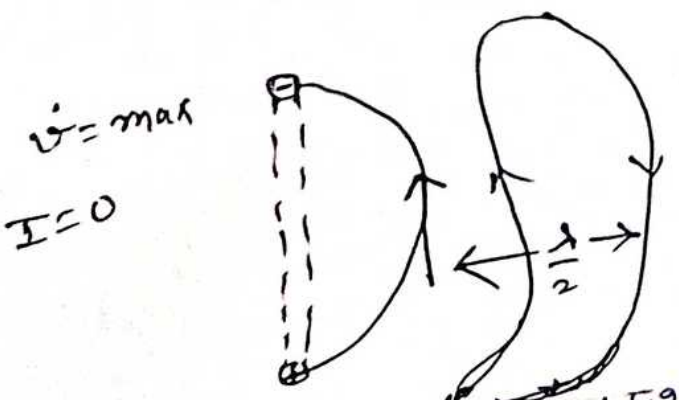
$t = \frac{T}{4}$

As the pass the mid point the field line cut loose



$t = \frac{3}{8} T$

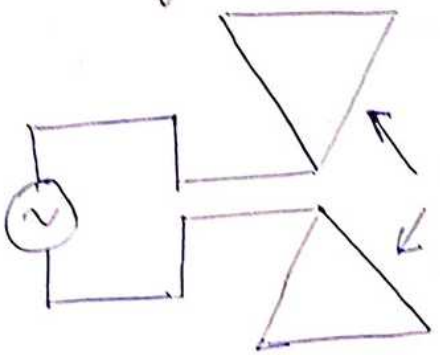
wavefronts moving outs



$t = \frac{T}{2}$

Periodic Antena :-

A log periodic antena is an antena having a structual geometry such that its impedance and radiation characteristics repeat periodically as the logarithm of frequency



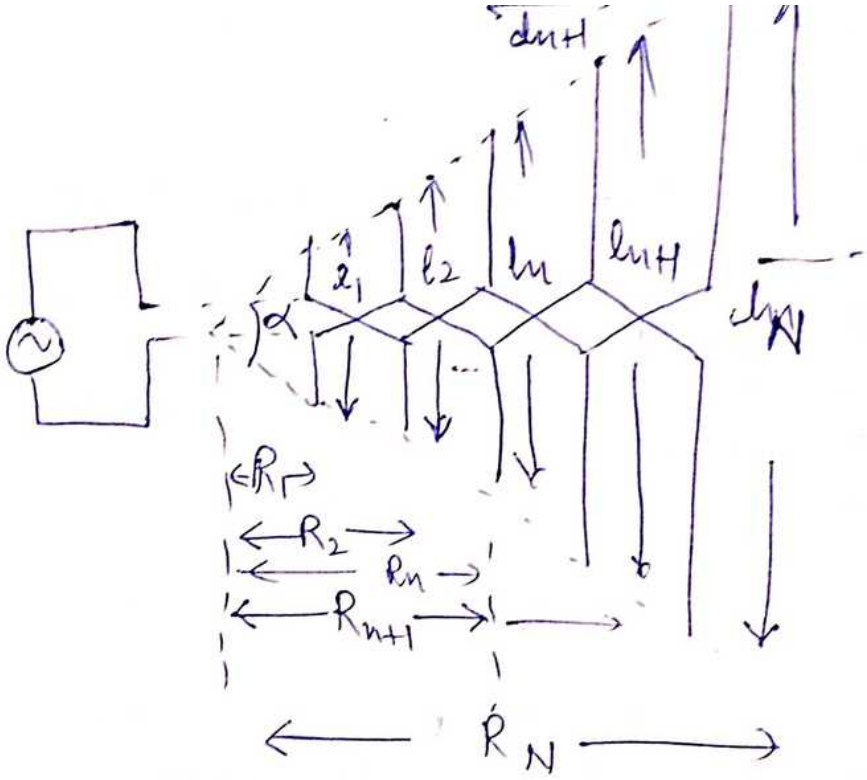
[Bow tie antena]

For understanding log periodic antena array we studied Bow-tie antena. In this antena current is decreases slowly as we go down from the input ends but this current is abruptly terminated at the fins. This reason is limited BW so it is not truely ~~fitted~~ frequency independent antena. But the structure of it can be modified to have frequency independence.

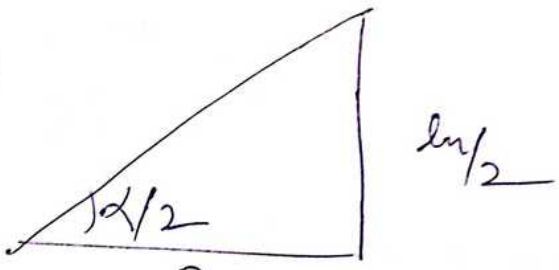
Log Periodic structure:- By modifying the structure of Bow-tie antena different geometries are possible.

(1) The log-Periodic toothed planer ~~array~~ antena
(2) The log-Periodic toothed wire antena
(3) The log Periodic toothed trapezoid antena
(4) The log-Periodic zig zag wire antena
The log-Periodic dipole array geometries

(15)



Scale factor = $\frac{R_n}{R_{n+1}}$



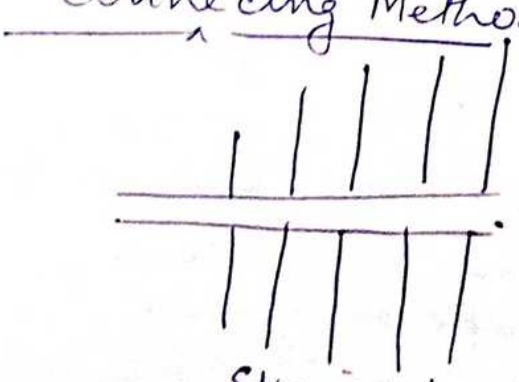
$$\tan \alpha/2 = \frac{l_n/2}{R_n} = \frac{l_{n+1}/2}{R_{n+1}}$$

This is true for every element

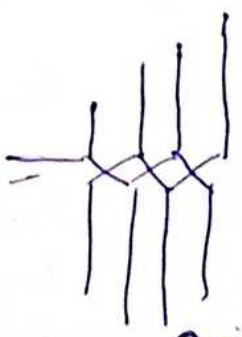
$$\frac{l_1}{R_1} = \frac{l_2}{R_2} = \frac{l_n}{R_n} = \frac{l_{n+1}}{R_{n+1}} = \frac{l_N}{R_N}$$

$$\text{So, } \tau = \frac{R_n}{R_{n+1}} = \frac{l_n}{l_{n+1}}$$

Connecting Methods :—



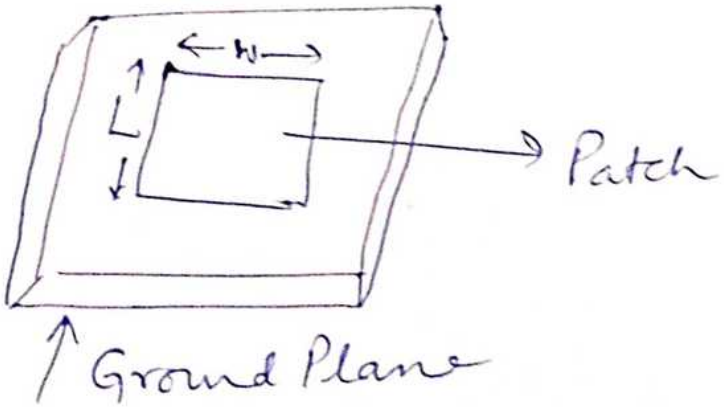
Straight here



Criss cross connection

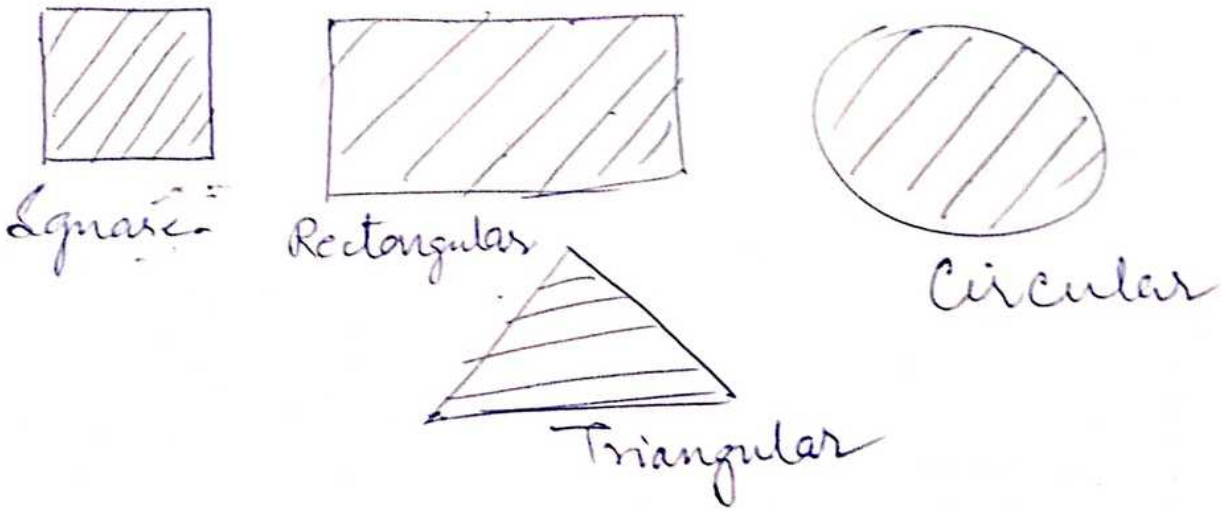
ideband, simplicity, high directivity
d circular Polarization are properties which
akes it useful for telemetry applications

Microstrip Patch antenna



$t << \lambda_0$. $2.2 \leq \varepsilon_r \leq 12$

Shapes of Patches



Unit-IV

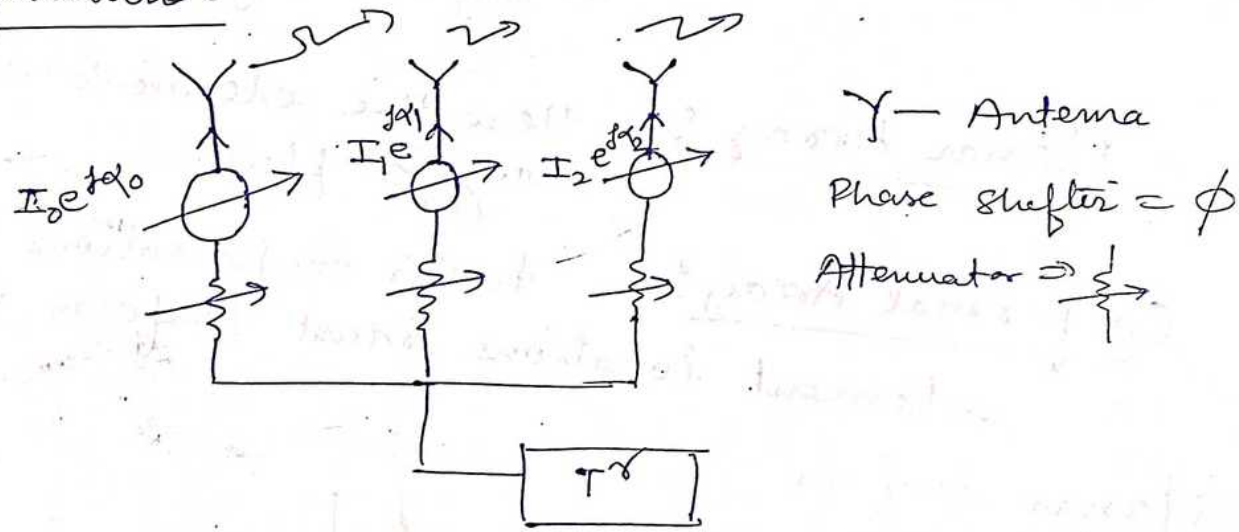
(5)

Antena Array :- Usually the radiation pattern of a single element is relatively wide and provides low value of directivity (gain). For many application like point-to-point communication the desired radiation pattern is a single narrow beam.

"Antena formed by multi elements arranged to produce particular radiation pattern is called an array"

So to Provide directive pattern, from the elements of an array interfere constructively in the desired direction and interfere destructively in the remaining direction

Configurations :-



The currents supplied to the antenas are expressed as

$I_0 e^{j\alpha_0}$, $I_1 e^{j\alpha_1}$, ---

I_0, I_1 --- are amplitudes of the currents

and α_0, α_1 --- are the phase angles

the amplitude of the current determine the amplitude of the field radiated and phase angle of the current determine the phase of the field

The fields radiated by the antena is given as

$E_0 e^{j\psi_0}$, $E_1 e^{j\psi_1}$, $E_2 e^{j\psi_2}$ - - -

E_0, E_1 are the magnitudes and ψ_1, ψ_2 --- are the

so the total field

$$E_{Total} = E_0 e^{j\psi_0} + E_1 e^{j\psi_1} + \cdots$$

if $I_0 e^{j\alpha_0} = I_1 e^{j\alpha_1} = \cdots$

then $E_0, E_1 \cdots$ are equal but ψ_0, ψ_1 are not equal. this is not only decided by α. but also by distance (d)

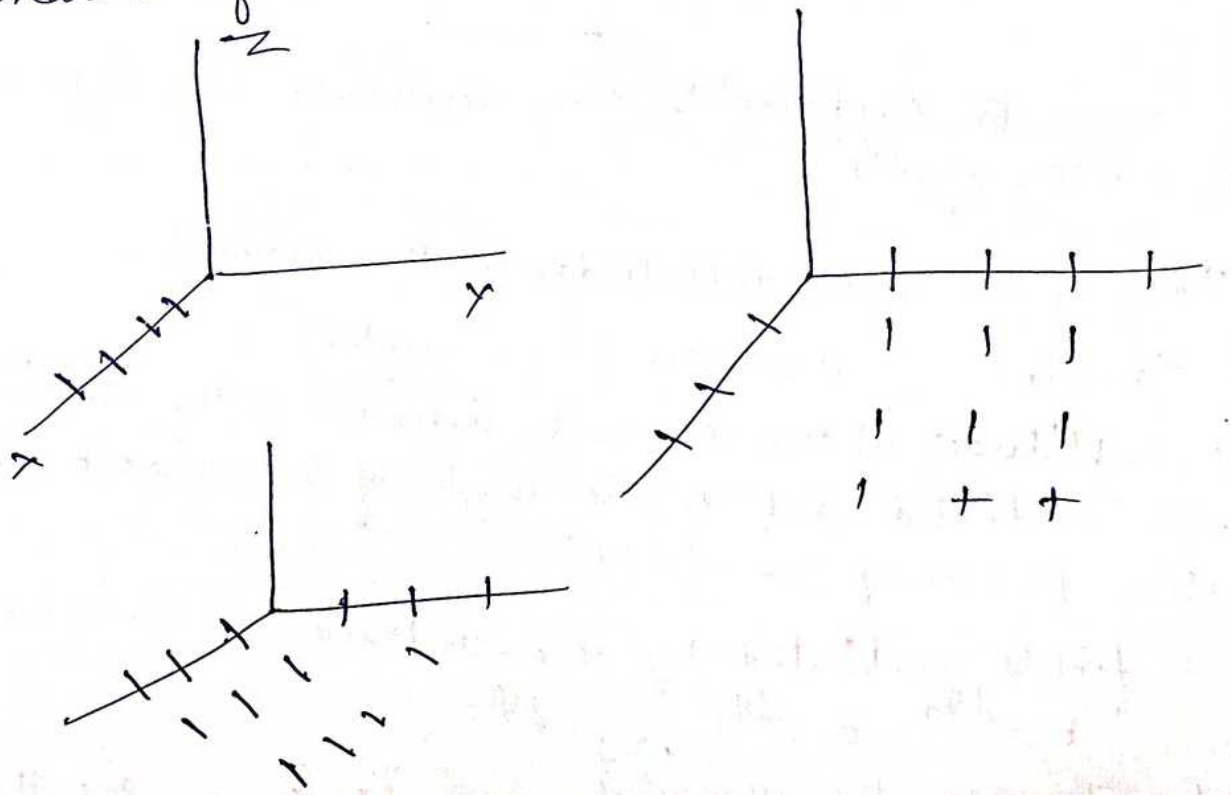
$$\boxed{\psi = \beta d + \alpha}$$

Array configurations :-

(I) Linear Array :- In this the array elements are arranged along a straight line

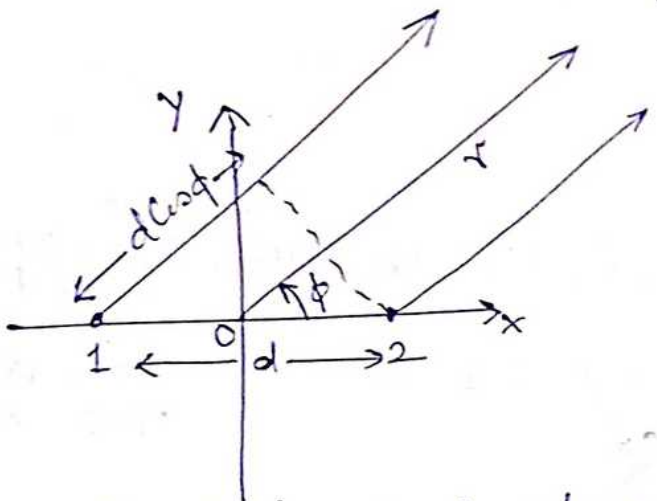
(II) Planar Array :- Here the elements are centre along a plane

(III) Conformal Array :- In this configurations the element locations must conform to some nonplanar surface



50 element Array :-

1) Two isotropic source of same amplitude and phase :-



a) Origin as a reference:

$(E_1 = E_2 = E_0)$

$$\psi = \beta d \cos\phi + 0$$

fields due to antenna 2 leads reference by $\frac{\psi}{2}$ and antenna 1 fields lags reference by $\frac{\psi}{2}$.

So field due to Source 1 $E_1 = E_0 e^{-j\psi/2}$

" " " " 2 $E_2 = E_0 e^{+j\psi/2}$

Total field $E = E_1 + E_2$

$$\Rightarrow E = E_0 e^{-j\psi/2} + E_0 e^{j\psi/2}$$

$$E = 2E_0 \left[\frac{e^{-j\psi/2} + e^{j\psi/2}}{2} \right]$$

$$\Rightarrow \boxed{E = 2E_0 \cos\frac{\psi}{2}}$$

$$\Rightarrow \boxed{E_n = \cos\psi/2 = \text{Normalized Field}}$$

$$\Rightarrow E_n = \cos\left[\frac{\beta d \cos\phi}{2}\right] = \cos\left[\frac{2\pi}{\lambda} \cdot \frac{d\cos\phi}{2}\right]$$

$$\Rightarrow \boxed{E_n = \cos\left(\frac{\pi d}{\lambda}\cos\phi\right)}$$

Case I if $d = \frac{\lambda}{2}$, $\alpha = 0$

$$\Rightarrow E_n = \cos\left[\frac{\pi}{\lambda}\cdot\frac{\lambda}{2}\cos\phi\right]$$

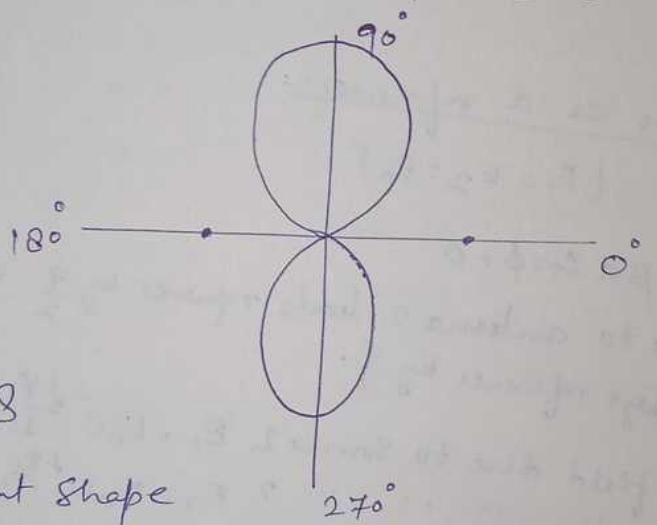
$$\boxed{E_n = \cos\left(\frac{\pi}{2}\cos\phi\right)}$$

$\phi = 0^\circ, 30^\circ, 60^\circ, 90^\circ, 120^\circ, 150^\circ, 180^\circ, 210^\circ, 240^\circ, 270^\circ$

0 0.2 .707 1 .707 0.2 0 0.2 .707 1

$300^\circ, 330^\circ$

.707 0.2



→ Figure 8

→ Doughnut Shape

→ Broad side

Broad side antenna array :— If the maximum is perpendicular to the axis of antenna array then it is called broadside array.

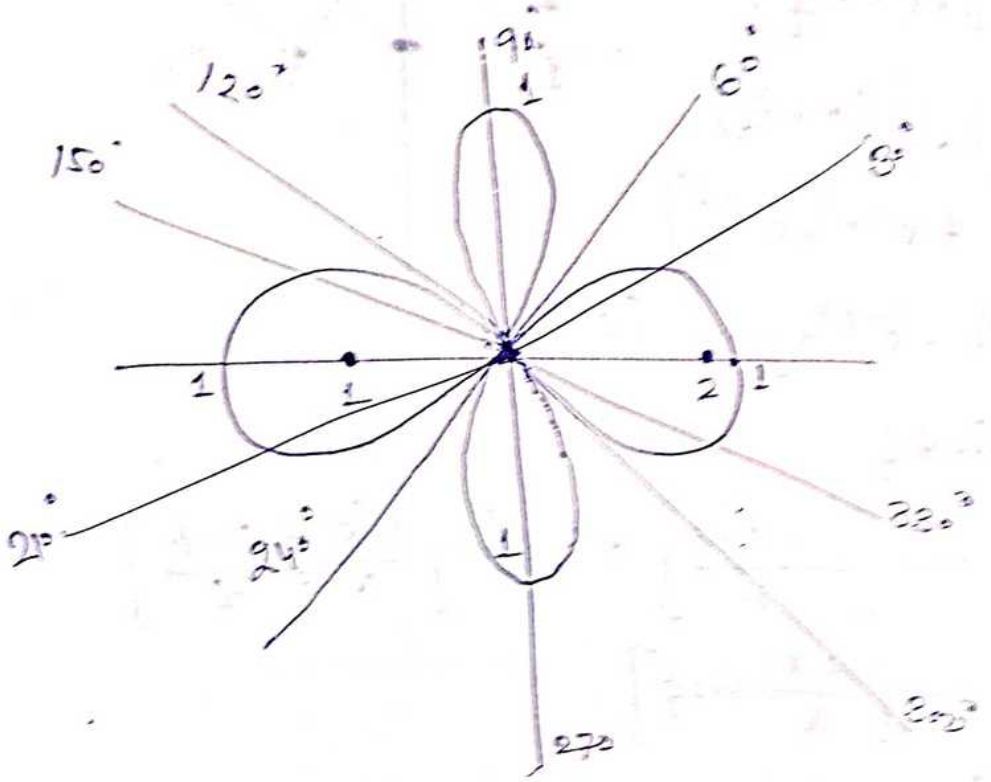
End Fire antenna Array :— If the maxima of the radiation is along the direction of array axis then it is called the end fire array.

Case 2 , $d = \lambda$, $\alpha = 0^\circ$

$$E_n = \cos\left(\frac{\pi}{\lambda}\lambda\cos\phi\right) = \cos(\pi\cos\phi)$$

ϕ = 0, 30°, 60°, 90°, 120°, 150°, 180°, 210°, 240°, 270°, 300°

E_n = 1, 0.91, 0, 1, 0, 0.91, 1, 0.91, 0, 1, 0



b) One antenna as a reference (1 One)

$$\psi = \beta d \cos\phi$$

$$\vec{E}_1 = E_0 e^{j0}$$

$$E_2 = E_0 e^{j\psi}$$

Total field - $E = E_0 e^{j0} + E_0 e^{j\psi}$

$$E = E_0 \left(1 + e^{j\psi}\right)$$

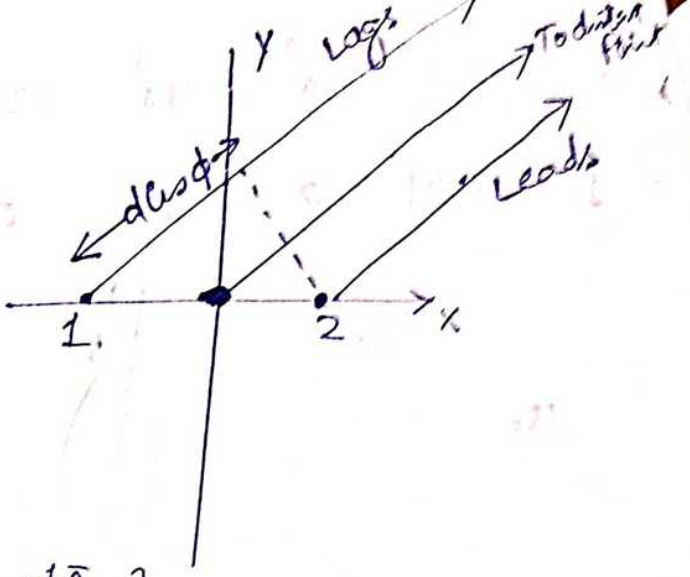
$$\Rightarrow E = E_0 e^{j\frac{\psi}{2}} \left[e^{-j\frac{\psi}{2}} + e^{j\psi/2}\right]$$

$$\Rightarrow E = 2E_0 \cos\frac{\psi}{2} \angle\psi/2$$

$$\Rightarrow E_n = \cos\left(\frac{\psi}{2}\right) = \cos\left(\frac{\beta d \cos\phi}{2}\right)$$

Changing reference point will only effect on ... radiation Pattern

Case II Two isotropic source of same amplitude and different Phase



$$E_1 = E_0 e^{j\psi/2}$$

$$E_2 = E_0 e^{-j(\pi + \frac{\psi}{2})}$$

$$\Rightarrow E = E_0 e^{j\psi/2} + E_0 e^{-j(\pi + \psi/2)}$$

$$= E_0 \left[e^{j\frac{\psi}{2}} + e^{-j\pi} \cdot e^{-j\frac{\psi}{2}} \right]$$

$$= E_0 \left[e^{j\psi/2} - e^{-j\psi/2} \right] \quad \left[e^{-j\pi} = -1 \right]$$

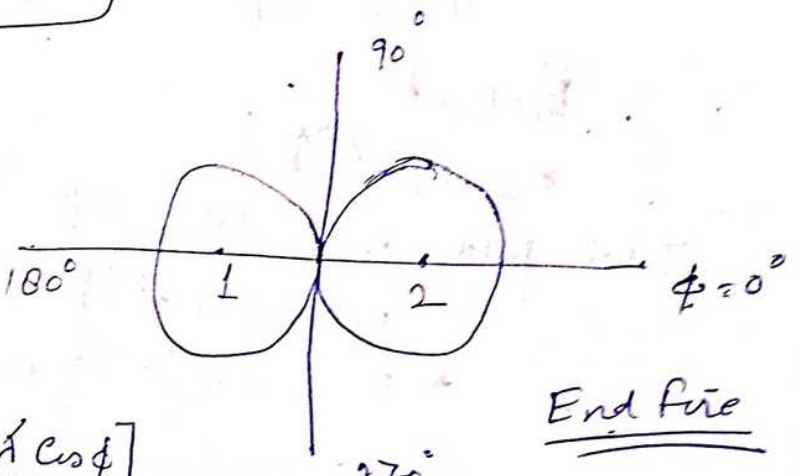
$$\Rightarrow \boxed{E = 2jE_0 \sin \psi/2}$$

$$E_n = \sin \frac{\psi}{2} = \sin \left[\frac{\beta d \cos\phi}{2} \right] = \sin \left[\frac{2\pi}{\lambda} \frac{d \cos\phi}{2} \right]$$

$$\Rightarrow E_n = \sin \left[\frac{d\pi}{\lambda} \cos\phi \right]$$

if $d = \frac{\lambda}{2}$

$$\Rightarrow \boxed{E_n = \sin \left(\pi/2 \cos\phi \right)}$$

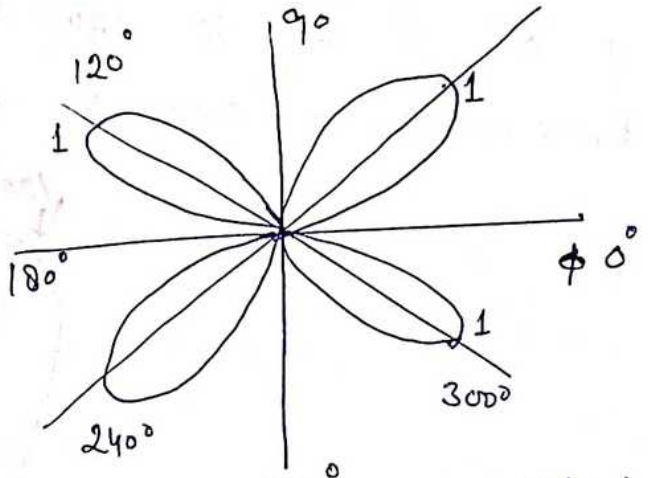


End fire

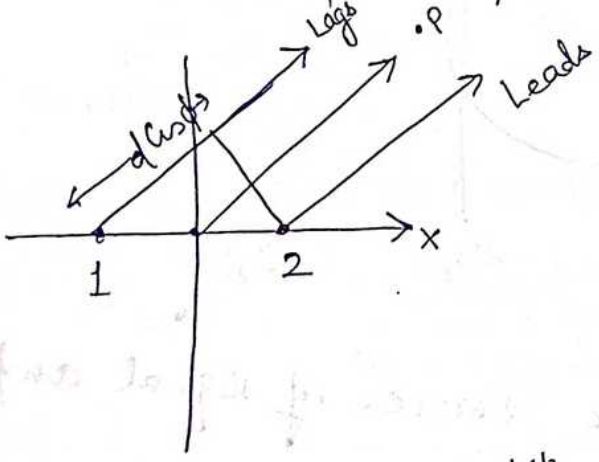
if $d = \lambda$ $\quad E_n = \sin \left[\frac{2\pi}{\lambda} \cdot \frac{\lambda \cos\phi}{2} \right]$

$$\Rightarrow E_n = \sin \left[\pi \cos\phi \right]$$

ϕ	0	30°	60°	90°	120°	150°	180°	210°	240	270	300	330
E_n	0	0.41	1	0	1	0.41	0	0.41	1	0	1	0.41



Case III :- Two isotropic source of same amplitude and in phase quadrature:



$$E = E_1 + E_2 = E_0 e^{j\frac{\psi}{2}} + E_0 e^{-j\frac{\psi}{2}} = E_0 e^{j\left[\frac{\beta d \cos\phi + \alpha}{2}\right]} + E_0 e^{-j\left[\frac{\beta d \cos\phi + \alpha}{2}\right]}$$

$$\Rightarrow E = E_0 e^{j\left[\frac{\beta d \cos\phi}{2} + \frac{\alpha}{2}\right]} + E_0 e^{-j\left[\frac{\beta d \cos\phi}{2} + \frac{\alpha}{2}\right]}$$

$\alpha = \pi/2$

$$\Rightarrow E = E_0 e^{j\left[\frac{\beta d \cos\phi}{2} + \frac{\pi}{4}\right]} + E_0 e^{-j\left[\frac{\beta d \cos\phi}{2} + \frac{\pi}{4}\right]}$$

if $d = \frac{\lambda}{2}$

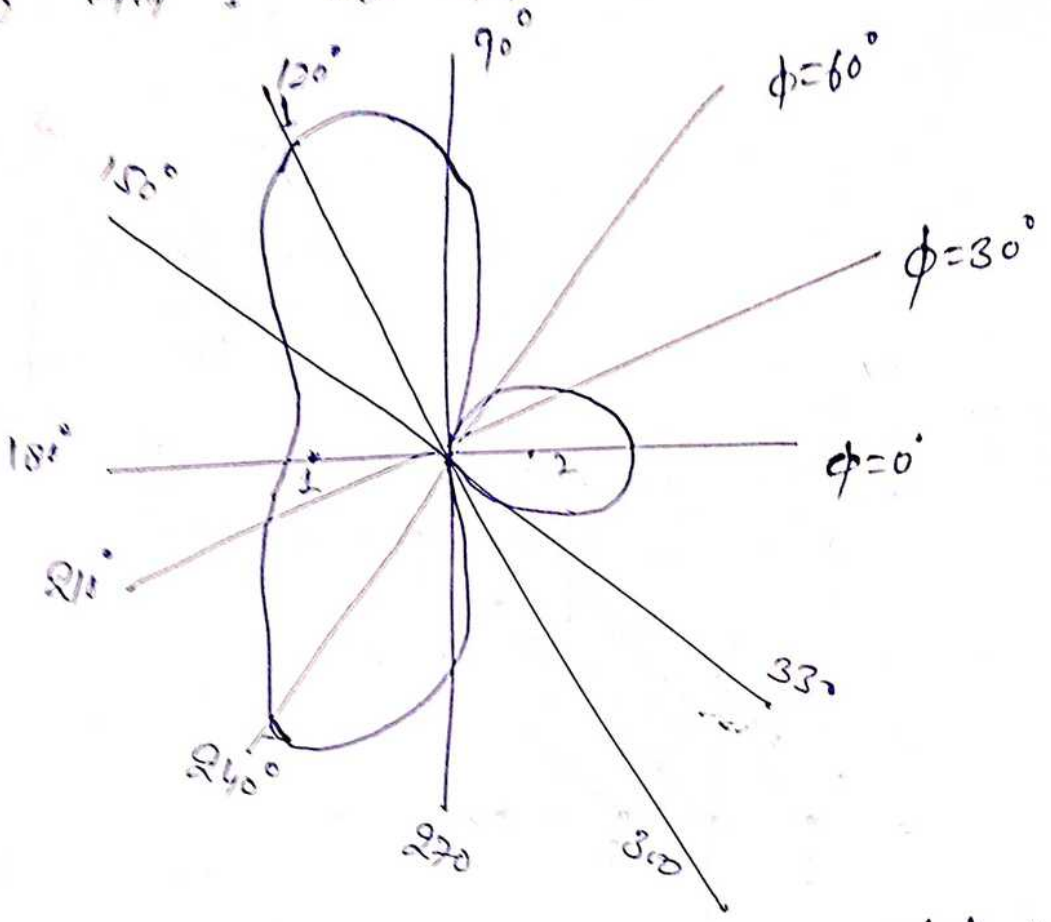
$$\Rightarrow E = E_0 e^{j\left[\frac{2\pi}{\lambda}\cdot\frac{\lambda}{2}\cos\phi + \pi/4\right]} + E_0 e^{-j\left[\frac{2\pi}{\lambda}\cdot\frac{\lambda}{2}\cos\phi + \frac{\pi}{4}\right]}$$

$$= E_0 e^{j\left[\pi/2\cos\phi + \pi/4\right]} + E_0 e^{-j\left[\frac{\pi}{2}\cos\phi + \pi/4\right]}$$

$$\Rightarrow \boxed{E_n = \cos\left[\pi/4 + \pi/2\cos\phi\right]}$$

ϕ = 0° 30° 60° 90 120° 150 180° 210 240 270 300 330

E_n = 0.707 0.84 0 0.707 1 0.83 0.707 0.83 1 0.707 1 0.83



Case IV: Two isotropic source of equal amplitude and any phase difference.

$$\psi = \beta d \cos\phi + \alpha$$

$$E = E_0 e^{j\psi/2} + E_0 e^{-j\psi/2} = 2E_0 \left[\frac{e^{j\psi/2} + e^{-j\psi/2}}{2}\right]$$

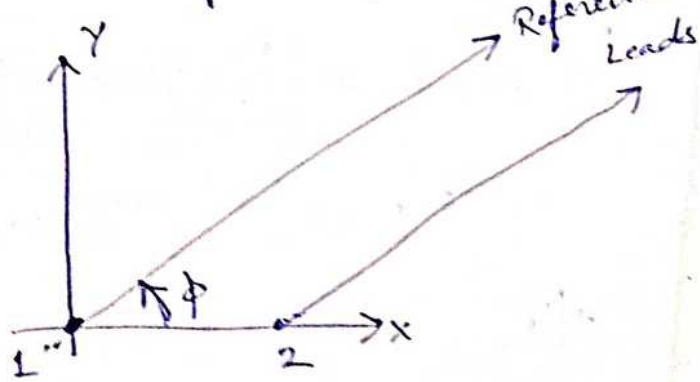
$$= 2E_0 \cos\left[\frac{\psi}{2}\right] = 2E_0 \cos\left[\frac{\beta d \cos\phi + \alpha}{2}\right]$$

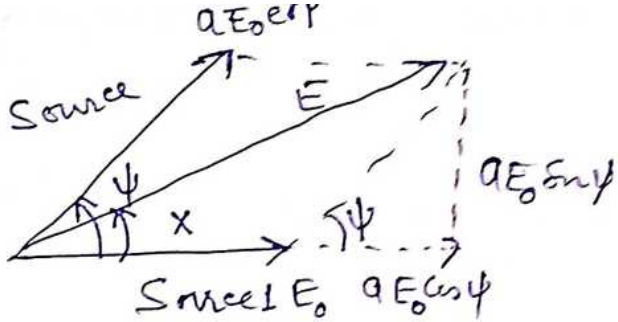
$$\text{So } E_n = \cos\left[\frac{\beta d \cos\phi + \alpha}{2}\right]$$

Case V: Two isotropic point source of unequal amplitude and any phase diff

Point source 1 E_0

2 – aE_0 [0 < a < 1]





$$|E|^2 = (E_0 + aE_0\cos\psi)^2 + (aE_0\sin\psi)^2$$

$$|E|^2 = E_0^2\left[(1+a\cos\psi)^2 + (a\sin\psi)^2\right]$$

$$\Rightarrow |E|^2 = E_0\sqrt{(1+a\cos\psi)^2 + (a^2\sin^2\psi)}$$

The angle of resultant with respect to reference (Source 1) is

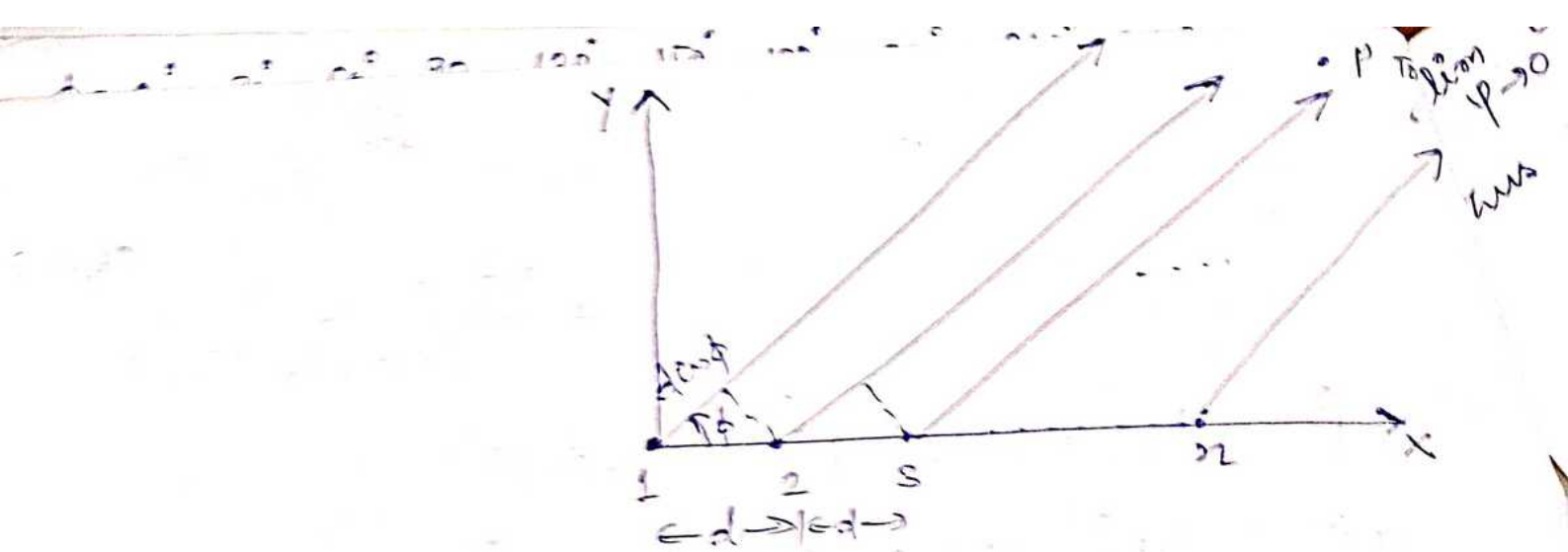
$$\tan x = \frac{aE_0\sin\psi}{E_0 + aE_0\cos\psi} = \frac{a\sin\psi}{1+a\cos\psi}$$

$$\vec{E} = E_0\sqrt{(1+a\cos\psi)^2 + a^2\sin^2\psi}\ \angle \tan^{-1}\left[\frac{a\sin\psi}{1+a\cos\psi}\right]$$

n-element array of equal amplitude and spacing :-

(I) Linear Array :- An array is said to be linear if the individual elements of the array are spaced equally along a line.

(II) Uniform Array :- A linear array is said to be uniform if the elements of the array are fed with currents of equal amplitude and having a progressive phase shift along the line



$$E = E_0 e^{i\psi} + E_0 e^{i2\psi} + E_0 e^{i3\psi} + \cdots + E_0 e^{i(n-1)\psi} \quad \text{---(1)}$$

$$\psi = \frac{2\pi}{\lambda} d \cos\theta + \alpha$$

$$\Rightarrow E = E_0 \left[1 + e^{i\psi} + e^{i2\psi} + \cdots + e^{i(n-1)\psi} \right]$$

$$\Rightarrow E e^{i\psi} = E_0 \left[e^{i\psi} + e^{i2\psi} + \cdots + e^{in\psi} \right] \quad \text{---(II)}$$

$$\Rightarrow E\left[1 - e^{i\psi}\right] = E_0 \left[1 - e^{in\psi}\right]$$

$$\Rightarrow E = E_0 \left[\frac{1 - e^{in\psi}}{1 - e^{i\psi}} \right] = E_0 \frac{e^{in\psi/2} \left[e^{-in\psi/2} - e^{in\psi/2} \right]}{e^{i\psi/2} \left[e^{-i\psi/2} - e^{i\psi/2} \right]}$$

$$\Rightarrow E = E_0 e^{i(n-1)\psi/2} \frac{\left[\sin n\psi/2 \right]}{\left[\sin \psi/2 \right]}$$

$$\Rightarrow E = E_0 \left[\frac{\sin n\psi/2}{\sin \psi/2} \right] \angle (n-1)\psi/2$$

If reference is shifted to the centre of arrangement

$$\Rightarrow E = E_0 \left[\frac{\sin n\psi/2}{\sin \psi/2} \right] \angle 0^\circ$$

When $\psi = 0$

$$\lim_{\psi \to 0} E = E_0 \lim_{\psi \to 0} \frac{\frac{d}{d\psi}\left(\sin n\psi/2\right)}{\frac{d}{d\psi}\left(\sin \psi/2\right)} = E_0 \lim_{\psi \to 0} \frac{\frac{n}{2}\cos(n\psi/2)}{\frac{1}{2}\cos(\psi/2)}$$

$$\lim_{\psi \to 0} E = nE_0$$

Thus maximum of n times the field from a single source

$$E_n = \frac{E}{E_{max}} = \frac{1}{n} \frac{\sin(n\psi/2)}{\sin(\psi/2)}$$

$$\Rightarrow \text{Array Factor}, AF = \frac{1}{n} \frac{\sin(n\psi/2)}{\sin(\psi/2)}$$

Pattern Multiplication Rule

Total field of non-isotropic array = (AF) isotropic × Field of single non-isotropic element

Null directions :-

$$\sin\left(\frac{n\psi}{2}\right) = 0 \text{ , but } \sin\left(\frac{\psi}{2}\right) \neq 0$$

$$\frac{n}{2}\left[\beta d \cos\phi + \alpha\right]_{\phi_{ON}} = \pm N\pi$$

$$\Rightarrow \cos\phi_{ON} = \frac{1}{\beta d}\left[\pm\frac{2N\pi}{n} - \alpha\right]$$

$$\Rightarrow \phi_{ON} = \cos^{-1}\left[\frac{1}{\beta d}\left(-\alpha \pm \frac{2N\pi}{n}\right)\right]$$

where $N = 1, 2, 3,$ but $N \neq n, 2n, 3n$

if $N = 0, \rightarrow \psi = 0, \rightarrow \sin\left(\frac{\psi}{2}\right) = 0$

if $N = n, 2n, 3n$

$$\frac{n}{2}\psi = \pm n\pi \Rightarrow \psi = \pm 2\pi$$

So $\sin\left(\frac{\psi}{2}\right) = \sin(\pm\pi) = 0$

Direction of Major lobe maxima

$$\sin\left(\frac{\psi}{2}\right) = 0$$

$$\Rightarrow \frac{\psi}{2} = \pm m\pi$$

$$\Rightarrow \frac{1}{2}\left[\beta d\cos\phi + \alpha\right]_{\phi=\phi_{om}} = \pm m\pi$$

$$\Rightarrow \phi_{om} = \cos^{-1}\left[\frac{1}{\beta d}(-\alpha \pm 2m\pi)\right], \quad m = 0, 1, 2 \ldots$$

for $m=0$ $\quad \phi_{om} = \cos^{-1}\left[\frac{\alpha}{\beta d}\right]$

Side lobe Maxima

Numerator attains Max value

$$\sin\left(\frac{n\psi}{2}\right) = \pm 1$$

$$\frac{n}{2}\psi\Big|_{\phi=\phi_s} = \pm(2s+1)\pi/2$$

$$\Rightarrow \frac{n}{2}\left[\beta d\cos\phi_s + \alpha\right] = \pm(2s+1)\pi/2$$

$$\Rightarrow \phi_s = \cos^{-1}\left[\frac{1}{\beta d}\left(-\alpha \pm \left(\frac{2s+1}{n}\right)\pi\right)\right]$$

$s = 1, 2, 3 \ldots$

Array of n isotropic Point source of equal amplitude and Spacing (Broadside case) :-

$$E_\psi = E_0 e^{-j\psi/2} + E_0 e^{j\psi/2}$$

$\Rightarrow$ For max radiation

$\psi = 0$

$$\Rightarrow \psi = \beta d\cos\phi + \alpha$$

$$\Rightarrow 0 = \beta d\cos(90^\circ, \text{or } 270^\circ) + \alpha$$

$$\Rightarrow \boxed{\alpha = 0}$$

For Broadside array all the elements in the array should have the same phase excitation ($\alpha = 0$) in addition to same amplitude excitation.

ay of n isotropic point source of equal amplitude and spacing (End fire case):—

An array is said to be endfire if the maximum radiation is along one end of the axis of the array. $\phi = 0^\circ$ or 180°, with $\psi = 0$

condition for $\phi = 0^\circ$

$$\psi = \beta d \cos\phi + \alpha$$

$$\Rightarrow 0 = \beta d \cos(0^\circ) + \alpha$$

$$\Rightarrow \boxed{\alpha = -\beta d}$$

Thus end along $\phi = 0$ radiates provided there is a successive phase lag b/w the elements by an amount of βd.

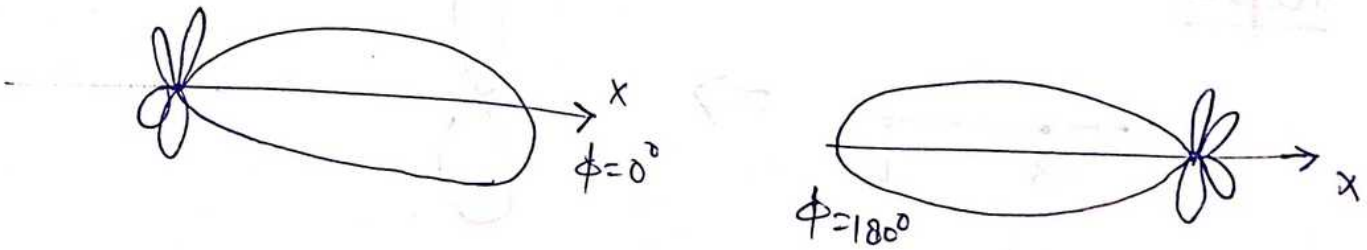
condition for $\phi = 180^\circ$

$$\Rightarrow \psi = \beta d \cos\phi + \alpha$$

$$\Rightarrow \phi = \beta d \cos(180^\circ) + \alpha$$

$$\Rightarrow \boxed{\alpha = \beta d}$$

Thus end along $\phi = 180^\circ$ radiates provided there is a successive phase lead bet



The Hansen Woodyard Endfire Array:—

$$\alpha = -\left(\beta d + \frac{2.94}{n}\right) \simeq \left(\beta d + \frac{\pi}{n}\right) \text{ for max in } \phi = 0^\circ$$

$$\alpha = \left(\beta d + \frac{2.94}{n}\right) = +\left(\beta d + \frac{\pi}{n}\right) \text{ for max } \phi = 180^\circ$$

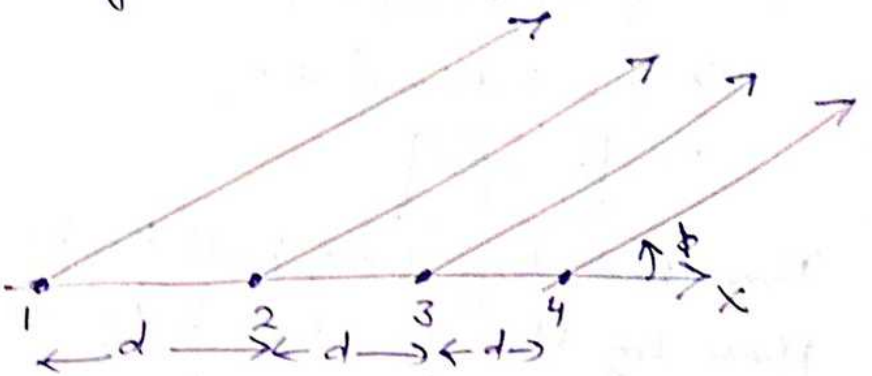
$\psi = -\beta d$, $\psi = \beta d$ Phase shift firing along $\phi = 0^\circ$ or $\phi = 180^\circ$

This produces a max field in the $\phi = 0^\circ$ or 180°. but does not give the maximum directivity.

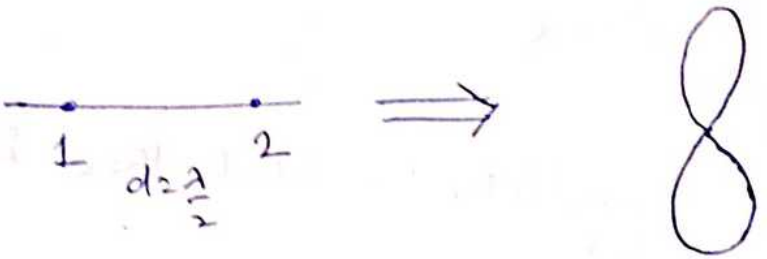
Pattern Multiplications :-

Pattern Multiplication using isotropic sources :-

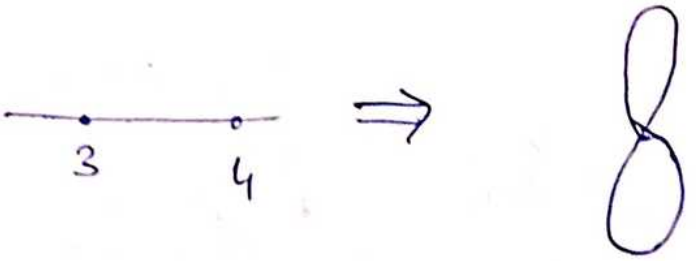
$d = \frac{\lambda}{2}$, $\psi = 0$



Step 1 :- Consider antenna 1 and 2 as one unit

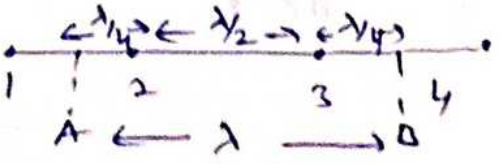


Step 2

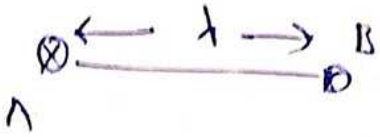


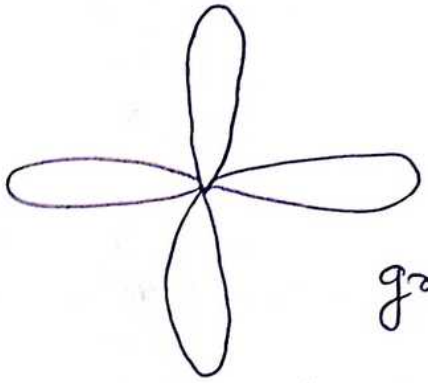
Step 3





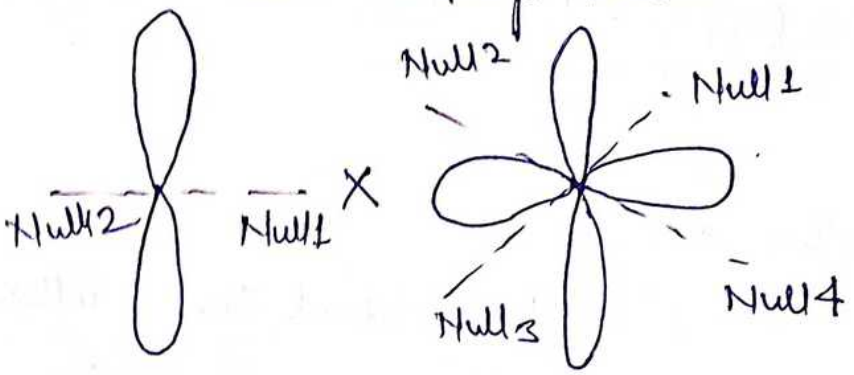
Step 3



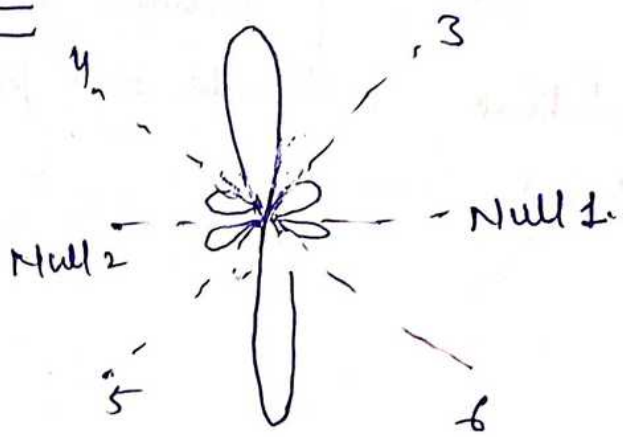


Group Pattern

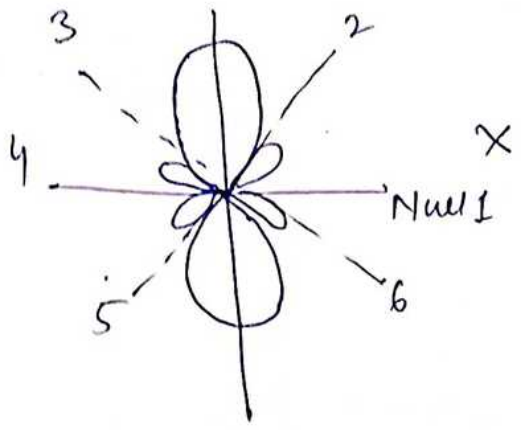
Total Pattern = Unit Pattern × Group Pattern



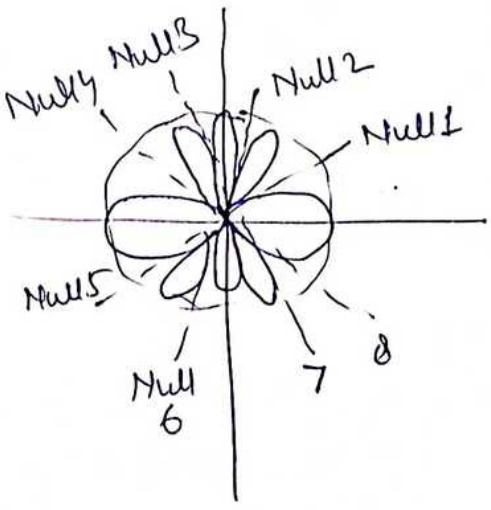
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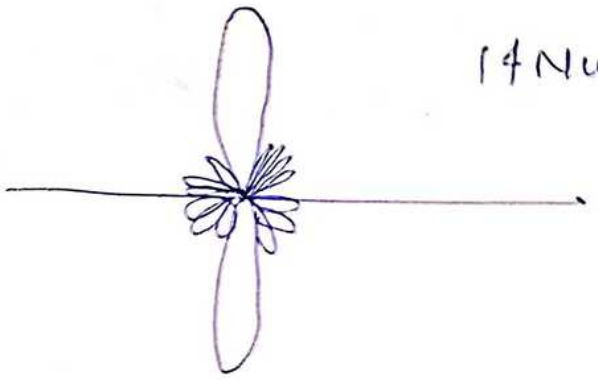
For 8-isotropic Source



Unit Pattern



=



Pattern Multiplication Using Non-isotropic but similar Source

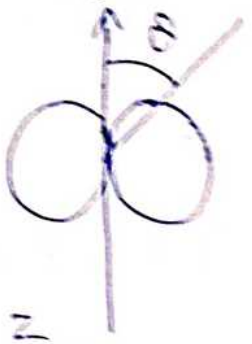
Total Pattern source

$E(\theta, \phi)$ = (Individual Source Pattern) × (Pattern of an array of isotropic source having same locations relative amplitude and phase as the non-isotropic point source)

Size and Pattern of some of Non-isotropic source

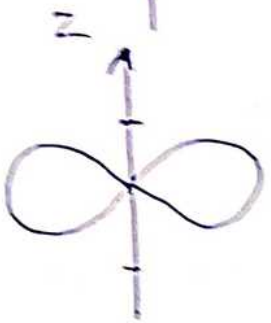
1. Infinitesimal $\left(\ell \leq \frac{\lambda}{50}\right)$

$$E(\theta) = \sin\theta$$



2. Half wave $\left(\ell = \frac{\lambda}{2}\right)$

$$E(\theta) = \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}$$



3. Finite length

$$E(\theta) = \frac{\cos\left(\frac{\beta\ell}{2}\cos\theta\right) - \cos\left(\frac{\beta\ell}{2}\right)}{\sin\theta}$$

For dipole along x-axis

$$\theta \rightarrow \gamma$$

$$\cos\gamma = \sin\theta\cos\phi$$

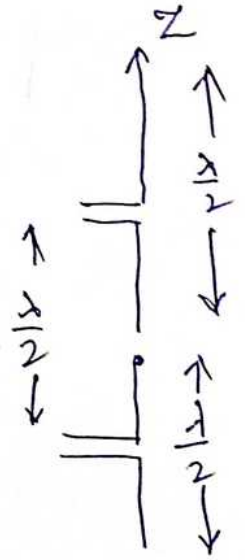
For dipole along y-axis

$$\cos\gamma = \sin\theta\sin\phi$$

For dipole along x-axis

$$E(\gamma) = E(\theta,\phi) = \frac{\cos\left(\frac{\pi}{2}\cos\gamma\right)}{\sin\gamma}$$

$$= \frac{\cos\left(\frac{\pi}{2}\sin\theta\cos\phi\right)}{\sqrt{1-\cos^2\gamma}} = \frac{\cos\left[\frac{\pi}{2}\sin\theta\cos\phi\right]}{\sqrt{1-\sin^2\theta\cos^2\phi}}$$





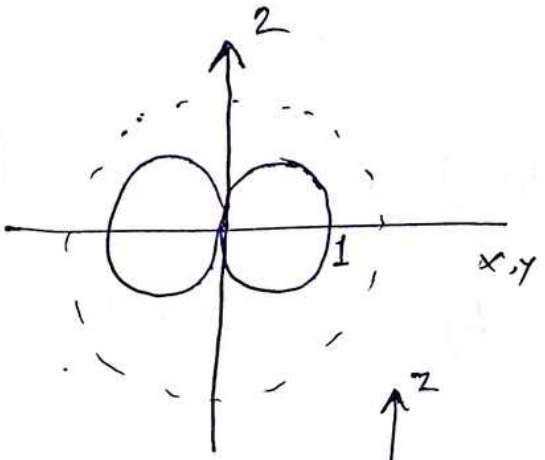
Isotropic Source

$$f(\theta,\phi) = f(\theta) = \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}$$

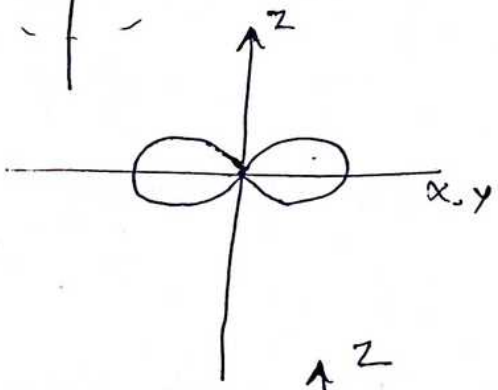
(AF) isotropic $F(\theta,\phi) = \cos\left(\frac{\pi}{2}\cos\theta\right)$

So $E(\theta,\phi) = f(\theta,\phi) \times F(\theta,\phi) = \frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}$

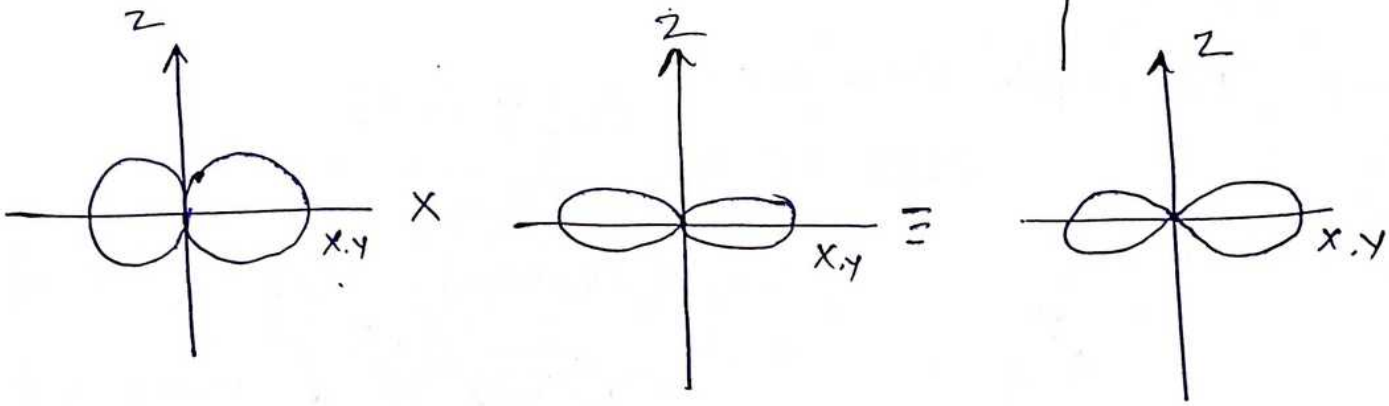
$$f(\theta,\phi) = \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}$$



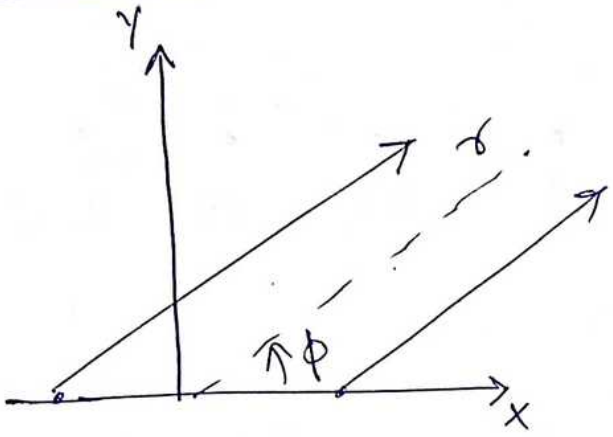
$$F(\theta,\phi) = AF = \cos\left(\frac{\pi}{2}\cos\theta\right)$$

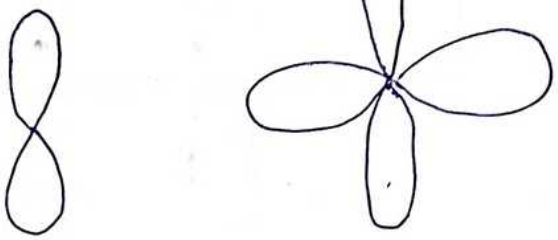


Resultant Pattern = $f(\theta,\phi) \times F(\theta,\phi)$



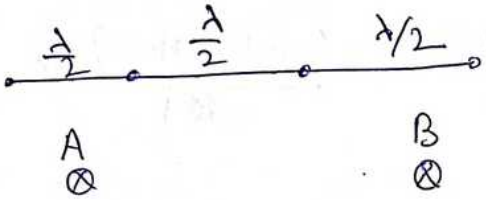
Linear Broadside Arrays with Non-Uniform Amplitude distributions :-

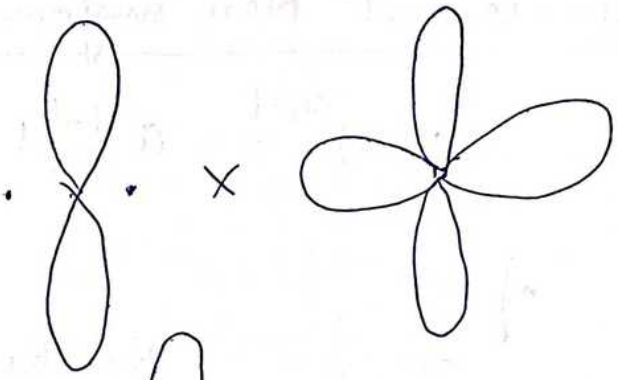




$d \geq \frac{\lambda}{2}$ $\quad$ $d \geq \lambda$

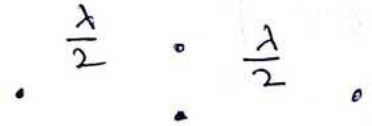
$$E_n = \cos\left[\frac{\pi d}{\lambda}\cos\theta\right]$$

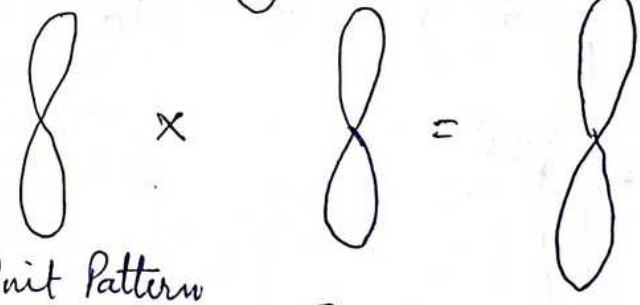




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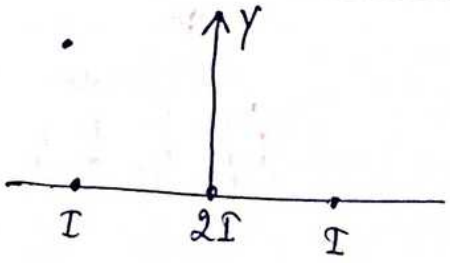






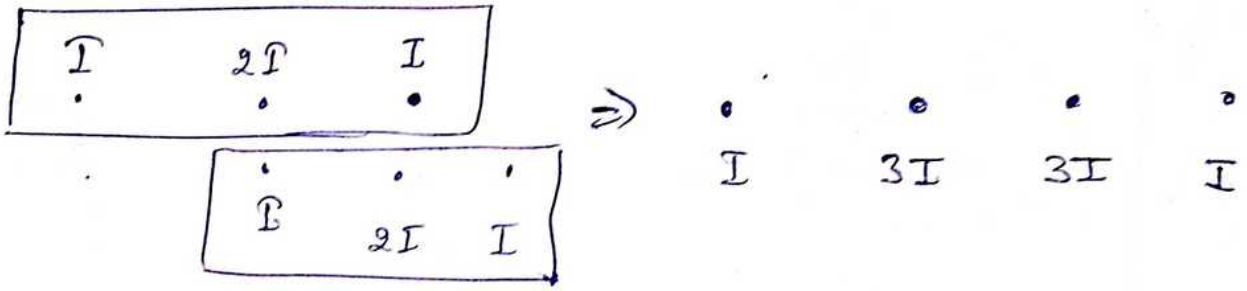
Unit Pattern $\quad$ Group Pattern $\quad$ Resultant

Three element Non-Uniform Array





Four Element Non-uniform Array :-



1 : 3 ! 3 ! 1

Field of unit Pattern : $Cos\left(\frac{\pi d}{\lambda} Cos\theta\right)$

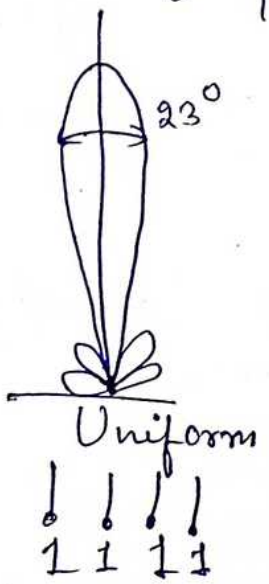
n-element Non uniform array :-

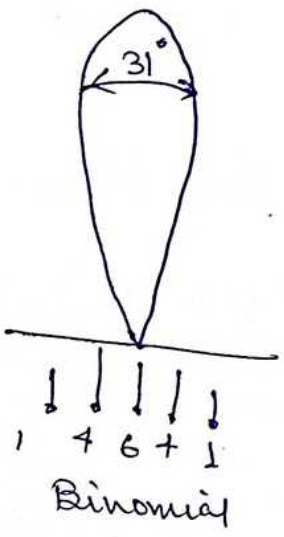
$$(a+b)^{n-1} = a^{n-1} + (n+1)a^{n-2}b + \frac{(n-1)(n-2)}{2!}a^{(n-3)}b^2$$

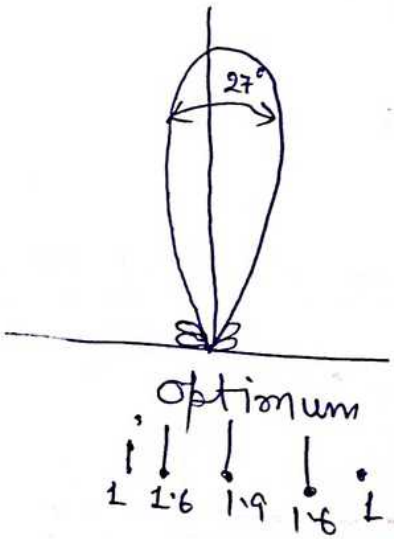
If

n	Relative Amp					
3		1	2	1		
4	1	3	3	1		
5	1	4	6	4	1	
6	1	5	10	10	5	1

Pascal's Triangles









Linear Arrays with Optimum or Dolph-Tchebyscheff or Chebyshev Arrays

Tchebyscheff Polynomial

$$T_m(x) = \cos(m \cos^{-1} x) \quad \text{for } |x| < +1$$

and $$T_m(x) = \cosh(m \cosh^{-1} x) \quad \text{for } |x| > +1$$

Let $\beta = \cos^{-1} x = \frac{\psi}{2}$

$\Rightarrow x = \cos\beta = \frac{\psi}{2}$

For $m=0 \Rightarrow T_0(x) = \cos(m\beta) = \cos(0) = 1$

$$\Rightarrow \boxed{T_0(x) = 1}$$

For $m=1$

$$T_1(x) = \cos(m\beta) = \cos\beta = x$$

$$\Rightarrow \boxed{T_1(x) = x}$$

For $m=2$ $\quad T_2(x) = \cos(2\beta) = 2\cos^2\beta - 1 = 2x^2 - 1$

$$\Rightarrow \boxed{T_2(x) = 2x^2 - 1}$$

$$T_3(x) = \cos(3\beta) = 4\cos^3\beta - 3\cos\beta = 4x^3 - 3x$$

Recursion formula

$$\boxed{T_{m+1}(x) = 2xT_m(x) - T_{m-1}(x)}$$

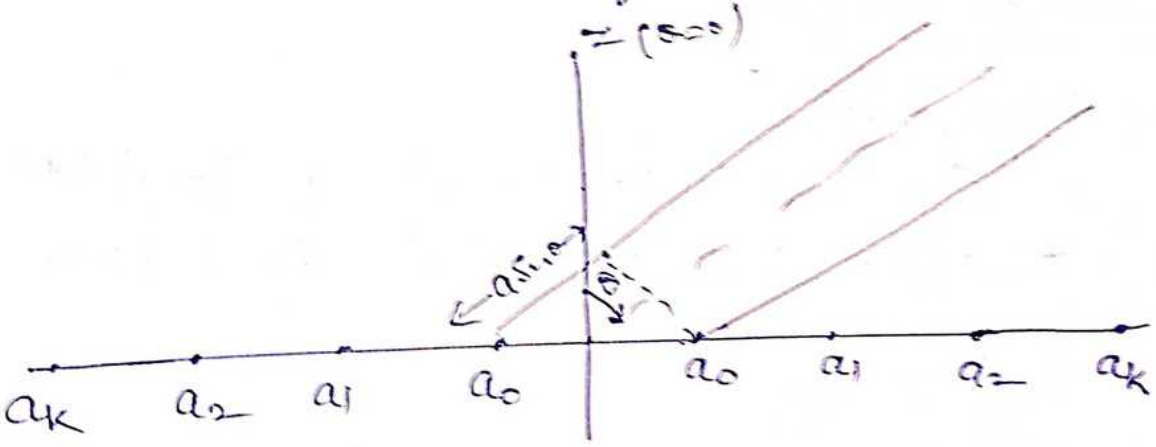
The roots of polynomial ~~occurs when~~ corresponds to null and occurs when

$$\cos(m\delta) = 0$$

$$\Rightarrow m\delta = (2K+1)\frac{\pi}{2} \Rightarrow \delta = \frac{(2K+1)\pi}{2m}, \text{ where}$$

$K = 1, 2, 3.$

(1) For even number of isotropic sources:



$$E_t = 2a_0\cos\left[\frac{\psi}{2}\right] + 2a_1\cos\left[\frac{3\psi}{2}\right] + 2a_2\cos\left[\frac{5\psi}{2}\right] + \cdots + 2a_K\cos\left[\frac{\eta_e - 1}{2}\right]$$

Here $\psi = \beta d \sin\theta$

$\eta_e = 2, 4, 6 \cdots 2(K+1)$, $K = 0, 1, 2 \cdots$

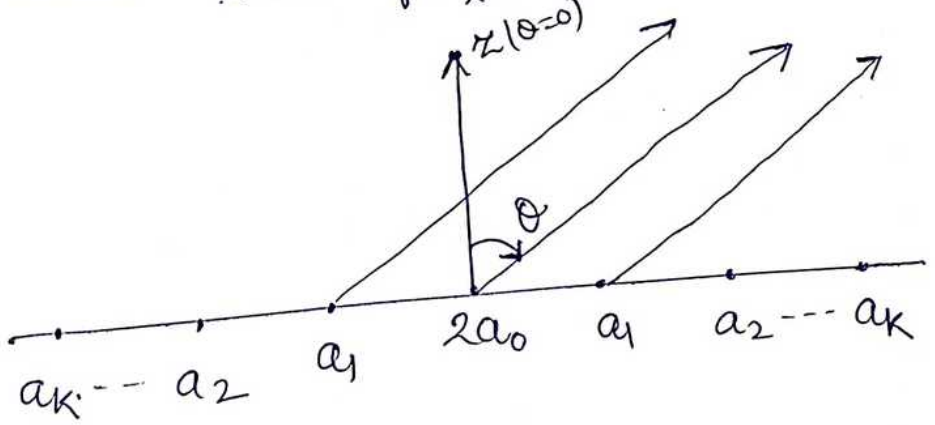
$\Rightarrow \eta_e - 1 = 2(K+1) - 1 = 2K+1$

$\Rightarrow \frac{\eta_e - 1}{2} = \frac{2K+1}{2}$

So $E_t = 2\sum_{K=0}^{N-1} a_K \cos\left(\frac{2K+1}{2}\right)\psi$

Here $N = \frac{\eta_e}{2}$ = No of elements on one side

(ii) For odd number of isotropic sources :-



$$E_{to} = 2a_0 + 2a_1 \cos\psi + 2a_2 \cos 2\psi + \cdots + 2a_K \cos\left(\frac{n_0-1}{2}\right)\psi$$

where $n_0 = 1, 3, 5, \ldots (2K+1)$ and $K = 0, 1, 2, 3$

or $\frac{n_0-1}{2} = K$

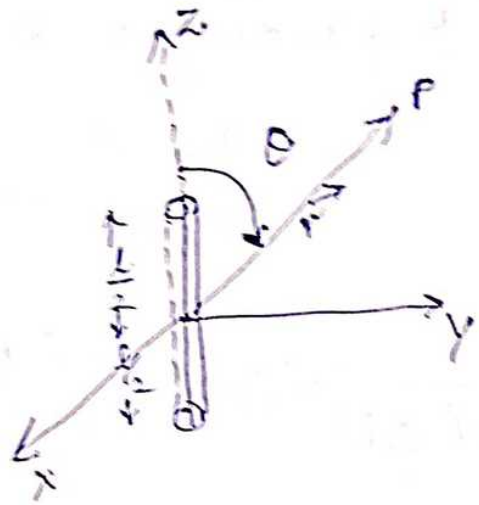
$$\Rightarrow E_{to} = 2 \sum_{K=0}^{N} a_K \cos\left(\frac{n_0-1}{2}\right)\psi$$

where $N = \frac{n_0-1}{2}$ = No of elements on either side

$$E_{te} = \sum_{K=0}^{N-1} a_K \cos(2K+1)\frac{\psi}{2}$$ [Factor 2 is dropped]

$$E_{to} = \sum_{K=0}^{N} a_K \cos(2K)\psi/2$$

Field due short dipole:-



$\vec{I}(z') = I_0 \vec{a_z}$

Step I, Find $\vec{A}$

$$\vec{A}(x,y,z) = \frac{\mu}{4\pi} \int I(x',y',z') \frac{e^{-jkR}}{R} dl'$$

(x,y,z) – Observation Point

(x',y',z') coordinates of source

$dl' = dz'$

$$\boxed{R = r}$$

$$\vec{A}(x,y,z) = \frac{\mu}{4\pi} \int_{-l/2}^{l/2} I_0 \vec{a_z} \frac{e^{-jkr}}{r} dz' = \frac{\mu I_0 l}{4\pi r} e^{-jkr} \vec{a_z}$$

$$\vec{A} = A_z \vec{a_z}$$

Step II Find $\vec{H}$

$$\vec{B} = \mu \vec{H} = \nabla \times \vec{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \vec{a_r} & r\vec{a_\theta} & r\sin\theta\,\vec{a_\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & A_\theta & r\sin\theta A_\phi \end{vmatrix}$$

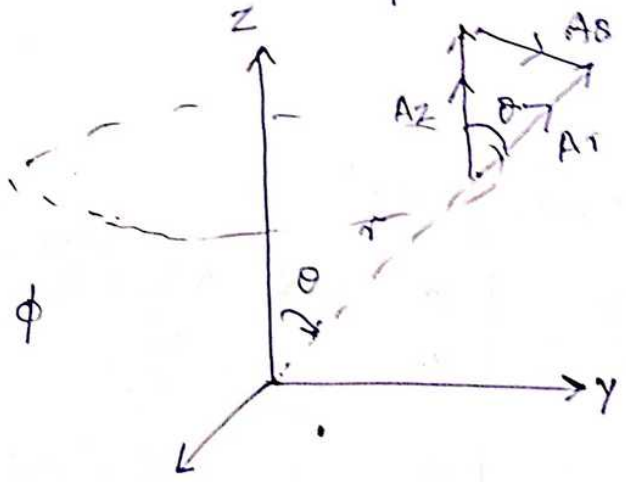
$A_r = A_z \cos\theta$

$A_\theta = -A_z \sin\theta$

$A_\phi = 0$

A_z is same in all ϕ

so $\frac{\partial}{\partial \phi} = 0$



$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \vec{a_r} & r\vec{a_\theta} & r\sin\theta\,\vec{a_\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & 0 \\ A_r & rA_\theta & 0 \end{vmatrix} = \mu \left[H_r \vec{a_r} + H_\theta \vec{a_\theta} + H_\phi \vec{a_\phi} \right]$$

Expanding and comparing

$H_r = 0$

$H_\theta = 0$

$$\frac{1}{r^2 \sin\theta}\begin{vmatrix} \vec{a_r} & r\vec{a_\theta} & r\sin\theta\,\vec{a_\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & 0 \\ 0 & 0 & r\sin\theta H_\phi \end{vmatrix} = \mu H_\phi \vec{a_\phi}$$

$$\frac{1}{r^2 \sin\theta}\left(r\sin\theta\,\vec{a_\phi}\right)\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial}{\partial \theta}(A_r)\right] = \mu H_\phi \vec{a_\phi}$$

$$\Rightarrow H_\phi = \frac{1}{\mu r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial}{\partial \theta}(A_r)\right]$$

$$= \frac{1}{\mu r}\left[\frac{\partial}{\partial r}\left[-rA_z \sin\theta\right] - \frac{\partial}{\partial \theta}(A_z \cos\theta)\right]$$

Putting A_z

$$H_\phi = \frac{1}{\mu r}\left[\frac{\mu I_0 l}{4\pi}\left(\frac{\partial}{\partial r}(-e^{-jkr}\sin\theta)\right] - \frac{\partial}{\partial \theta}\left(\frac{e^{-jkr}}{r}\cos\theta\right)\right]$$

$$H_\phi = \frac{I_0 l}{4\pi r}\sin\theta\,(jk)\left[1 + \frac{1}{jkr}\right]e^{-jkr}$$

$H_r = 0$

$H_\theta = 0$

Step II $\quad \nabla \times \vec{H} = j\omega\varepsilon\vec{E}$

$$\nabla \times \vec{H} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \vec{a_r} & r\vec{a_\theta} & r\sin\theta\,\vec{a_\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & 0 \\ 0 & 0 & r\sin\theta H_\phi \end{vmatrix} = j\omega\varepsilon\left[E_r\vec{a_r} + E_\theta\vec{a_\theta} + E_\phi\vec{a_\phi}\right]$$

$$j\omega\varepsilon E_\theta = \frac{1}{r^2\sin\theta}\left\{-r\frac{\partial}{\partial r}r\sin\theta H_\phi\right\}$$

$$j\omega\varepsilon E_r = \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}(r\sin\theta H_\phi)$$

$K = \omega\sqrt{\mu\varepsilon}$

$$\Rightarrow \frac{K}{\omega\varepsilon} = \frac{\omega\sqrt{\mu\varepsilon}}{\omega\varepsilon} = \sqrt{\frac{\mu}{\varepsilon}} = \eta$$

$$E_r = \frac{I_0 l}{2\pi r^2}\cos\theta(\eta)\left[1 + \frac{1}{jkr}\right]e^{-jkr}$$

$$E_\theta = \frac{I_0 l}{4\pi r}\sin\theta(\eta)(jk)\left[1 + \frac{1}{jkr} + \frac{1}{(jkr)^2}\right]e^{-jkr}$$

$E_\phi = 0$

$H_r = 0$

$H_\theta = 0$

$$H_\phi = \frac{I_0 l}{4\pi r}\sin\theta(jk)\left[1 + \frac{1}{jkr}\right]e^{-jkr}$$

$\frac{1}{r^3}$ — electrostatics field [due to dipole]

$\frac{1}{r^2}$ [Inductive field]

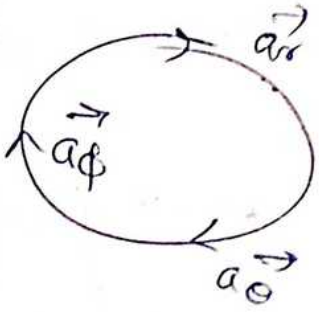
$\frac{1}{r}$ [Far field or Radiation field]

Radiation Resistance of small dipole :-

$$\vec{E} = E_r \vec{a_r} + E_\theta \vec{a_\theta}$$

$$\vec{H} = H_\phi \vec{a_\phi}$$

$\vec{W}$ (Complex Poynting Vector) $= \frac{1}{2}(\vec{E} \times \vec{H}^*)$



$$= \frac{1}{2}\left[E_r \vec{a_r} + E_\theta \vec{a_\theta}\right] \times \left[H_\phi^* \vec{a_\phi}\right]$$

$$= \frac{1}{2}\left[-E_r H_\phi^* \vec{a_\theta} + E_\theta H_\phi^* \vec{a_r}\right]$$

$$\vec{W} = W_r \vec{a_r} + W_\theta \vec{a_\theta} + W_\phi \vec{a_\phi}$$

$$\Rightarrow W_r = \frac{1}{2} E_\theta H_\phi^*$$

$$W_\theta = -\frac{1}{2} E_r H_\phi^*$$

$$W_\phi = 0$$

So $W_r = \frac{1}{2} E_\theta H_\phi^*$

$$= \frac{1}{2} \frac{I_0 l}{4\pi r} \sin\theta(\eta)\, jk \left[1 + \frac{1}{jkr} + \frac{1}{(jkr)^2}\right] e^{-jkr} \times \frac{I_0 l}{4\pi r} \sin\theta(-jk) \left[1 - \frac{1}{jkr}\right] e^{jkr}$$

$$= -\frac{1}{2} \left(\frac{I_0 l}{4\pi r}\right)^2 \sin^2\theta\, \eta\, (jk)^2 \left[1 + \frac{1}{jkr} + \frac{1}{(jkr)^2} - \frac{1}{jkr} - \frac{1}{(jkr)^2} - \frac{1}{(jkr)^3}\right]$$

$$= \frac{I_0^2 l^2}{32\pi^2 r^2} \sin^2\theta(\eta) \left(\frac{2\pi}{\lambda}\right)^2 \left[1 - \frac{j}{(kr)^3}\right]$$

$$W_r = \frac{\eta}{8} \left|\frac{I_0 l}{\lambda}\right|^2 \frac{\sin^2\theta}{r^2} \left[1 - \frac{j}{(kr)^3}\right]$$

and $W_\theta = -\frac{1}{2} E_r H_\phi^* = -\frac{1}{2} \frac{I_0 l}{2\pi r^2} \cos\theta(\eta) \left[1 + \frac{1}{jkr}\right] e^{-jkr} \times \frac{I_0 l}{4\pi r} \sin\theta(-jk) \left[1 - \frac{1}{jkr}\right] e^{+jkr}$

$$W_\theta = \frac{\eta k |I_0 l|^2}{16\pi^2 r^3} \cos\theta \sin\theta \left[1 + \frac{1}{(kr)^2}\right]$$

Power $P = \oint \vec{W} \cdot d\vec{s} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} (W_r \vec{a_r} + W_\theta \vec{a_\theta}) \cdot (r^2 \sin\theta\, d\theta\, d\phi\, \vec{a_r})$

$$P = \int_0^{2\pi} \int_0^{\pi} W_r r^2 \sin\theta\, d\theta\, d\phi$$

$$P = \phi \Big|_0^{2\pi} \int_0^{\pi} \frac{\eta}{8} \left|\frac{I_0 l}{\lambda}\right|^2 \frac{\sin^2\theta}{r^2} \left[1 - \frac{j}{(kr)^3}\right] r^2 \sin\theta\, d\theta$$

$$= 2\pi \frac{\eta}{8} \left|\frac{I_0 l}{\lambda}\right|^2 \left[1 - \frac{j}{(kr)^3}\right] \int_0^{\pi} \sin^3\theta\, d\theta$$

$$= 2\pi \cdot \frac{\eta}{8} \left| \frac{I_0 l}{\lambda} \right|^2 \left[1 - \frac{j}{(kr)^3} \right] \times \frac{4}{3} = \frac{\eta \pi}{3} \left| \frac{I_0 l}{\lambda} \right|^2 \left[1 - \frac{j}{(kr)^3} \right]$$

$$P_{rad} = \eta \frac{\pi}{3} \left| \frac{I_0 l}{\lambda} \right|^2$$

$$P_{rad} = \frac{1}{2} |I_0|^2 R_r$$

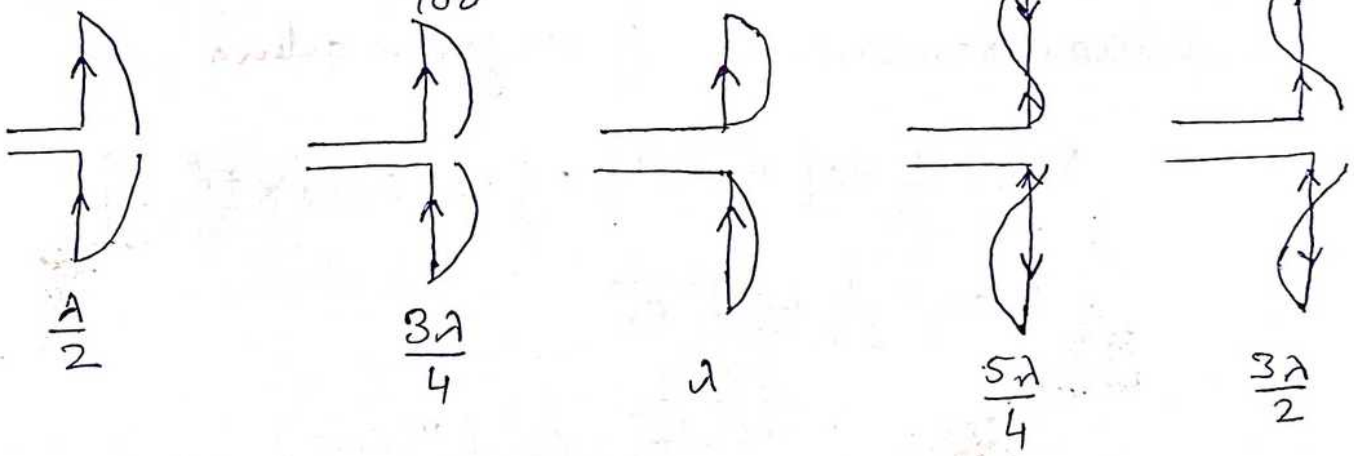
$$\Rightarrow \frac{1}{2} |I_0|^2 R_r = \eta \left(\frac{\pi}{3} \right) \left| \frac{I_0 l}{\lambda} \right|^2$$

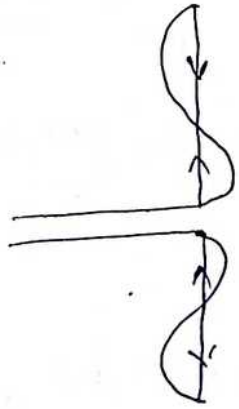
$$R_r = 2\eta \frac{\pi}{3} \left(\frac{l}{\lambda} \right)^2$$

$$\Rightarrow R_r = 2 \times 120 \pi \cdot \frac{\pi}{3} \left(\frac{l}{\lambda} \right)^2 = 80\pi^2 \left(\frac{l}{\lambda} \right)^2$$

The Thin Linear Antenna's

$d << \frac{\lambda}{100}$



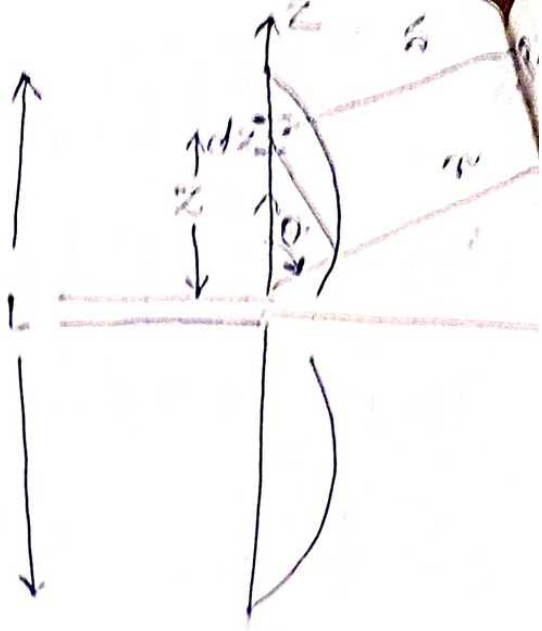


Retarded value of current at any point z

$$|I| = I_0 \sin\left[\frac{2\pi}{\lambda}\left(\frac{L}{2} \pm z\right)\right] e^{j\omega\left[t - \frac{r}{c}\right]}$$

$\frac{L}{2} + z$ when $z < 0$

$\frac{L}{2} - z$ when $z > 0$



$$H_\phi = \frac{j I_0}{2\pi r}\left[\frac{\cos\left(KL\cos\theta/2\right) - \cos\left(\frac{KL}{2}\right)}{\sin\theta}\right]$$

$$E_\theta = \frac{j\omega\eta[I_0]}{r}\left[\frac{\cos\left(KL\cos\theta/2\right) - \cos\left(\frac{KL}{2}\right)}{\sin\theta}\right]$$

Radiation Resistance of $\frac{\lambda}{2}$ Half Wave dipole

$$W_{av} = \frac{1}{2}\mathrm{Re}\left[\vec{E} \times \vec{H}^*\right] = \frac{1}{2}\mathrm{Re}\left[E_\theta \vec{a}_\theta * \frac{E_\theta^*}{\eta}\vec{a}_\phi\right]$$

$$W_{av} = \frac{1}{2\eta}|E_\theta|^2 \vec{a}_r$$

$$= \frac{1}{2\eta}\frac{60^2 I_0^2}{r^2}\frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}\vec{a}_r$$

$$= \frac{15 I_0^2}{\pi r^2}\frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}\vec{a}_r$$

$$P_{rad} = \oint \vec{W}_{av} \cdot d\vec{s}$$

$$= \int_0^{2\pi}\int_0^{\pi}\left(\frac{15 I_0^2}{\pi r^2}\frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}\vec{a}_r\right)\cdot\left(r^2\sin\theta\, d\theta\, d\phi\, \vec{a}_r\right)$$

$$P_{rad} = \frac{15 I_0^2}{\pi} [2\pi] \int_0^{\pi} \frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta} d\theta$$

$$= 30 I_0^2 \int_0^{\pi} \frac{\cos^2\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta} d\theta$$

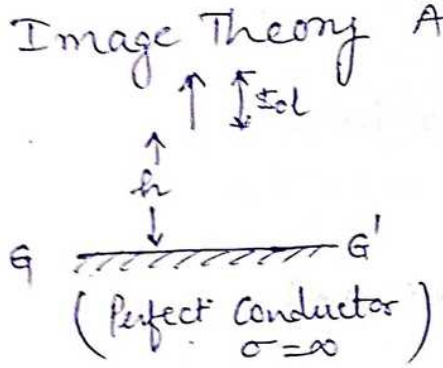
$$P_{rad} \cong 36.56\, I_0^2 = I_{rms}^2 R_r \left(\frac{I_0}{\sqrt{2}}\right)^2 R_r$$

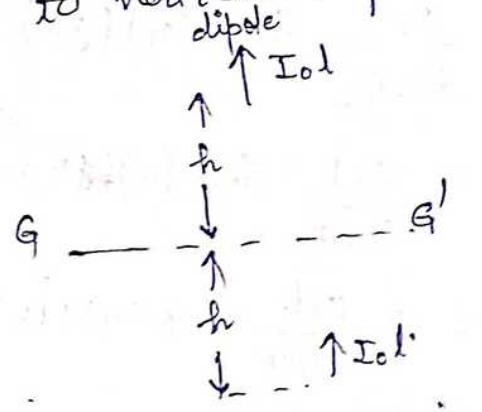
$$\Rightarrow$$

$$\Rightarrow \boxed{R_r = 36.56 \times 2 = 73.2}$$

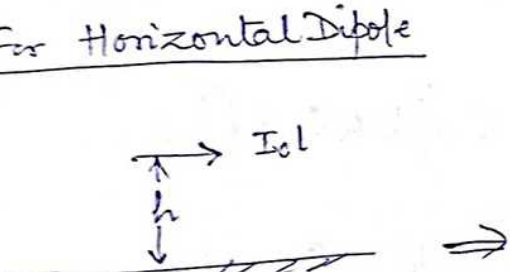
Antenas above ground plane:-

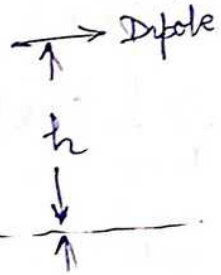
(i) Image Theory Applied to vertical Dipole

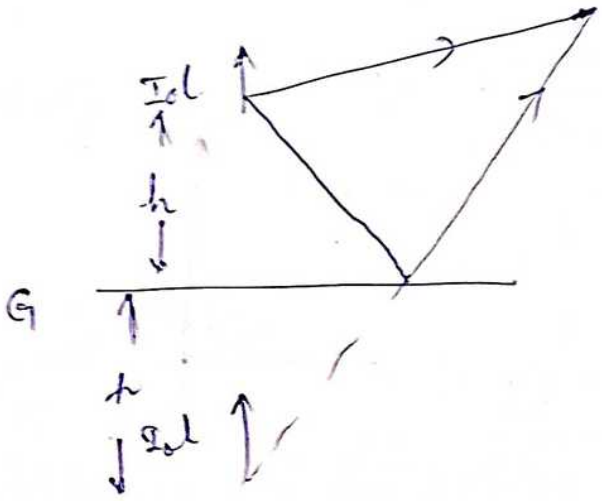


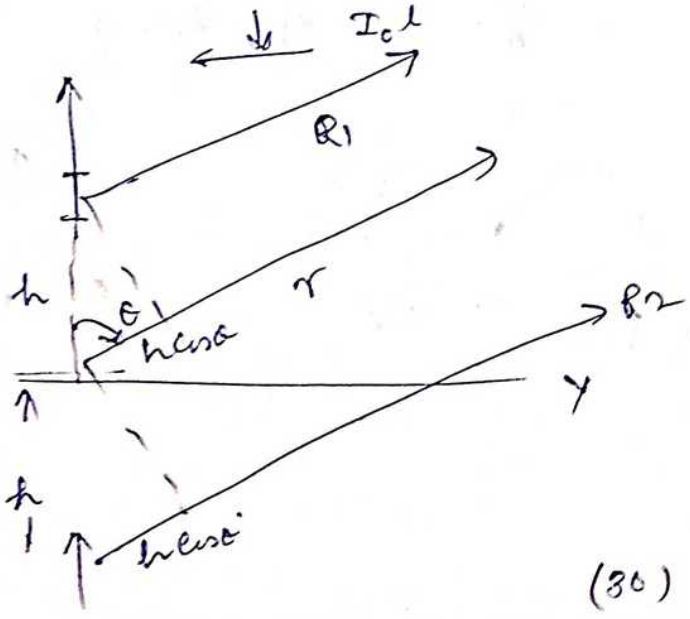


(ii) For Horizontal Dipole









$$E_\theta = \frac{I_0 l}{4\pi r} \sin\theta(\eta)(jk) e^{-jkr}$$

$R_1 = r - h\cos\theta$

$R_2 = r + h\cos\theta$

if $r \gg h$, $R_1 \approx R_2 \approx r$

$$E_{\theta 1} = \frac{I_0 l}{4\pi R_1} \sin\theta(\eta)(jk) e^{-jkR_1}$$

$$E_{\theta 2} = \frac{I_0 l}{4\pi R_2} \sin\theta(\eta)(jk) e^{-jkR_2}$$

$$\Rightarrow E_{\theta 1} = \frac{I_0 l}{4\pi r} \sin\theta(\eta)(jk) e^{-jk(r - h\cos\theta)}$$

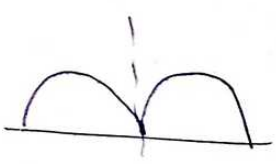
and $E_{\theta 2} = \frac{I_0 l}{4\pi r} \sin\theta(\eta)(jk) e^{-jk(r + h\cos\theta)}$

$E_\theta = E_{\theta 1} + E_{\theta 2}$

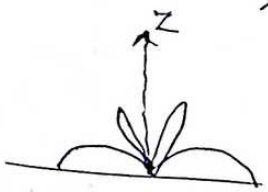
$$= \frac{I_0 l}{4\pi r}(jk)\eta \sin\theta\left[2\cos\left[kh\cos\theta\right]\right]$$

No of side lobe on each side $= \frac{2h}{\lambda} + 1$

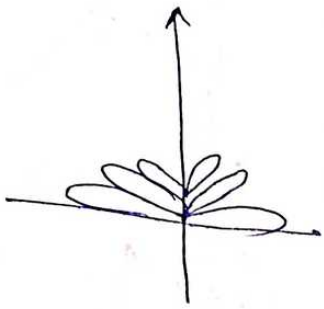
if $h = \frac{\lambda}{2}$, $\frac{2\times\lambda}{2\lambda} + 1 = 2$



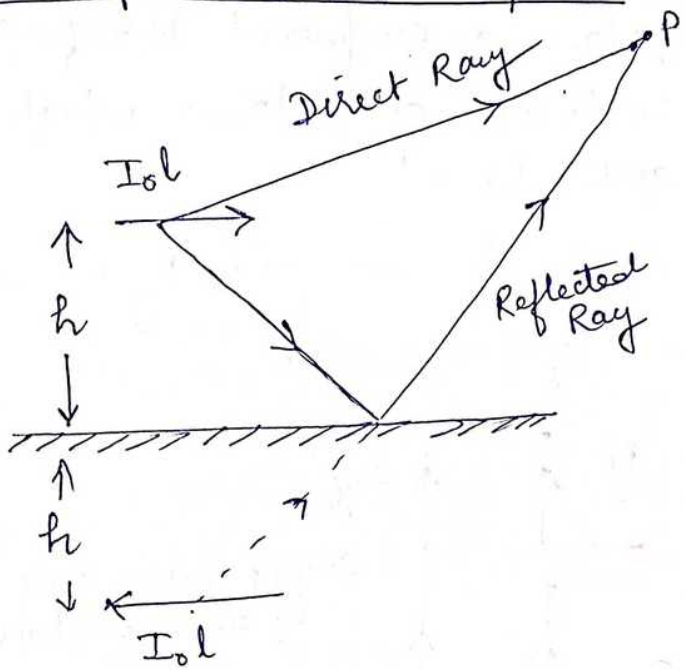
if $h = 0$

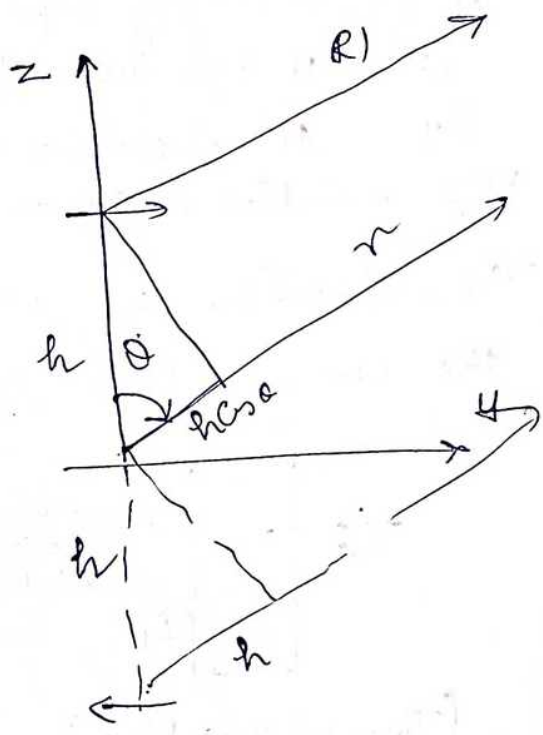


if $h = \lambda$, $\frac{2\times\lambda}{2\lambda} + 1$ (3)



...lysis of Horizontal dipoles :-





$$E_\theta = \frac{I_0 l}{4\pi r} \sin\theta(\eta)(jk) e^{-jkr}$$

$$E_{\psi_1} = \frac{I_0 l}{4\pi R_1} \sin\theta\psi(\eta) e^{-jkR_1}$$

$$E_{\psi_2} = |\Gamma| \frac{I_0 l}{4\pi R_2} \sin\psi(\eta) e^{-jkR_2}$$

$$R_1 \cong r - h\cos\theta$$

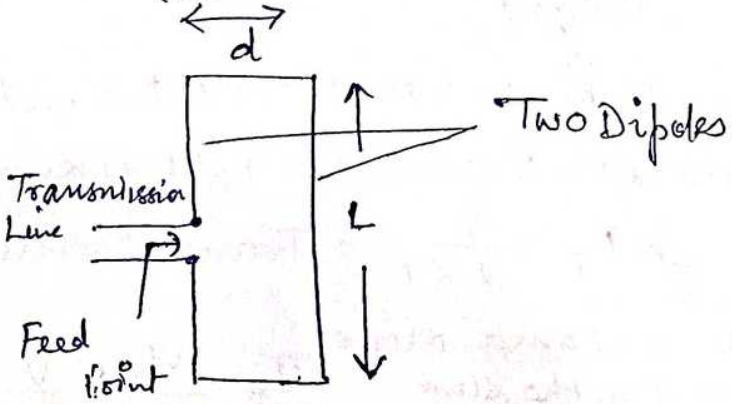
$$R_2 \cong r + h\cos\theta$$

$$\Rightarrow \boxed{R_1 \sim R_2 \sim r}$$

$$E = E_{\psi_1} + E_{\psi_2} = \frac{I_0 l}{4\pi r} \sin\psi(\eta)(jk) e^{-jkr} \left[2j \sin(kh\cos\theta)\right]$$

$$\text{No of lobes} \cong 2\left(\frac{h}{\lambda}\right)$$

Folded Dipole:

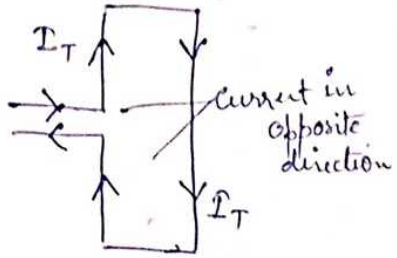


It has two very important Properties

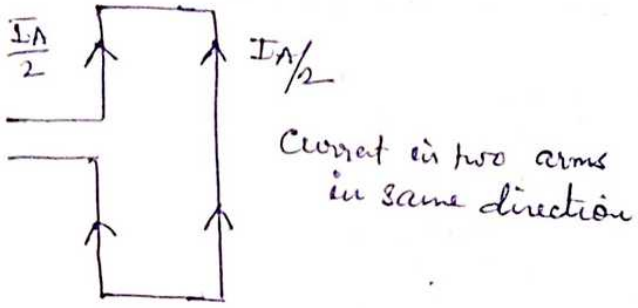
(I) It has high I/P impedance as compared to half wave dip

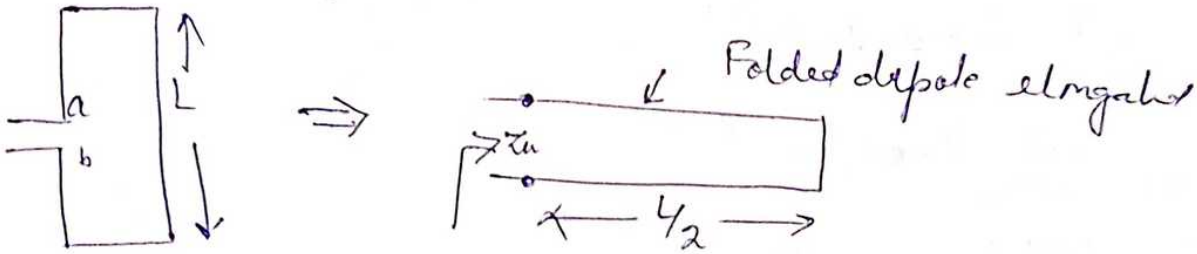
(II) It increases the bandwidth of antenna which is very essential in FM and TV applications.

The operation of folded dipole is analysed by considering the current to be composed of two modes

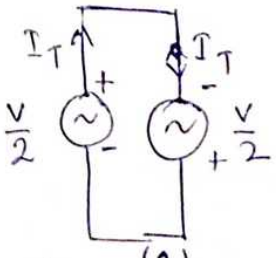


(Transmission Line Mode)



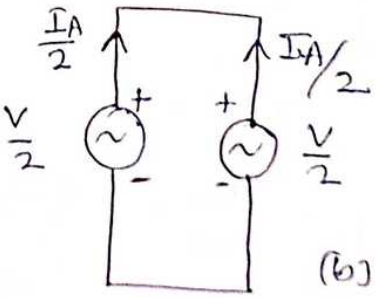


$$Z_T = j Z_0 \tan(\beta \ell/2)$$



(a)

(i) Transmission line Current



(b)

(b) Antenna Mode

if we superimposed these two mode then total voltage on left element is V while on right is zero.

$$I_T = \frac{V}{2 Z_T}$$: Transmission line current

For the antenna mode total current is the sum of current in each arm

$$I_A = \frac{V/2}{Z_D} + \frac{V}{2 Z_D}$$

The length of a resonant dipole

$$L = \frac{\lambda}{2} = \frac{c}{2f} = \frac{3\times10^8}{2\times f_{MHz}\times10^6} = \frac{150}{f_{MHz}}\,(m) = \frac{150\times3.28}{f_{MHz}}$$

$$L = \frac{492}{f_{MHz}}\ feet$$

When we consider a velocity factor of 95%.

$$L = \frac{\lambda}{2} = \frac{492\times0.95}{f_{MHz}} = \frac{492(1-0.05)}{f_{MHz}}\ (feet)$$

for $L = \frac{2\lambda}{2}$

$$L = \frac{492(2-0.05)}{f_{MHz}}\ feet$$

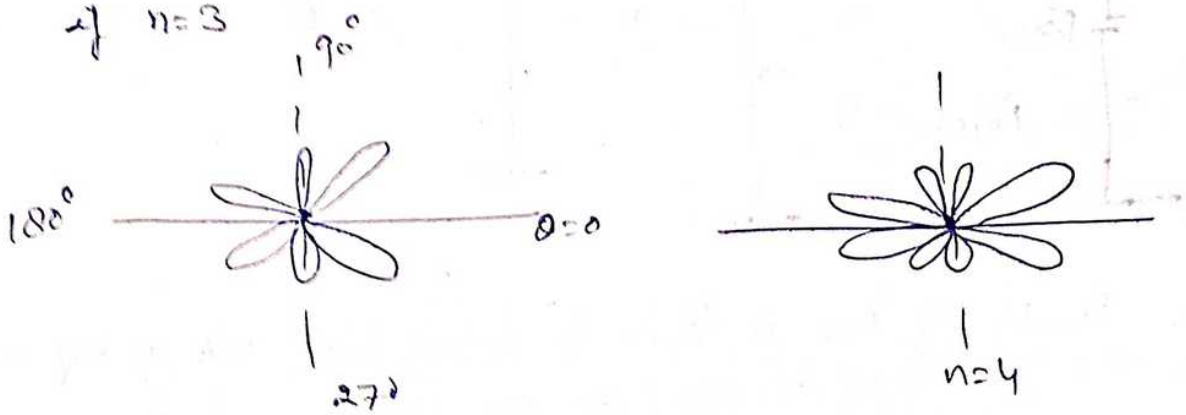
For a long wire antenna of $n\left(\frac{\lambda}{2}\right)$ length

$$\boxed{L = n\left(\frac{\lambda}{2}\right) = \frac{492(n-0.05)}{f_{MHz}}\ feet}$$

Field $E(r,\theta,\phi) = \frac{60\,I_{rms}}{r}\,\frac{\left[\cos\frac{n\pi}{2}\cos\theta\right]}{\sin\theta}$, n is odd

$E(r,\theta,\phi) = \frac{60\,I_{rms}}{r}\,\frac{\left[\sin\frac{n\pi}{2}\cos\theta\right]}{\sin\theta}$, n is even

If n=3



The total voltage on left element is V and total current is $I_{Total} = I_T + \frac{I_A}{2}$

$$Z_{in} = \frac{V}{I_{Total}} = \frac{V}{I_T + \frac{I_A}{2}} = \frac{V}{\frac{V}{2Z_T} + \frac{V}{4Z_D}} = \frac{4Z_T Z_D}{Z_T + 2Z_D}$$

if $L = \frac{\lambda}{2}$

$$Z_T = jZ_0 \tan(\beta l) = jZ_0 \tan\left(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{2}\right) = \infty$$

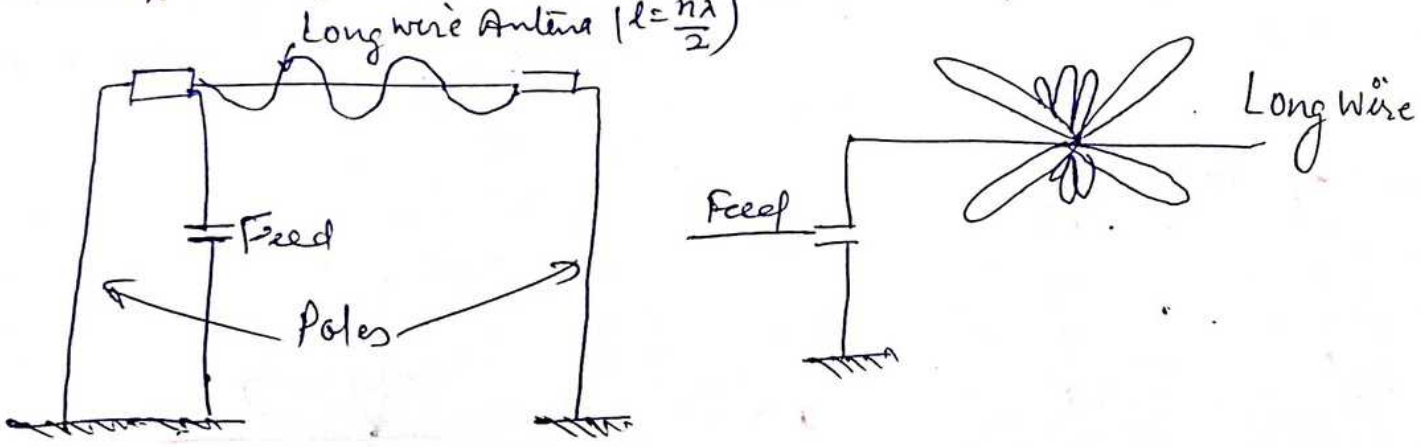
$$\Rightarrow Z_{in} = \frac{4Z_T Z_D}{Z_T + 2Z_D} = \frac{4Z_D}{1 + \frac{2Z_D}{Z_T}} = 4Z_D$$

$Z_D = 73\,\Omega$

$$\Rightarrow \boxed{Z_{in} = 4 \times 73 = 292\,\Omega}$$

Long Wire Antena :- The long wire antena is a generally single long wire, generally two or more wavelength ($4\frac{\lambda}{2}$ or more) long at the operating wavelength.

Resonant Long Wire Antena :-



When the length of the antena is integral multiple of half wavelengths. the antena is said to be resonat antena